



TIA Exam P Online Course

Lesson Handouts

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Exam P Lesson Handouts

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Fundamentals of Probability

1

Basic Principles

Unions and Intersections

Exercise

Basic Principles of Probability

2

1. $0 \leq P[A] \leq 1$
2. $P[S] = 1 \quad P[\emptyset] = 0$
3. If $A_1 \cap A_2 = \emptyset$, then $P[A_1 \cup A_2] = P[A_1] + P[A_2]$

Example 1

Roll a fair six sided die

$S = \text{"Sample Space"} = \text{all possible outcomes} = \{1, 2, 3, 4, 5, 6\}$

$$A = \{3, 4, 5, 6\} \quad P[A] = \frac{4}{6}$$

$$A_1 = \{3, 4, 5\} \quad A_2 = \{6\} \quad A_1 \cup A_2 = A \quad A_1 \cap A_2 = \emptyset$$

$$P[A_1] = \frac{3}{6} \quad P[A_2] = \frac{1}{6} \quad P[A_1] + P[A_2] = P[A]$$

Basic Principles of Probability

3

1. $0 \leq P[A] \leq 1$
2. $P[S] = 1 \quad P[\emptyset] = 0$
3. If $A_1 \cap A_2 = \emptyset$, then $P[A_1 \cup A_2] = P[A_1] + P[A_2]$

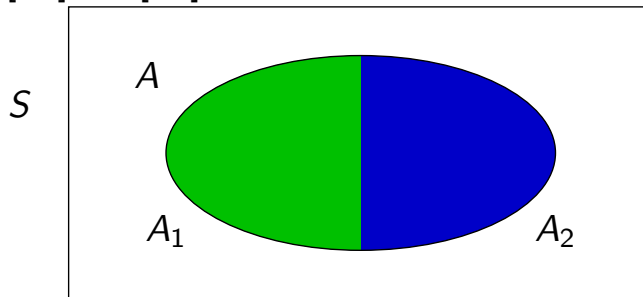
Example 2

Pick a point “at random” from S , where total area of $S = 1$

$P[A] = \text{area of } A \quad A_1 \cap A_2 = \emptyset, A_1 \cup A_2 = A$

Area of $A = \text{area of } A_1 + \text{area of } A_2$

$P[A] = P[A_1] + P[A_2]$



Unions and Intersections

4

1. $0 \leq P[A] \leq 1$
2. $P[S] = 1 \quad P[\emptyset] = 0$
3. $P[A \cup B] = P[A] + P[B] - P[A \cap B]$

Example 1

Roll a fair six sided die

$S = \text{“Sample Space”} = \text{all possible outcomes} = \{1, 2, 3, 4, 5, 6\}$

$A = \text{“roll an odd number”} = \{1, 3, 5\} \quad P[A] = \frac{3}{6}$

$B = \text{“roll a 3 or less”} = \{1, 2, 3\} \quad P[B] = \frac{3}{6}$

$A \cup B = \{1, 2, 3, 5\} \quad A \cap B = \{1, 3\}$

$$P[A \cup B] = \frac{4}{6} = \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = P[A] + P[B] - P[A \cap B]$$

Unions and Intersections

5

1. $0 \leq P[A] \leq 1$
2. $P[S] = 1 \quad P[\emptyset] = 0$
3. $P[A \cup B] = P[A] + P[B] - P[A \cap B]$

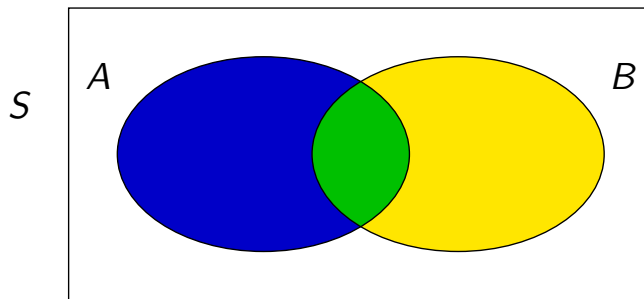
Example 2

Pick a point “at random” from S , where total area of $S = 1$

$P[A] = \text{area of } A$.

Area of $A \cup B = \text{area of } A + \text{area of } B - \text{area of } A \cap B$

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$



Exercise

6

The probability that a visit to a primary care physician's (PCP) office results in either lab work or referral to a specialist is 60%. Of those coming to a PCP's office, 35% are referred to specialists and 50% require lab work. Determine the probability that a visit to a PCP's office results in both lab work and referral to a specialist.

Exercise

6

The probability that a visit to a primary care physician's (PCP) office results in either lab work or referral to a specialist is 60%. Of those coming to a PCP's office, 35% are referred to specialists and 50% require lab work. Determine the probability that a visit to a PCP's office results in both lab work and referral to a specialist.

Let L denote the event that the trip requires lab work, and S the event that it results in a referral to a specialist.

We are given:

$$P[L \cup S] = 0.60 \quad P[S] = 0.35 \quad P[L] = 0.50$$

and we want $P[L \cap S]$

$$P[L \cup S] = P[S] + P[L] - P[L \cap S]$$

$$0.60 = 0.35 + 0.50 - P[L \cap S]$$

$$P[L \cap S] = \boxed{0.25}$$

Complements

1

Recap

Complements

Exercise

Recap

2

Previously, saw that

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$

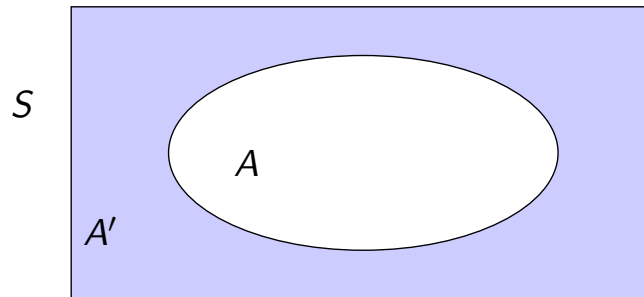
Also had special case: If $A \cap B = \emptyset$

$$P[A \cup B] = P[A] + P[B]$$

Want now to look at another important special case.

Complements

3



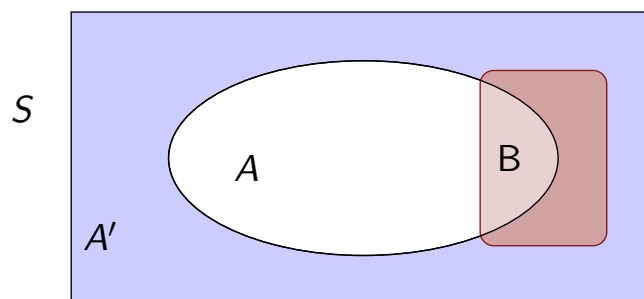
$A^c = A' =$ everything in S but not in A .

$$A \cap A' = \emptyset \quad A \cup A' = S$$

E.g., if we roll a fair six sided die and $A =$ roll a 3 or higher,
 then $S = \{1, 2, 3, 4, 5, 6\}$, $A = \{3, 4, 5, 6\}$ and $A' = \{1, 2\}$

Complements

4



$$A \cap A' = \emptyset \quad A \cup A' = S$$

$$P[A] + P[A'] = P[A \cup A'] = P[S] = 1$$

$$P[A'] = 1 - P[A]$$

Also, $(A')' = A$ so $P[A] = 1 - P[A']$

$B = (A \cap B) \cup (A' \cap B)$ and $(A \cap B) \cap (A' \cap B) = \emptyset$

so $P[B] = P[A \cap B] + P[A' \cap B]$

Exercise

5

In a group of patients with shoulder injuries, 20% visit a physical therapist but not a chiropractor, and 75% visit at least one of these. The probability that a patient visits a chiropractor exceeds by 0.1 the probability of visiting a physical therapist. Find the probability that a randomly chosen member of this group visits a physical therapist.

Exercise

5

In a group of patients with shoulder injuries, 20% visit a physical therapist but not a chiropractor, and 75% visit at least one of these. The probability that a patient visits a chiropractor exceeds by 0.1 the probability of visiting a physical therapist. Find the probability that a randomly chosen member of this group visits a physical therapist.

Let T = visit a physical therapist, and C = visit a chiropractor.

$$\begin{aligned}
 P[T \cap C'] &= 0.20 & P[T \cup C] &= 0.75 \\
 P[C] &= 0.1 + P[T] \\
 P[T \cup C] &= P[C] + P[T] - P[T \cap C] \\
 0.75 &= 0.1 + P[T] + P[T] - (P[T] - P[T \cap C']) \\
 0.75 &= 0.1 + P[T] + P[T \cap C'] \\
 P[T] &= 0.75 - 0.1 - 0.2 \\
 P[T] &= \boxed{0.45}
 \end{aligned}$$

Venn Diagrams

1

Venn Diagrams

Examples

Mutually Exclusive Events

Exercises

Venn Diagrams

2

Venn diagrams are a visual way to represent unions and intersections.

They allow us to easily deal with more complicated cases than what we have seen so far.

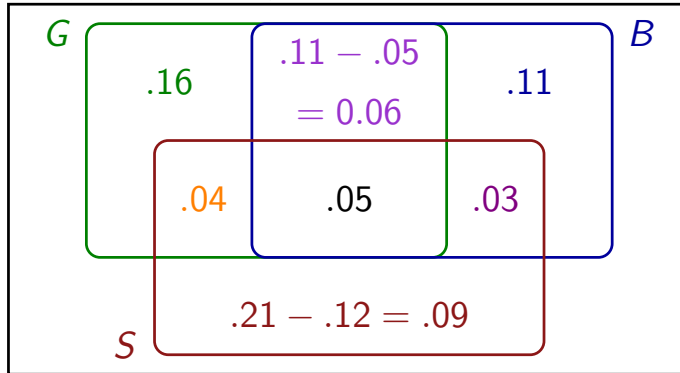
Example 1

3

A survey of a group's viewing habits found the following:

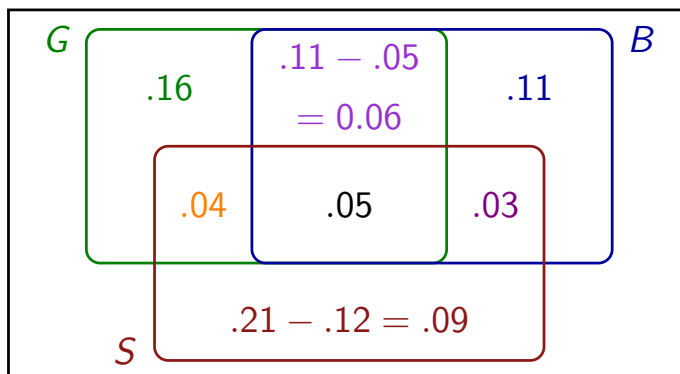
1. 31% watched gymnastics, 25% watched baseball, and 21% watched soccer
2. 11% watched gymnastics and baseball, 8% watched baseball and soccer, and 9% watched gymnastics and soccer
3. 5% watched all three sports.

Find the percentage of the group that watched none of the three sports.



Example 1

4



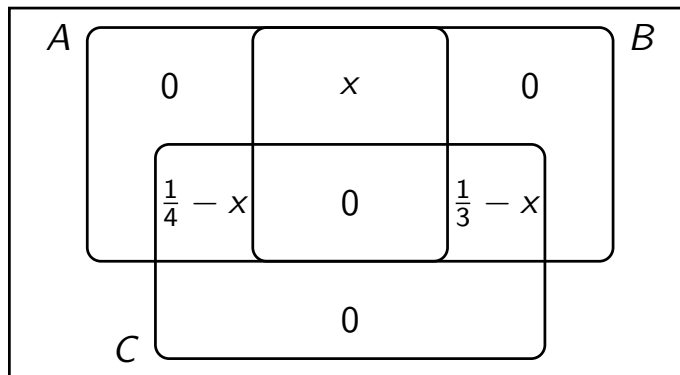
We want the probability that someone watches none of these sports, so we want

$$1 - 0.16 - 0.06 - 0.11 - 0.04 - 0.05 - 0.03 - 0.09 = \boxed{0.46}$$

Example 2

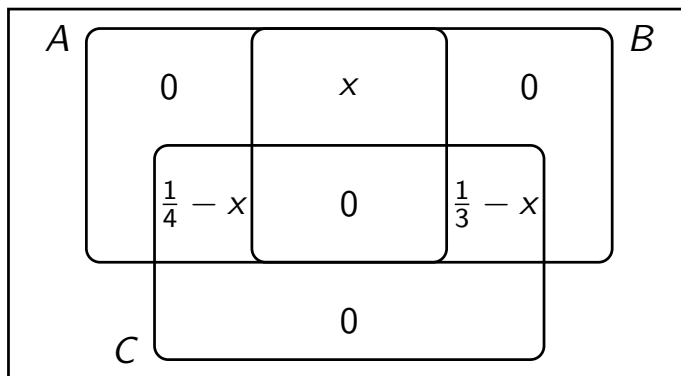
5

In a company's health care plan, employees may choose exactly two of the coverages A, B, and C, or they may choose none of them. The proportions of the employees that choose A, B, and C are $\frac{1}{4}$, $\frac{1}{3}$, and $\frac{5}{12}$, respectively. Find the probability that a randomly chosen employee will choose none of A, B, or C.



SOA #15; S.01.31

6



$$P[C] = \frac{5}{12} = \left(\frac{1}{4} - x\right) + \left(\frac{1}{3} - x\right) = \frac{7}{12} - 2x, \quad x = \frac{1}{12}$$

$$\begin{aligned} P[(A \cup B \cup C)'] &= 1 - x - \left(\frac{1}{4} - x\right) - \left(\frac{1}{3} - x\right) \\ &= 1 - \frac{1}{12} - \frac{3-1}{12} - \frac{4-1}{12} = \frac{1}{2} \end{aligned}$$

Example 2

7

Alternatively, $P[A] + P[B] + P[C]$ counts each person who buys coverage twice since each person buys exactly 2 coverages. $P[A \cup B \cup C]$ counts each person who buys coverage once, so

$$P[A] + P[B] + P[C] = 2 \cdot P[A \cup B \cup C]$$

$$\frac{1}{4} + \frac{1}{3} + \frac{5}{12} = 2 \cdot P[A \cup B \cup C]$$

$$\frac{1}{2} = P[A \cup B \cup C]$$

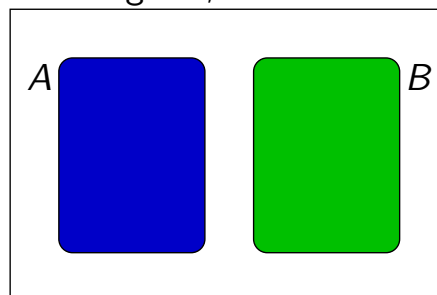
$$P[(A \cup B \cup C)'] = 1 - \frac{1}{2} = \boxed{\frac{1}{2}}$$

Mutually Exclusive Events

8

A and B are said to be mutually exclusive (or mutually disjoint) events if $A \cap B = AB = \emptyset$. Note that this means that $P[A \cap B] = P[AB] = 0$.

From the point of view of a Venn diagram, we can draw them as non-overlapping sets.



If we have 4 or more events, typically some of them will be mutually exclusive to simplify any Venn diagrams we need to draw.

Exercise

9

In a survey of gaming habits:

1. 30% played TLOU, 25% played Celeste, 10% played FFVII and 29% played BG3.
2. No one played both TLOU and FFVII, and no one played both Celeste and BG3.
3. 13% played only Celeste, 26% played only BG3, but no one played only FFVII.

Find the percentage of the group that played none of the four.

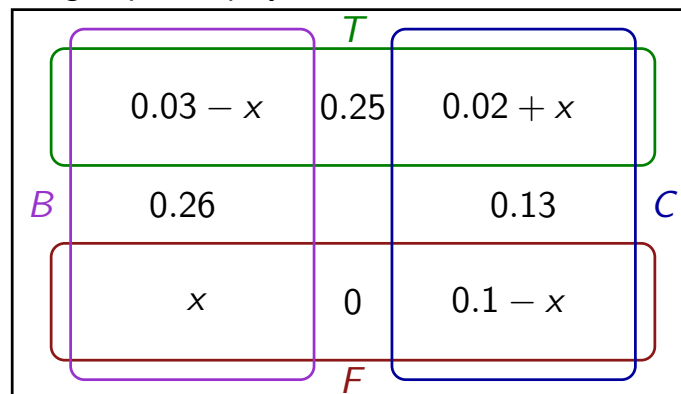
Exercise

9

In a survey of gaming habits:

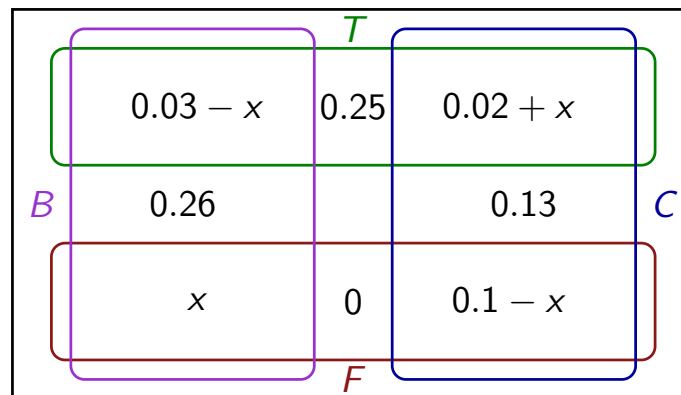
1. 30% played TLOU, 25% played Celeste, 10% played FFVII and 29% played BG3.
2. No one played both TLOU and FFVII, and no one played both Celeste and BG3.
3. 13% played only Celeste, 26% played only BG3, but no one played only FFVII.

Find the percentage of the group that played none of the four.



Exercise (cont)

10



We want the probability that someone plays none of these 4, which is

$$\begin{aligned}
 & 1 - (0.03 - x + 0.25 + 0.02 + x) - 0.26 - 0.13 - (x + 0.1 - x) \\
 &= 1 - 0.30 - 0.26 - 0.13 - 0.1 \\
 &= \boxed{0.21}
 \end{aligned}$$

De Morgan's Laws

1

Key Words

De Morgan's Laws

Exercise

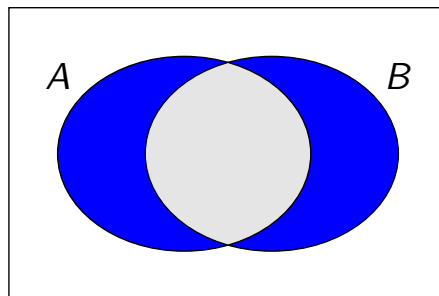
Key Words

2

For us, A or B means union ($A \cup B$), i.e., $A \cup B$ means A or B or both.
In words, *or* is the inclusive or.

A and B means intersection ($A \cap B$, also denoted as AB)

A or B but not both means we want to start with $A \cup B$ but exclude the intersection.

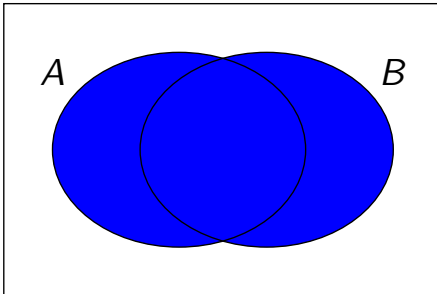


Algebraically, that means we want $(A \cap B') \cup (A' \cap B)$
What about neither A nor B ?

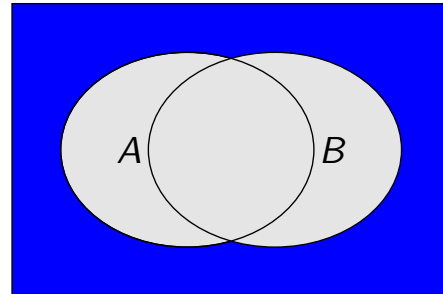
De Morgan's Laws: 2 Sets

3

$$A \cup B$$



$$A' \cap B' = (A \cup B)'$$

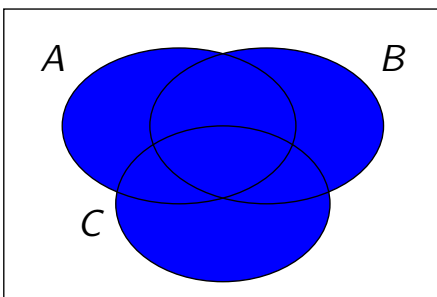


The left hand diagram is A or B , while the right hand diagram is neither A nor B .

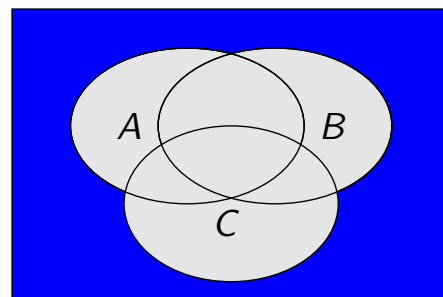
De Morgan's Laws: 3 Sets

4

$$A \cup B \cup C$$



$$A' \cap B' \cap C' = (A \cup B \cup C)'$$



De Morgan's Laws

5

General version of De Morgan's Laws:

$$\left[\bigcup_{i=1}^k A_i \right]' = \left[\bigcap_{i=1}^k A_i' \right]$$

$$\left[\bigcap_{i=1}^k A_i \right]' = \left[\bigcup_{i=1}^k A_i' \right]$$

If you bring the complement inside the brackets, unions and intersections get flipped.

Example

6

An auto insurance company has 10,000 policyholders. Each policyholder is classified as

1. young or old; and
2. male or female;

2,000 policyholders are neither young nor male, and 6,000 are either young or female.

How many policyholders are young?

Let Y denote young policyholders, and M male policyholders.

Neither young nor male is $(Y \cup M)' = Y' \cap M'$.

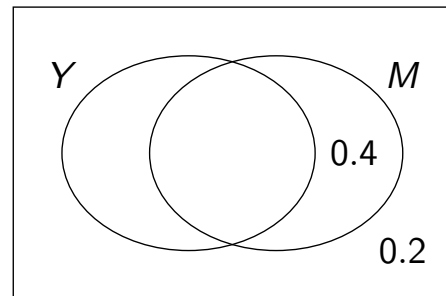
We are given $P[(Y \cup M)'] = P[Y' \cap M'] = 0.2$ and $P[Y \cup M'] = 0.6$.

We want $P[Y]$.

Example

7

$$\begin{aligned}
 P[Y' \cap M'] &= 0.2 \\
 P[Y \cup M'] &= 0.6 \\
 P[(Y \cup M')'] &= 1 - 0.6 \\
 P[Y' \cap M] &= 0.4 \\
 P[Y'] &= P[Y' \cap M'] + P[Y' \cap M] \\
 P[Y'] &= 0.2 + 0.4 = 0.6 \\
 P[Y] &= 1 - 0.6 = \boxed{0.4}
 \end{aligned}$$



Exercise

8

You are given:

- $P[A' \cup B' \cup C] = 0.7$
- $P[A \cap B \cap C] = 0.1$

Find $P[A \cap B]$

Exercise

8

You are given:

- $P[A' \cup B' \cup C] = 0.7$
- $P[A \cap B \cap C] = 0.1$

Find $P[A \cap B]$

$$P[A \cap B \cap C] = 0.1$$

$$A \cap B \cap C' = (A' \cup B' \cup C)'$$

$$P[A \cap B \cap C'] = 1 - 0.7 = 0.3$$

$$P[A \cap B] = P[A \cap B \cap C] + P[A \cap B \cap C']$$

$$= 0.1 + 0.3$$

$$= \boxed{0.4}$$

Inclusion-Exclusion

1

Review of Key Formulas

Inclusion-Exclusion

Exercise

Formulas from before

2

S = sample space, $P[S] = 1$, $P[\emptyset] = 0$

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$

Special case: if $A_1 \cap A_2 = \emptyset$ then $P[A_1 \cap A_2] = 0$
 so $P[A_1 \cup A_2] = P[A_1] + P[A_2] - 0 = P[A_1] + P[A_2]$

Generalization: if $A_i \cap A_j = \emptyset$ for all $i \neq j$ then

$$P\left[\bigcup_{i=1}^k A_i\right] = \sum_{i=1}^k P[A_i]$$

Complements: $A \cap A' = \emptyset$, $A \cup A' = S$

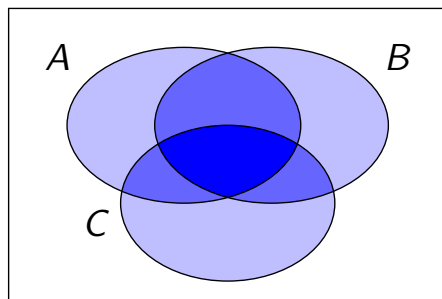
$$1 = P[S] = P[A] + P[A']$$

$$P[A] = 1 - P[A'] \quad P[A'] = 1 - P[A]$$

Inclusion-Exclusion

3

$$\begin{aligned}
 P[A \cup B] &= P[A] + P[B] - P[AB] \\
 P[A \cup B \cup C] &= P[A] + P[B] + P[C] \\
 &\quad - P[AB] - P[AC] - P[BC] \\
 &\quad + P[ABC]
 \end{aligned}$$



Inclusion-Exclusion

4

$$\begin{aligned}
 P[A \cup B \cup C \cup D] &= P[A] + P[B] + P[C] + P[D] \\
 &\quad - P[AB] - P[AC] - P[AD] - P[BC] - P[BD] - P[CD] \\
 &\quad + P[ABC] + P[ABD] + P[ACD] + P[BCD] \\
 &\quad - P[ABCD]
 \end{aligned}$$

Exercise

5

A survey of a group's viewing habits found the following:

1. 31% watched gymnastics, 25% watched baseball, and 21% watched soccer
2. 11% watched gymnastics and baseball, 8% watched baseball and soccer, and 9% watched gymnastics and soccer
3. 5% watched all three sports.

Find the percentage of the group that watched none of the three sports.

Exercise

5

A survey of a group's viewing habits found the following:

1. 31% watched gymnastics, 25% watched baseball, and 21% watched soccer
2. 11% watched gymnastics and baseball, 8% watched baseball and soccer, and 9% watched gymnastics and soccer
3. 5% watched all three sports.

Find the percentage of the group that watched none of the three sports.

Again, let G , B and S denote those who watch gymnastics, baseball, and soccer respectively. Then our inclusion-exclusion formula gives us

$$\begin{aligned}
 P[G \cup B \cup S] &= P[G] + P[B] + P[S] - P[GB] - P[BS] - P[GS] + P[GBS] \\
 &= 0.31 + 0.25 + 0.21 - 0.11 - 0.08 - 0.09 + 0.05 \\
 &= 0.54
 \end{aligned}$$

$$P[(G \cup B \cup S)'] = 1 - 0.54 = \boxed{0.46}$$

Conditional Probability

1

Die Rolling Example

Examples

Definitions

Key Words

Exercise

Die Rolling Example

2

Suppose you roll a fair 6-sided die. *Given that the result is odd*, what is the probability that it is 3 or less?

$$\begin{aligned} P[\text{roll 3 or less} \mid \text{odd}] &= \frac{\# \text{ of ways to roll 3 or less and odd}}{\# \text{ ways to roll an odd number}} \\ &= \frac{\#\{1, 3\}}{\#\{1, 3, 5\}} = \frac{2}{3} \end{aligned}$$

$$\begin{aligned} P[\text{roll 3 or less} \mid \text{odd}] &= \frac{\# \text{ of ways to roll 3 or less and odd}/6}{\# \text{ ways to roll an odd number}/6} \\ &= \frac{P[\text{roll 3 or less and odd}]}{P[\text{odd}]} \\ &= \frac{2/6}{3/6} = \frac{2}{3} \end{aligned}$$

Example 1

3

In a group of 635 men who died in 1999, 160 of the men died from causes related to heart disease. Moreover, 275 of the 635 men had at least one parent who suffered from heart disease, and, of these 275 men, 95 died from causes related to heart disease.

Find the probability that a man randomly selected from this group died of causes not related to heart disease and that neither of his parents suffered from heart disease.

95	180	At least 1 parent with heart disease 275
65	295	Neither parent with heart disease 360
HD 160	No HD 475	

Answer: $\boxed{295/635}$

Example 2

4

In a group of 635 men who died in 1999, 160 of the men died from causes related to heart disease. Moreover, 275 of the 635 men had at least one parent who suffered from heart disease, and, of these 275 men, 95 died from causes related to heart disease.

Find the probability that a man randomly selected from this group died of causes not related to heart disease, **given** that neither of his parents suffered from heart disease.

95	180	At least 1 parent with heart disease 275
65	295	Neither parent with heart disease 360
HD 160	No HD 475	

Answer: $\boxed{295/360}$

Definitions

5

Definition (Conditional Probability)

The conditional probability of A given B is

$$P[A | B] = \frac{P[A \cap B]}{P[B]} = \frac{P[AB]}{P[B]}$$

Note that rearranging terms gives

$$P[B] \cdot P[A | B] = P[AB] = P[A] \cdot P[B | A]$$

Key words

6

“Given that a person is in college, there is an 80% chance that they use Facebook”

means $P[\text{Facebook} | \text{college}] = 0.80$.

But they might not always use the word “given.”

“80% of college students use Facebook”

also means $P[\text{Facebook} | \text{college}] = 0.80$.

as does “College students have an 80% chance of using Facebook, while non-college students . . .”

Language that restricts possible outcomes to one group/case means condition on that case.

Exercise

7

The blood pressure (high, low, or normal) and heartbeats (regular or irregular) of a random sample of patients are measured. Of the patients,

1. 36% have high blood pressure and 16% have low blood pressure.
2. 21% have an irregular heartbeat.
3. Of those with an irregular heartbeat, one-third have high blood pressure.
4. Of those with normal blood pressure, one-eighth have an irregular heartbeat.

What portion have a regular heartbeat and low blood pressure?

Exercise

7

The blood pressure (high, low, or normal) and heartbeats (regular or irregular) of a random sample of patients are measured. Of the patients,

1. 36% have high blood pressure and 16% have low blood pressure.
2. 21% have an irregular heartbeat.
3. Of those with an irregular heartbeat, one-third have high blood pressure.
 $(1/3) \cdot 0.21 = 0.07$
4. Of those with normal blood pressure, one-eighth have an irregular heartbeat.
 $(1/8) \cdot 0.48 = 0.06$

What portion have a regular heartbeat and low blood pressure?

0.08	0.42	0.29	0.79 regular
0.08	0.06	0.07	0.21 irregular
0.16 low	0.48 normal	0.36 high	

Independence

1

Independence

Examples

Exercise

Independence

2

Previously saw

$$P[A | B] = \frac{P[A \cap B]}{P[B]} = \frac{P[AB]}{P[B]}$$
$$P[AB] = P[B] \cdot P[A | B] = P[A] \cdot P[B | A]$$

Definition (Independence)

A and B are *independent* if $P[AB] = P[A] \cdot P[B]$.

Intuitively, independence means $P[A] = P[A | B]$ and $P[B] = P[B | A]$ so knowing if A or B occurred gives no information on whether or not the other event occurred.

Examples

3

1. Suppose A and B are independent events with $P[A] = 0.6$ and $P[AB] = 0.3$. Find $P[B]$ and $P[A | B]$.

$$P[AB] = P[A] \cdot P[B] \quad \text{by independence}$$

$$0.3 = 0.6 \cdot P[B]$$

$$P[B] = \frac{0.3}{0.6} = 0.5$$

$$P[A | B] = P[A] = 0.6 \quad \text{by independence}$$

2. A and B are events such that $P[A] = 0.4$, $P[B] = 0.1$ and $P[AB] = 0.05$. Are they independent? What is $P[B | A]$?

$$P[A] \cdot P[B] = 0.4 \cdot 0.1 = 0.04 \neq 0.05 = P[AB]$$

so not independent

$$P[B | A] = \frac{P[AB]}{P[A]} = \frac{0.05}{0.4} = 0.125$$

Exercise

4

If $P[A] = 0.2$ and $P[B] = 0.3$, find $P[A \cup B]$ if:

- a) A and B are independent
- b) A and B are mutually exclusive

Exercise

4

If $P[A] = 0.2$ and $P[B] = 0.3$, find $P[A \cup B]$ if:

- a) A and B are independent
- b) A and B are mutually exclusive

$$\begin{aligned}\text{a) } P[A \cup B] &= P[A] + P[B] - P[A \cap B] \\ &= 0.2 + 0.3 - 0.2 \cdot 0.3 \\ &= 0.44\end{aligned}$$

$$\begin{aligned}\text{b) } P[A \cup B] &= P[A] + P[B] - P[A \cap B] \\ &= 0.2 + 0.3 - 0 \\ &= 0.50\end{aligned}$$

Sequences of Events

1

Probability of a Flush

Urn Problem

Exercises

Key Idea

2

The definition of conditional probability was that

$$P[B | A] = \frac{P[AB]}{P[A]}$$

We can use that to find $P[AB]$. Clearing the denominator gives

$$P[A] \cdot P[B | A] = P[AB]$$

This equation can be thought of as a sequence of events: first we need A to occur, and then second we need B to also occur, taking into account the fact that A occurred.

Probability of a flush

3

Find the probability of having a flush after being dealt five cards from a standard deck?

(Flush: at least 5 cards of one same suit. Standard deck: 4 suits, each with 13 cards)

We want: $P[\text{all 5 cards have the same suit}]$.

Method 1:

$$\begin{aligned} P[\text{all 5 cards have the same suit}] &= P[\text{all 5 are spades}] \\ &\quad + P[\text{all 5 are hearts}] + P[\text{all 5 are diamonds}] \\ &\quad + P[\text{all 5 are clubs}] = 4 \cdot P[\text{all 5 are spades}] \\ &= 4 \cdot \frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48} = 0.198\% \end{aligned}$$

$$\text{Method 2: } \frac{52}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48} = 0.198\%$$

Probability of a flush: Variation 1

4

If exactly three of the first 5 cards dealt are spades, what is the probability of being dealt a flush in the first 7 cards?

$$\begin{aligned} &P[\text{next 2 cards are spades}] \\ &= P[6\text{th card is a spade}] \cdot P[7\text{th is a spade} \mid 6\text{th is a spade}] \\ &= \frac{13 - 3}{52 - 5} \cdot \frac{13 - 4}{52 - 6} \\ &= \frac{10}{47} \cdot \frac{9}{46} \\ &= 0.0416 \\ &= \boxed{4.16\%} \end{aligned}$$

Probability of a flush: Variation 2

5

If exactly four of the first 5 cards dealt are spades, what is the probability of being dealt a flush in the first 7 cards?

We want $P[\text{at least one of next two is a spade}]$

Method 1:

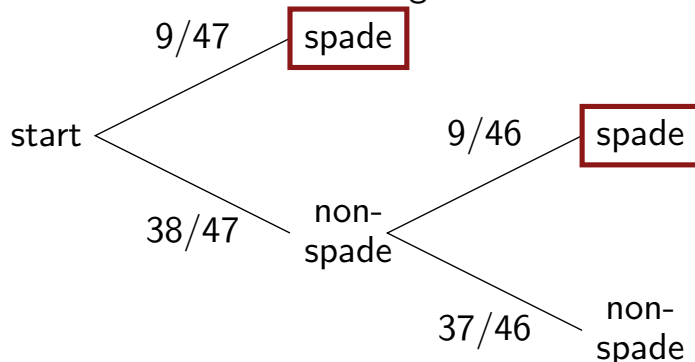
$$\begin{aligned}
 P[\text{at least one of next two is a spade}] &= 1 - P[\text{neither is a spade}] \\
 &= 1 - \frac{38}{47} \cdot \frac{37}{46} \\
 &= 1 - 0.65 \\
 &= 0.35
 \end{aligned}$$

Probability of a flush: Variation 2 Tree Diagram

6

If exactly four of the first 5 cards dealt are spades, what is the probability of being dealt a flush in the first 7 cards?

Method 2: Draw a tree diagram



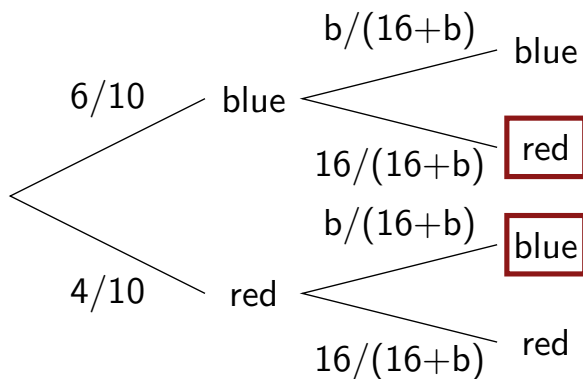
$$\text{So } P[\text{flush}] = \frac{9}{47} + \frac{38}{47} \cdot \frac{9}{46} = \boxed{0.35}$$

Urn Problem

7

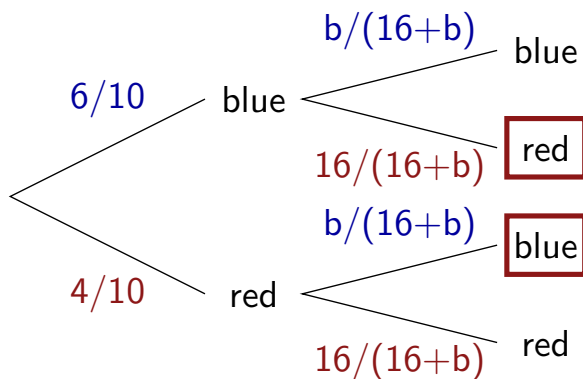
An urn contains 10 balls: 4 red and 6 blue. A second urn contains 16 red balls and an unknown number of blue balls. A single ball is drawn from each urn. The probability that both balls are different colors is 0.528.

Calculate the number of blue balls in the second urn.



Urn Problem

8



$$\begin{aligned}
 0.528 &= \frac{6}{10} \cdot \frac{16}{16+b} + \frac{4}{10} \cdot \frac{b}{16+b} \\
 &= \frac{96 + 4b}{160 + 10b} \\
 b &= \boxed{9}
 \end{aligned}$$

Exercise 1

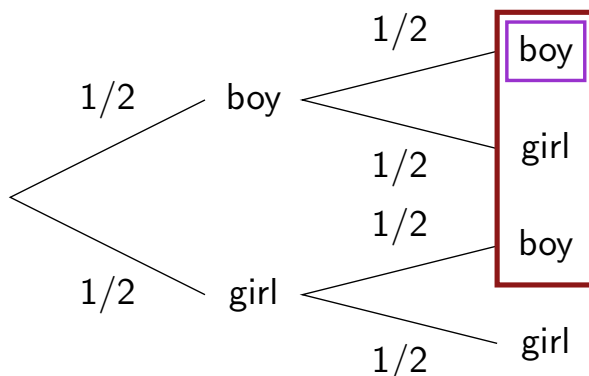
9

A family has two children, and they are not twins. Given that at least one of the children is a boy, what is the probability that both children are boys?

Exercise 1

9

A family has two children, and they are not twins. Given that at least one of the children is a boy, what is the probability that both children are boys?



$$P[2 \text{ boys} \mid \text{at least 1 boy}] = \frac{1/4}{3/4} = 1/3$$

Exercise 2

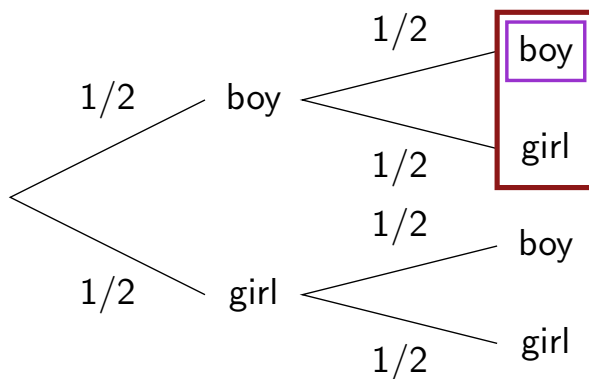
10

A family has two children, and they are not twins. Given that the oldest child is a boy, what is the probability that both children are boys?

Exercise 2

10

A family has two children, and they are not twins. Given that the oldest child is a boy, what is the probability that both children are boys?



$$P[2 \text{ boys} \mid \text{oldest is a boy}] = \frac{1/4}{2/4} = 1/2$$

Bayes' Theorem

1

Example 1

Statement of Bayes' Theorem

Example 2

Exercises

Bayes' Theorem

2

Often want $P[A | B]$ but are given $P[B | A]$

Can use

$$P[A | B] = \frac{P[AB]}{P[B]} = \frac{P[A] \cdot P[B | A]}{P[B]}$$

Typically then find $P[B]$ by breaking into cases.

Example 1

3

An auto insurance company insures drivers of all ages. An actuary compiled the following statistics on the company's insured drivers:

Age of Driver	Probability of Accident	Portion of Company's Insured Drivers
16-20	0.06	0.08
21-30	0.03	0.15
31-65	0.02	0.49
66-99	0.04	0.28

A randomly selected driver that the company ensures has an accident. Calculate the probability that the driver was 31-65.

We are given $P[\text{Accident} \mid \text{Age}]$, are reversing condition to $P[\text{Age} \mid \text{Accident}]$

Example 1 (cont.)

4

Age of Driver	Probability of Accident	Portion of Company's Insured Drivers
16-20	0.06	0.08
21-30	0.03	0.15
31-65	0.02	0.49
66-99	0.04	0.28

$$\begin{aligned}
 P[\text{age 31-65} \mid \text{accident}] &= \frac{P[\text{age 31-65, accident}]}{P[\text{accident}]} \\
 &= \frac{P[\text{age 31-65}] \cdot P[\text{accident} \mid \text{age 31-65}]}{P[\text{accident}]} = \frac{(0.49)(0.02)}{P[\text{accident}]}
 \end{aligned}$$

$$\begin{aligned}
 P[\text{accident}] &= \sum_{\text{age groups}} P[\text{accident, age group}] \\
 &= (.08)(.06) + (.15)(.03) + (.49)(.02) + (.28)(.04) = 0.0303
 \end{aligned}$$

$$P[\text{age 31-65} \mid \text{accident}] = \frac{0.49 \cdot 0.02}{0.0303} = \boxed{0.3234}$$

The Law of Total Probability

5

Theorem (Law of Total Probability)

If $A_1, A_2, \dots, A_k$ are disjoint and $P[A_1] + P[A_2] + \dots + P[A_k] = 1$ then

- $P[B] = P[BA_1] + P[BA_2] + \dots + P[BA_k]$
- $P[B] = P[A_1] \cdot P[B | A_1] + \dots + P[A_k] \cdot P[B | A_k]$

The sets $A_1, \dots, A_k$ are called a partition of the sample space. We will often refer to them as a list of all possible cases.

In the previous example, the age groups were the A_i , and B was the event of an accident.

Bayes' Theorem

6

Theorem (Bayes' Theorem)

Suppose $A_1, \dots, A_k$ are a partition of the sample space. Then

$$\begin{aligned}
 P[A_1 | B] &= \frac{P[A_1 B]}{P[B]} \\
 &= \frac{P[A_1] \cdot P[B | A_1]}{\sum_{i=1}^k P[BA_i]} \\
 P[A_1 | B] &= \frac{P[A_1] \cdot P[B | A_1]}{\sum_{i=1}^k P[A_i] \cdot P[B | A_i]}
 \end{aligned}$$

The final denominator is $\sum_{\text{cases}} P[\text{case}] \cdot P[B | \text{case}]$

Example 2

7

Life insurance policy holders are categorized as **standard**, **preferred**, and **ultra-preferred**. Of a company's policyholders, **50% are standard**, **40% are preferred**, and **10% are ultra-preferred**. The probability of dying in the next year is **0.010** for each **standard policyholder**, **0.005** for **preferred policyholders**, and **0.001** for **ultra-preferred**.

A policyholder dies in the next year. What is the probability that the deceased policyholder was standard?

Let S denote someone who is standard.

$$\begin{aligned}
 P[S \mid \text{died}] &= \frac{P[S \text{ and died}]}{P[\text{died}]} \\
 &= \frac{0.50 \cdot 0.010}{0.50 \cdot 0.010 + 0.40 \cdot 0.005 + 0.10 \cdot 0.001} \\
 &= \frac{0.005}{0.0071} = \boxed{70.4\%}
 \end{aligned}$$

Exercise 1

8

Taxicabs in Crobuzon are all either green or blue. On Tuesday, a taxicab got into an accident. A witness to the accident thought that the cab involved was blue, and further tests showed that the witness has an 80% chance of correctly identifying the color of a taxicab, independently of its color.

If 100% of the taxicabs on the streets on Tuesday were green, what was the probability that the taxicab involved in the accident was blue?

Exercise 1

8

Taxicabs in Crobuzon are all either green or blue. On Tuesday, a taxicab got into an accident. A witness to the accident thought that the cab involved was blue, and further tests showed that the witness has an 80% chance of correctly identifying the color of a taxicab, independently of its color.

If 100% of the taxicabs on the streets on Tuesday were green, what was the probability that the taxicab involved in the accident was blue?

None of the taxicabs were blue that night. The witness is wrong and the cab was not blue. So the answer is 0% (and in particular, not 80%)!

Exercise 2

9

Taxicabs in Crobuzon are all either green or blue. On Tuesday, a taxicab got into an accident. A witness to the accident thought that the cab involved was blue, and further tests showed that the witness has an 80% chance of correctly identifying the color of a taxicab, independently of its color.

If 85% of the taxicabs on the streets on Tuesday were green, what was the probability that the taxicab involved in the accident was blue?

Exercise 2

9

Taxicabs in Crobuzon are all either green or blue. On Tuesday, a taxicab got into an accident. A witness to the accident thought that the cab involved was blue, and further tests showed that the witness has an 80% chance of correctly identifying the color of a taxicab, independently of its color.

If 85% of the taxicabs on the streets on Tuesday were green, what was the probability that the taxicab involved in the accident was blue?

$$\begin{aligned} P[\text{Blue cab} \mid \text{witness said blue}] &= \frac{P[\text{Blue cab and witness said blue}]}{P[\text{witness said blue}]} \\ &= \frac{0.15 \cdot 0.80}{0.15 \cdot 0.80 + 0.85 \cdot 0.20} \\ &= \frac{0.12}{0.29} = \boxed{41\%} \end{aligned}$$

Mode

1

Overview

Random Variables

Mode

Example

Exercise

Overview

2

3 ways of describing “typical” value of a random variable:

- Mode = most common value
- Median = “middle” value
- Mean = average value

Median and mode are relatively easy to find, usually appear as part of a longer problem. Mean can be either a step or an entire problem.

Will focus on mode in this lesson.

Random Variables

3

X is a random variable if it is a number whose value depends on chance. Formally,

$X: S \rightarrow \mathbb{R}$ where $S =$ sample space.

Typically we will use capital letters for random variables and lower case letters for possible (non-random!) values.

X is a discrete random variable if we can list all the possible values.

$1 = \sum_x P[X = x]$ for discrete variables.

Mode

4

For a discrete random variable X , y is the mode of X if $P[X = y] \geq P[X = x]$ for all x .

I.e., the mode y is the input that maximizes $P[X = y]$

It is possible that the max is not unique so X can have multiple modes.

Mode Example

5

Suppose I roll an otherwise fair 7 sided die whose faces are 1, 1, 1, 2, 4, 4, and 6. Find the mode.

Let X be the result of the roll

$$P[X = 1] = \frac{3}{7}$$

$$P[X = 2] = \frac{1}{7}$$

$$P[X = 4] = \frac{2}{7}$$

$$P[X = 6] = \frac{1}{7}$$

When $y = 1$, $P[X = y]$ reaches its max of $3/7$, so 1 is the mode.

Exercise

6

Find the mode of a Poisson random variable with mean 2.8, meaning that

$$P[N = n] = \frac{2.8^n}{n!} \cdot e^{-2.8} \text{ for } n = 0, 1, 2, \dots$$

where $n! = n(n-1)(n-2) \dots (2)(1)$

Exercise

6

Find the mode of a Poisson random variable with mean 2.8, meaning that

$$P[N = n] = \frac{2.8^n}{n!} \cdot e^{-2.8} \text{ for } n = 0, 1, 2, \dots$$

where $n! = n(n-1)(n-2)\dots(2)(1)$

Using the TI-30XS MultiView Data function:

n	$P[N = n]$
0	0.0608
1	0.1703
2	0.2384
3	0.2225
4	0.1557
5	0.0872
6	0.0407

So the mode is 2.

Medians

1

Definitions

Example

Percentiles

Exercise

Median

2

Definition (Cumulative Distribution Function)

$F(x) = P[X \leq x]$ is the cumulative distribution function of X .

For example, $F(2) = P[X \leq 2]$

Definition (Median)

Exam definition: The median of X is the smallest m such that $P[X \leq m] = F(m) \geq 1/2$.

Remark: there are more complicated definitions because the median is not uniquely defined for some random variables. Such cases will not appear on the exam, this simplified definition is equivalent when the median is uniquely defined.

Median Example

3

Suppose I roll an otherwise fair 7 sided die whose faces are 1, 1, 1, 2, 4, 4, and 6. Find the median.

Let X be the result of the roll

$$P[X = 1] = \frac{3}{7}$$

$$P[X = 2] = \frac{1}{7}$$

$$P[X = 4] = \frac{2}{7}$$

$$P[X = 6] = \frac{1}{7}$$

$$P[X \leq 1] = \frac{3}{7}$$

$$P[X \leq 2] = \frac{4}{7}$$

$$P[X \leq 4] = \frac{6}{7}$$

$$P[X \leq 6] = \frac{7}{7}$$

When $y = 2$, $P[X \leq y]$ first reaches / exceeds $1/2$, so 2 is the median.

Percentiles

4

Definition (Percentiles)

The $100\% \cdot p^{th}$ percentile π_p is the smallest possible x such that $P[X \leq x] \geq p$.

Example: If X is our die rolling example from before:

x	0	1	2	4	6
$F(x)$	0	$3/7$	$4/7$	$6/7$	1

5th percentile = 1 since $P[N \leq 1] \geq 0.05$ but $P[N \leq 0] < 0.05$

10th percentile = 1 since $P[N \leq 1] \geq 0.10$ but $P[N \leq 0] < 0.10$

Median = 50th percentile = 2

90th percentile = 6 since $P[N \leq 4] < 0.9$ but $P[N \leq 6] \geq 0.9$

More Complex Example

5

Roll a fair 6-sided die. What is the median outcome?

n	$P[N = n]$	$F(n) = P[N \leq n]$
1	1/6	1/6
2	1/6	2/6
3	1/6	3/6=1/2
4	1/6	4/6
5	1/6	5/6
6	1/6	1

By our definition, 3 is the first time $F(n) \geq 1/2$ so the median is 3.

But $F(x) = 1/2$ for any x such that $3 \leq x < 4$ (e.g., $F(3.5) = 1/2$.)

Some texts say that any x with $F(x) = 1/2$ is a median, giving infinitely many medians (i.e., anything in the interval $3 \leq x < 4$).

This situation will not occur on the exam.

Exercise

6

Suppose that $P[N = n] = n/15$ for $n = 1, 2, 3, 4$ or 5 . Find the median of N .

Exercise

6

Suppose that $P[N = n] = n/15$ for $n = 1, 2, 3, 4$ or 5 . Find the median of N .

Evaluating the CDF:

$$P[N \leq 1] = P[N = 1] = 1/15 < 1/2$$

$$P[N = 2] = 2/15$$

$$P[N \leq 2] = 1/15 + 2/15 = 3/15 < 1/2$$

$$P[N = 3] = 3/15$$

$$P[N \leq 3] = 3/15 + 3/15 = 6/15 < 1/2$$

$$P[N = 4] = 4/15$$

$$P[N \leq 4] = 6/15 + 4/15 = 10/15 \geq 1/2$$

So the median is 4.

Moments: Expected Value / Mean

1

Example

Definitions

Example

Exercise

Example

2

In a group of 10 people, I owe \$4 to two of them, \$2 to one of them, \$1 to two of them, and nothing to the others. On average, how much do I owe to these 10 people?

Method 0: The average debt is

$$\frac{\text{Total Amount Owed}}{\text{Number of People}} = \frac{4 \cdot 2 + 2 \cdot 1 + 1 \cdot 2 + 0 \cdot 5}{10} = \frac{12}{10} = 1.20$$

Method 1: Pick one of the 10 people at random. Let X be the amount owed to that person.

$$\begin{aligned} E[X] &= \sum_x x \cdot P[X = x] = 4 \cdot \frac{2}{10} + 2 \cdot \frac{1}{10} + 1 \cdot \frac{2}{10} + 0 \cdot \frac{5}{10} \\ &= 0.8 + 0.2 + 0.2 + 0 = 1.2 \end{aligned}$$

Definitions

3

Definition (Expected Value)

If X is a discrete random variable then

$$E[X] = \sum_x x \cdot P[X = x]$$

Definition (Generalizations)

$$E[X^2] = \sum_x x^2 \cdot P[X = x]$$

$$E[g(X)] = \sum_x g(x) \cdot P[X = x]$$

Example

4

An insurance policy pays 100 per day for up to 3 days of hospitalization and 50 per day of hospitalization thereafter. Find the expected payment for hospitalization if the number of days of hospitalization, X , is a discrete random variable with

$$P(X = k) = \begin{cases} \frac{6-k}{15} & \text{for } k = 1, 2, 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$$

Let $g(k)$ = payment for k days in the hospital.

$$E[g(X)] = \sum g(k) \cdot P[X = k]$$

k	1	2	3	4	5
$g(k)$	100	200	300	350	400
$P[X = k]$	5/15	4/15	3/15	2/15	1/15

$$E[g(X)] = 100 \cdot \frac{5}{15} + 200 \cdot \frac{4}{15} + 300 \cdot \frac{3}{15} + 350 \cdot \frac{2}{15} + 400 \cdot \frac{1}{15} = 220$$

Exercise

5

Suppose that $P[N = n] = n/15$ for $n = 1, 2, 3, 4$ or 5 . Find $E[N]$.

Exercise

5

Suppose that $P[N = n] = n/15$ for $n = 1, 2, 3, 4$ or 5 . Find $E[N]$.

$$\begin{aligned} E[N] &= \sum_n n \cdot P[N = n] \\ &= 1 \cdot \frac{1}{15} + 2 \cdot \frac{2}{15} + 3 \cdot \frac{3}{15} + 4 \cdot \frac{4}{15} + 5 \cdot \frac{5}{15} \\ &= \frac{55}{15} \\ &= \frac{11}{3} = 3.67 \end{aligned}$$

Recall that the median was 4. The mode is 5. The median and mode must be possible values of the variable, the expected value doesn't have to be.

Tools for Finding Means

1

Overview

Law of Total Expectation

Linear Combinations

Exercise

Overview

2

Previously had

$$E[X] = \sum_x x \cdot P[X = x]$$
$$E[g(X)] = \sum_x g(x) \cdot P[X = x]$$

Sometimes can relate to other means to find what we want.

Will look at 2 methods that do so.

Law of Total Expectation

3

Definition (Law of Total Expectation)

If $A_1, \dots, A_n$ are a partition of the sample space, that is, $S = \bigcup_{i=1}^n A_i$ and $A_i \cap A_j = \emptyset$ for $i \neq j$, then

$$E[X] = \sum_{i=1}^n E[X | A_i] \cdot P[A_i]$$

This often happens with the A_i being high / medium / low risk groups

The usual definition is the situation with $A_i = \{X = i\}$

Linear Combinations

4

Note that

$$\begin{aligned} E[aX + b] &= \sum_x (ax + b) \cdot P[X = x] \\ &= \sum_x ax \cdot P[X = x] + \sum_x b \cdot P[X = x] \\ &= a \cdot \left(\sum_x x \cdot P[X = x] \right) + b \cdot \sum_x P[X = x] \\ E[aX + b] &= a \cdot E[X] + b \end{aligned}$$

That is, means of linear functions distribute as you would expect

In general, $E[aX + bY] = aE[X] + bE[Y]$ for any two random variables X and Y .

Exercise

5

40% of an insurer's claims are from high risk individuals, with an average claim amount of 500, and 60% are from low risk individuals with an average claim amount of 100.

Due to administrative costs, a claim of X costs the insurer $1.05X + 10$. What is the expected cost to the insurer of a randomly selected claim?

Exercise

5

40% of an insurer's claims are from high risk individuals, with an average claim amount of 500, and 60% are from low risk individuals with an average claim amount of 100.

Due to administrative costs, a claim of X costs the insurer $1.05X + 10$. What is the expected cost to the insurer of a randomly selected claim?

$$\begin{aligned} E[\text{Cost}] &= E[1.05X + 10] = 1.05 E[X] + 10 \\ &= 1.05 (0.4 \cdot 500 + 0.6 \cdot 100) + 10 \\ &= 1.05 \cdot 260 + 10 \\ &= \boxed{283} \end{aligned}$$

Means: Survival Function Approach

1

Initial Example Revisited

Survival Function Method

Exercises

Initial Example Revisited

2

In a group of 10 people, I owe \$4 to two of them, \$2 to one of them, \$1 to two of them, and nothing to the others. On average, how much do I owe to these 10 people?

Saw how to use the definition previously.

Method 2: I need to repay the debt.

- Step 1: Pay \$1 to everyone who is owed money.
- Step 2: Pay \$1 more to everyone who is still owed money (i.e., people who were initially owed \$2 or \$4). etc.

Total payment: \$5 in step 1, \$3 in step 2, \$2 in step 3 and \$2 in step 4. After step 4, everyone has been paid back.

Total payment = # owed at least \$1 + # owed at least \$2 + ... = 5+3+2+2 = 12

Average per person = (total payment) / (# people) = 12/10 = 1.20 as before.

Survival function method

3

If $N \geq 0$ and N is an integer valued variable,

$$E[N] = P[N > 0] + P[N > 1] + P[N > 2] + \dots$$

$$E[N] = \sum_{n=0}^{\infty} P[N > n]$$

or letting $k = n + 1$ so $P[N > n] = P[N \geq n + 1] = P[N \geq k]$

$$E[N] = \sum_{k=1}^{\infty} P[N \geq k]$$

The name is because $P[N > n] = 1 - F(n)$ is called the survival function.

The discrete version doesn't extend easily to finding $E[g(X)]$ (but the continuous version of the survival function method will).

Exercise 1

4

Following a certain type of surgery, patients are hospitalized for N days, with $P[N \geq k] = \frac{5-k}{5}$ for $k = 0, 1, 2, 3, 4$ or 5 . Find $E[N]$ using the survival method.

Exercise 1

4

Following a certain type of surgery, patients are hospitalized for N days, with $P[N \geq k] = \frac{5-k}{5}$ for $k = 0, 1, 2, 3, 4$ or 5 . Find $E[N]$ using the survival method.

$$\begin{aligned}
 E[N] &= \sum_{n=0}^{\infty} P[N > n] \\
 &= P[N > 0] + P[N > 1] + P[N > 2] + P[N > 3] + P[N > 4] \\
 &= P[N \geq 1] + \cdots + P[N \geq 5] \\
 &= \frac{4}{5} + \frac{3}{5} + \frac{2}{5} + \frac{1}{5} + 0 \\
 &= \boxed{2}
 \end{aligned}$$

Exercise 2

5

Following a certain type of surgery, patients are hospitalized for N days, with $P[N \geq k] = \frac{5-k}{5}$ for $k = 0, 1, 2, 3, 4$ or 5 . Find $E[N]$ using the definition.

Exercise 2

5

Following a certain type of surgery, patients are hospitalized for N days, with $P[N \geq k] = \frac{5-k}{5}$ for $k = 0, 1, 2, 3, 4$ or 5 . Find $E[N]$ using the definition.

$$\begin{aligned} P[N = k] &= P[N \geq k] - P[N \geq k + 1] \\ &= \frac{5-k}{5} - \frac{5-(k+1)}{5} \\ &= \frac{1}{5} \quad \text{for } k = 0, 1, 2, 3 \text{ or } 4 \\ E[N] &= 0 \cdot \frac{1}{5} + 1 \cdot \frac{1}{5} + 2 \cdot \frac{1}{5} + 3 \cdot \frac{1}{5} + 4 \cdot \frac{1}{5} \\ &= \frac{10}{5} \\ &= \boxed{2} \end{aligned}$$

Variance

1

Basic Definition

Exercise

Example

2

Suppose that X , Y and Z are three random variables such that

$$\begin{array}{lll} P[X = 2] = 1 & & \\ P[Y = 1] = \frac{1}{3} & P[Y = 2] = \frac{1}{3} & P[Y = 3] = \frac{1}{3} \\ P[Z = 1] = \frac{1}{2} & & P[Z = 3] = \frac{1}{2} \end{array}$$

Then $E[X] = E[Y] = E[Z] = 2$. But intuitively, Y is more likely to differ from the mean than X , and Z is even more likely to do so.

The variance of a variable is a way to quantify how much it differs from its mean.

Definition (Variance)

$$\text{Var}[X] = E[(X - \mu_X)^2] \text{ where } \mu_X = E[X]$$

Example

3

We want to find the variance of X , Y , and Z , where

$$\begin{aligned} P[X = 2] &= 1 \\ P[Y = 1] &= \frac{1}{3} & P[Y = 2] &= \frac{1}{3} & P[Y = 3] &= \frac{1}{3} \\ P[Z = 1] &= \frac{1}{2} & & & P[Z = 3] &= \frac{1}{2} \end{aligned}$$

$$\text{Var}[X] = E[(X - 2)^2] = 1 \cdot (2 - 2)^2 = 0$$

$$\text{Var}[Y] = \frac{1}{3} \cdot (1 - 2)^2 + \frac{1}{3} \cdot (2 - 2)^2 + \frac{1}{3} \cdot (3 - 2)^2 = \frac{2}{3}$$

$$\text{Var}[Z] = \frac{1}{2} \cdot (1 - 2)^2 + \frac{1}{2} \cdot (3 - 2)^2 = 1$$

Terminology

4

$E[X^k]$ is the k th moment of X . Sometimes it is called the k th *raw* moment of X .

$\mu = E[X]$ = mean = average = first moment of X .

$E[X^2]$ is the second (raw) moment of X .

$\text{Var}[X] = E[(X - \mu)^2] = \sigma^2$ = 2nd central moment of X .

$E[(X - \mu)^k]$ = k th central moment.

$E[(X - a)^k]$ = k th moment about a .

Exercise

5

An insurance policy pays 100 per day for up to 3 days of hospitalization and 50 per day for each day of hospitalization thereafter. The number of days of hospitalization, X , is a discrete random variable with probability function

$$P(X = k) = \frac{6 - k}{15} \text{ for } k = 1, 2, 3, 4, 5 \text{ and } 0 \text{ otherwise.}$$

The mean payment amount is 220. Find the variance of a payment for hospitalization.

Exercise

5

An insurance policy pays 100 per day for up to 3 days of hospitalization and 50 per day for each day of hospitalization thereafter. The number of days of hospitalization, X , is a discrete random variable with probability function

$$P(X = k) = \frac{6 - k}{15} \text{ for } k = 1, 2, 3, 4, 5 \text{ and } 0 \text{ otherwise.}$$

The mean payment amount is 220. Find the variance of a payment for hospitalization.

Let Y denote the payment amount.

k	1	2	3	4	5
Y	100	200	300	350	400
$P[X = k]$	5/15	4/15	3/15	2/15	1/15

$$\begin{aligned} E[(Y - 220)^2] &= \frac{5(-120)^2}{15} + \frac{4(-20)^2}{15} + \frac{3 \cdot 80^2}{15} + \frac{2 \cdot 130^2}{15} + \frac{180^2}{15} \\ &= \boxed{10,600} \end{aligned}$$

Variance

1

Main Method for Finding Variance

Example Revisited

Linear Transformations

Standard Deviation

Exercise

A faster way (usually) to find the variance:

2

Let $\mu = E[X]$. Then

$$\begin{aligned}\text{Var}[X] &= E[(X - \mu)^2] \\ &= E[X^2 - 2\mu X + \mu^2] \\ &= E[X^2] - 2\mu \cdot E[X] + \mu^2 \\ &= E[X^2] - 2\mu \cdot \mu + \mu^2 \\ &= E[X^2] - \mu^2\end{aligned}$$

$$\text{Var}[X] = E[X^2] - (E[X])^2$$

$$E[X^2] = \text{Var}[X] + (E[X])^2$$

Remark: from the definition, $\text{Var}[X] \geq 0$, with $\text{Var}[X] = 0$ if and only if $P[X = \mu] = 1$, i.e, X is constant.

This implies that $E[X^2] \geq (E[X])^2$.

Example Revisited

3

An insurance policy pays 100 per day for up to 3 days of hospitalization and 50 per day thereafter. The number of days of hospitalization, X , has probability function

$$P(X = k) = \frac{6 - k}{15} \text{ for } k = 1, 2, 3, 4, 5 \text{ and } 0 \text{ otherwise.}$$

Find the variance of a payment Y for hospitalization.

k	1	2	3	4	5
Y	100	200	300	350	400
$P[X = k]$	5/15	4/15	3/15	2/15	1/15

$$E[Y] = 100 \cdot \frac{5}{15} + 200 \cdot \frac{4}{15} + 300 \cdot \frac{3}{15} + 350 \cdot \frac{2}{15} + 400 \cdot \frac{1}{15} = 220$$

$$E[Y^2] = 100^2 \cdot \frac{5}{15} + 200^2 \cdot \frac{4}{15} + 300^2 \cdot \frac{3}{15} + 350^2 \cdot \frac{2}{15} + 400^2 \cdot \frac{1}{15} = 59,000$$

$$\text{Var}[Y] = 59,000 - 220^2 = 10,600$$

Multiplying by a Constant

4

Suppose we know $\text{Var}[X]$. What is $\text{Var}(3X)$? What is $\text{Var}(cX)$?

$$\begin{aligned} \text{Var}[3X] &= E[(3X)^2] - (E[3X])^2 \\ &= E[9X^2] - (3E[X])^2 \\ &= 9[E(X^2) - (E[X])^2] \\ &= 9 \cdot \text{Var}[X] \end{aligned}$$

$$\text{Var}[cX] = c^2 \text{Var}[X]$$

Linear Transformations

5

Suppose we know $\text{Var}[X]$. What is $\text{Var}[X + b]$? What is $\text{Var}[aX + b]$?

$$\begin{aligned}\text{Var}[X + b] &= E[(X + b - E[X + b])^2] \\ &= E[(X + b - E[X] - b)^2] \\ &= E[(X - E[X])^2] \\ &= \text{Var}[X]\end{aligned}$$

I.e., shifting by a constant doesn't change distances from the mean so variance is same.

$$\begin{aligned}\text{Var}[aX + b] &= \text{Var}[aX] \\ &= a^2 \text{Var}[X]\end{aligned}$$

Standard Deviation and Coefficient of Variation

6

The standard deviation $\text{SD}[X]$ satisfies:

$$\begin{aligned}\text{SD}[X] &= \sigma_X = \sqrt{\text{Var}[X]} \\ \text{Var}[cX] &= c^2 \text{Var}[X] \\ \text{SD}[cX] &= |c| \cdot \text{SD}[X]\end{aligned}$$

Exam problems may ask about the Coefficient of Variation $\text{CV}[X]$

$$\begin{aligned}\text{CV}[X] &= \frac{\sigma}{\mu} = \frac{\text{SD}[X]}{E[X]} \\ \text{CV}[cX] &= \frac{\text{SD}[cX]}{E[cX]} \\ &= \frac{c \cdot \text{SD}[X]}{c \cdot E[X]} \quad \text{if } c > 0 \\ \text{CV}[cX] &= \text{CV}[X]\end{aligned}$$

Note: If $E[X]$ is held constant and σ increases, then $\text{CV}[X]$ also increases.

Exercise

7

A random variable X satisfies $E[X] = 5$ and $SD[X] = 3$. Find

- a) $E[X^2]$
- b) $\text{Var}[2X + 6]$
- c) $\text{CV}[2X + 6]$

Exercise

7

A random variable X satisfies $E[X] = 5$ and $SD[X] = 3$. Find

- a) $E[X^2]$
- b) $\text{Var}[2X + 6]$
- c) $\text{CV}[2X + 6]$

$$E[X^2] = \text{Var}[X] + (E[X])^2 = 3^2 + 5^2 = 34$$

$$\text{Var}[2X + 6] = \text{Var}[2X] = 2^2 \text{Var}[X] = 4 \cdot 9 = 36$$

$$E[2X + 6] = 2E[X] + 6 = 2 \cdot 5 + 6 = 16$$

$$\begin{aligned} \text{CV}[2X + 6] &= \frac{SD[2X + 6]}{E[2X + 6]} \\ &= \frac{\sqrt{36}}{16} = \frac{3}{8} \end{aligned}$$

Discrete uniform random variables

1

Standard Uniform

Generalizations

Exercise

Standard Uniform: Expected Value and Variance

2

Definition (Discrete Uniform)

X is a (discrete) uniform on $\{1, 2, \dots, n-1, n\}$ if $P[X = i] = \frac{1}{n}$ for those n choices.

Note that the average of 1 and n is $\frac{n+1}{2}$, as is the average of 2 and $n-1$, and 3 and $n-2$ and $\dots$, so

$$E[X] = \frac{n+1}{2}$$

We will see later that $\text{Var}[X] = \frac{n^2 - 1}{12}$

Example

3

Suppose X is uniform on $\{3, 4, 5, 6, 7, 8\}$. What is $E[X]$?

As before, we find $E[X]$ by pairing extremes (3 and 8, 4 and 7, ...). Each pair has an average value of $\frac{3+8}{2} = \frac{11}{2}$ so $E[X] = \frac{11}{2}$

Alternatively, X is not a standard uniform because it starts at 3, not 1. But $X - 2$ is a standard uniform on $\{1, 2, \dots, 6\}$

$$\begin{aligned} E[X - 2] &= \frac{1 + 6}{2} = \frac{7}{2} \\ E[X] &= E[X - 2] + 2 \\ &= \frac{7}{2} + 2 = \frac{11}{2} \end{aligned}$$

Example

4

Suppose X is uniform on $\{3, 4, 5, 6, 7, 8\}$. What is $\text{Var}[X]$?

As before, $X - 2$ is a standard uniform on $\{1, 2, \dots, 6\}$

$$\begin{aligned} \text{Var}[X - 2] &= \frac{n^2 - 1}{12} \\ &= \frac{36 - 1}{12} \\ &= \frac{35}{12} \\ \text{Var}[X] &= \text{Var}[X - 2] \\ &= \frac{35}{12} \end{aligned}$$

General case

5

Suppose that X is uniform on $\{a, a + 1, \dots, b\}$.

Then $E[X] = \frac{a+b}{2}$ which is the average of the first and last possible values.

$\text{Var}[X] = \text{Var}[X - (a - 1)]$ since shifting by constants doesn't change the variance.

But $X - (a - 1)$ is uniform on $\{1, 2, \dots, b - (a - 1)\}$ so

$$\begin{aligned}\text{Var}[X] &= \frac{[b - (a - 1)]^2 - 1}{12} \\ &= \frac{(\# \text{ of values in range})^2 - 1}{12}\end{aligned}$$

Exercise

6

The number of losses N is uniformly distributed on $\{5, 6, \dots, 20\}$. Each loss results in a payment of 100. Find the mean and standard deviation of the payment amount.

Exercise

6

The number of losses N is uniformly distributed on $\{5, 6, \dots, 20\}$. Each loss results in a payment of 100. Find the mean and standard deviation of the payment amount.

Let X denote the payment amount. $X = 100N$.

$$\begin{aligned}E[X] &= 100 \cdot E[N] \\&= 100 \cdot \frac{5 + 20}{2} = 1,250 \\ \text{Var}[X] &= 100^2 \cdot \text{Var}[N] \\&= 100^2 \cdot \frac{(20 - 5)^2 - 1}{12} = 212,500 \\ \text{SD}[X] &= 100 \cdot \sqrt{\frac{255}{12}} \\&= 461\end{aligned}$$

Permutations

1

Complete Lists

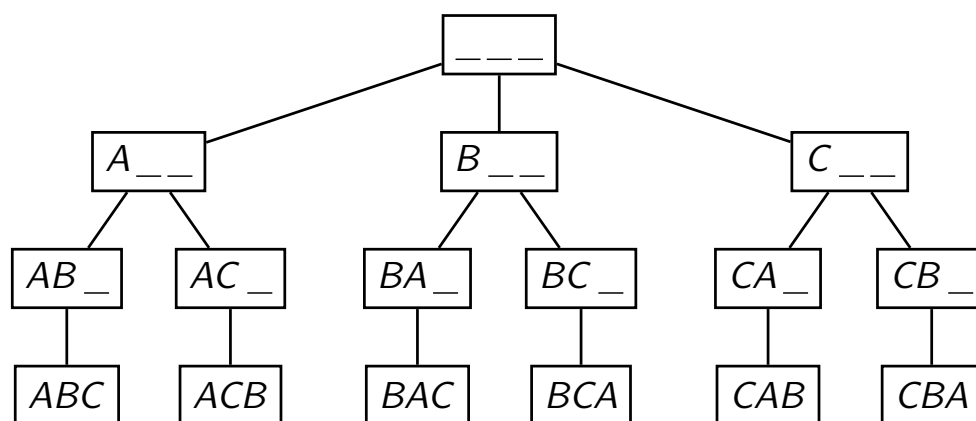
Partial Lists

Exercises

Complete Lists

2

How many possible rankings are there of a group of 3 people named A , B and C ?



There are 3 choices of who comes first, 2 people left who can be second, and then only 1 choice remaining for third. The total number of possibilities is $3 \cdot 2 \cdot 1$

Complete Lists

3

How many possible rankings are there of a group of 12 people?

$$\underline{12} \underline{11} \underline{10} \dots \underline{3} \underline{2} \underline{1}$$

There are 12 choices of who can be first.

There are 11 people left who can be second.

There are 10 people left who can be third.

$\vdots$ $\vdots$ $\vdots$ $\vdots$

There are 3 people left who can be tenth.

There are 2 people who can be eleventh, and that leaves just one who can be last.

Complete Lists

4

Key idea: at each step, multiply the number of choices for the new step with the choices so far. That gives an answer of

$$\begin{aligned} & 12 \cdot 11 \cdot 10 \cdots 2 \cdot 1 \\ & = 12! \\ & n! = n \cdot (n - 1) \cdot (n - 2) \cdots 2 \cdot 1 \\ & 1! = 1 \\ & 0! = 1 \end{aligned}$$

Note: The MultiView's "prb" menu has an $n!$ function

Each order is called a permutation, $n!$ is the number of permutations of n objects.

Partial Lists

5

A contest with 12 people gives out 3 distinct prizes. How many ways are there to give out these prizes?

$$\underline{12} \underline{11} \underline{10}$$

So there are $12 \cdot 11 \cdot 10$ ways to do this.

This is an example of a partial permutation, or a k -permutation.

There is a formula for the number of k -permutations of n objects, but it is better to think through them from scratch and multiply the number of choices at each step.

Exam questions will have complications so that formula doesn't apply but ideas do.

Exercise 1

6

How many 3 digit numbers are there with all even digits?

Exercise 1

6

How many 3 digit numbers are there with all even digits?

$$\underline{4} \cdot \underline{5} \cdot \underline{5} = \boxed{100}$$

The first digit can't be 0 because then our number would be at most 2 digits. So there are 4 choices of first digit (namely 2, 4, 6 or 8)

The second and third digits can be 0, so there are 5 choices for each of them (namely 0, 2, 4, 6 or 8).

Terminology: because I am allowed to repeat digits, I am choosing digits with replacement.

Exercise 2

7

How many 3 digit numbers are there with all even digits and no repeated digits?

Exercise 2

7

How many 3 digit numbers are there with all even digits and no repeated digits?

$$\underline{4} \cdot \underline{4} \cdot \underline{3} = \boxed{48}$$

The first digit can't be 0 because then our number would be at most 2 digits. There are 4 choices of first digit (namely 2, 4, 6 or 8)

The second digit can be 0, which should give 5 choices (0, 2, 4, 6 or 8). But we cannot repeat the first digit, leaving only $5 - 1 = 4$ choices.

The third digit can be 0, giving 5 potential choices (0, 2, 4, 6 or 8). But we cannot repeat either of the first 2 digits, leaving only $5 - 2 = 3$ choices.

Terminology: because I am not allowed to repeat digits, I am choosing digits without replacement.

Combinations

1

Example

Combinations

Partitions

Exercises

Partial Lists

2

A contest with 12 people gives out 3 distinct prizes. Previously counted the ways to give out these prizes.

$$\underline{12} \underline{11} \underline{10}$$

Giving $12 \cdot 11 \cdot 10$ ways to do this.

How many ways are there to give out the prizes if all 3 are the same?

The key difference in the second version is whether the top 3 in order are A, B, C or B, A, C, or C, B, A, or ... we give out the same prizes and there is no difference.

The answer will be less because each rearrangement of the top 3 gives the same prizes.

Partial Lists

3

A contest with 12 people gives out 3 prizes. How many ways are there to give out the prizes if all 3 are the same?

In writing $12 \cdot 11 \cdot 10$, each top 3 was counted $3 \cdot 2 \cdot 1$ times (since that is how many orders there are of 3 things), so our answer is

$$\frac{12 \cdot 11 \cdot 10}{3 \cdot 2 \cdot 1}$$

This comes up often enough to deserve notation.

Answer is $\binom{12}{3}$

Can be found on calculator by 12 nCr 3 under 'prb' menu

Key Ideas

4

$$\begin{aligned} n! &= n \cdot (n-1) \cdots 3 \cdot 2 \cdot 1 \\ &= \# \text{ ways to arrange } n \text{ items in a list} \end{aligned}$$

To choose r items from a group of n , if all items we choose are equal then the number of ways to do so is:

$$\frac{n \cdot (n-1) \cdot (n-2) \cdots (n-r+1)}{r \cdot (r-1) \cdot (r-2) \cdots (2) \cdot (1)} = \frac{n!}{(n-r)!r!}$$

$$\frac{n!}{(n-r)!r!} = \binom{n}{r} \text{ is called } n \text{ choose } r.$$

Notes: $\binom{n}{r} = \binom{n}{n-r} = nCr$ on calculators

$$\binom{n}{0} = \binom{n}{n} = \frac{n!}{0! \cdot n!} = 1 \text{ since } 0! = 1 \text{ as there is only 1 way to choose the entire set.}$$

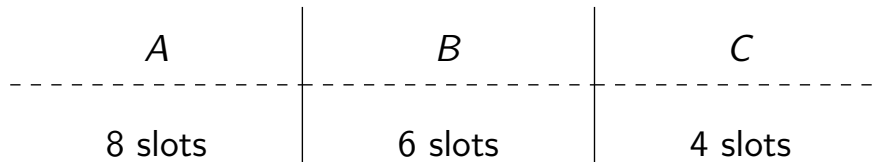
Partitions

5

18 people are to be divided into 3 groups, one with 8 people, one with 6, and one with 4. How many such divisions are possible?

Standard method: Initially assume that we have a complete rank, and then divide by the amount of “overcounting”

So first we rank all 18 people, and let group *A* be the top 8, group *B* the next 6, and group *C* the bottom 4.



There are $18!$ ways to rank everyone, but within each group people are equal, so we over counted by $8! \cdot 6! \cdot 4!$. Our final answer is thus $\frac{18!}{8! \cdot 6! \cdot 4!}$

Partitions

6

18 people are to be divided into 3 groups, one with 8 people, one with 6, and one with 4. How many such divisions are possible?

Second approach:

Step 1: Pick the group with 8 people.

We can do so in $\binom{18}{8} = \frac{18!}{8! \cdot 10!}$ ways.

Step 2: From the remaining $18 - 8 = 10$ people, pick the group with 6 people. This also determines the group with 4 people.

There are $\binom{10}{6} = \frac{10!}{6! \cdot 4!}$ ways to do so.

That gives a final answer of $\binom{18}{8} \cdot \binom{10}{6} = \frac{18!}{8! \cdot 10!} \cdot \frac{10!}{6! \cdot 4!} = \frac{18!}{8! \cdot 6! \cdot 4!}$

Exercise 1

7

4 distinct numbers are picked from the integers $\{1, 2, \dots, 30\}$. How many ways are there to draw them such that all 4 are divisible by 3?

Exercise 1

7

4 distinct numbers are picked from the integers $\{1, 2, \dots, 30\}$. How many ways are there to draw them such that all 4 are divisible by 3?

As 30 is the 10th multiple of 3, 10 of the possible numbers are divisible by 3.

We want to choose 4 of those 10, which can be done in

$$\binom{10}{4} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{4!} = 210$$

different ways.

Exercise 2

8

4 distinct numbers are picked from the integers $\{1, 2, \dots, 30\}$. How many ways are there to draw them such that 3 are divisible by 5 and the other is divisible by 7?

Exercise 2

8

4 distinct numbers are picked from the integers $\{1, 2, \dots, 30\}$. How many ways are there to draw them such that 3 are divisible by 5 and the other is divisible by 7?

6 numbers in the list are multiples of 5, while 4 are multiples of 7.

$5 \cdot 7 = 35 > 30$ so none are multiples of both.

There are $\binom{6}{3} = \frac{6 \cdot 5 \cdot 4}{3!} = 20$ ways to choose the 3 that are divisible by 5.

For each pick of those 3, there are $\binom{4}{1} = 4$ ways to choose the one divisible by 7.

That gives a total of $20 \cdot 4 = 80$ ways to do both.

The Binomial Distribution

1

Example

Bernoulli random variables

Binomial distributions

Exercises

Example

2

Avery is practicing free throws. If they make each shot with probability 0.7 and each shot is independent, what is the probability that they make the next 4 shots and then miss the 2 after that? What is the probability that they make exactly 4 of the next 6 shots?

4 makes then 2 misses is $(P[\text{Make}])^4 \cdot (P[\text{Miss}])^2 = 0.7^4 \cdot 0.3^2$

To make exactly 4 of 6, there are $\binom{6}{4}$ ways to choose which 4 shots are successful.

Each order has probability $0.7^4 \cdot 0.3^2$ for a final answer of $\binom{6}{4} \cdot 0.7^4 \cdot 0.3^2$

Bernoulli random variables

3

A Bernoulli(p) random variable, aka a Bernoulli 0-1 random variable, is a variable that can only be 0 or 1. Sometimes the case $X = 1$ is called a success, and $X = 0$ a failure.

$$P[X = 1] = p$$

$$P[X = 0] = 1 - p = q$$

$$E[X] = 1 \cdot p + 0 \cdot (1 - p) = p$$

$$E[X^2] = 1^2 \cdot p + 0^2 \cdot (1 - p) = p$$

$$\text{Var}[X] = E[X^2] - (E[X])^2$$

$$\text{Var}[X] = p - p^2 = p(1 - p)$$

$$\text{Var}[X] = pq$$

Bernoulli variance trick

4

If X is a random variable that can only take on two values a and b , with

$$P[X = b] = p \quad P[X = a] = 1 - p = q$$

then the mean and variance are

$$E[X] = a \cdot q + b \cdot p = a + p \cdot (b - a)$$

$$E[X^2] = a^2 \cdot q + b^2 \cdot p$$

$$\text{Var}[X] = E[X^2] - (E[X])^2$$

$$\text{Var}[X] = (b - a)^2 \cdot p \cdot q$$

Or: $X = (b - a)Y + a \quad Y \sim \text{Bernoulli}(p)$

$$E[X] = (b - a)E[Y] + a = p(b - a) + a$$

$$\text{Var}[X] = (b - a)^2 \text{Var}[Y]$$

$$\text{Var}[X] = (b - a)^2 \cdot p \cdot q$$

Final result arguably worth memorizing especially on later exams

Binomial distributions

5

X is a binomial (n, p) random variable if X is the number of successes in n independent trials, each of which is a success with the same probability p .

In the free throw example, $n = 6$, $p = 0.7$ and X was the number of made free throws.

$$P[X = k] = \binom{n}{k} p^k (1 - p)^{n-k}$$

The key pieces that you need are

- A fixed number of trials
- Trials are independent
- Success probability is the same in all trials

Binomial Distributions

6

To find the mean and variance, note that

$$X = \sum_{i=1}^n X_i, \quad X_i \sim \text{independent, Bernoulli}(p)$$

$$E[X] = \sum_{i=1}^n E[X_i] = \sum_{i=1}^n p = np$$

$$\text{Var}[X] = \sum_{i=1}^n \text{Var}[X_i] \text{ by independence}$$

$$\text{Var}[X] = \sum_{i=1}^n p(1 - p)$$

$$\text{Var}[X] = np(1 - p) = npq$$

Exercise 1

7

A commuter airline sells 32 tickets for a flight on a plane that has 30 seats. The probability that any particular passenger will not show up for a flight is 0.10, independent of other passengers. Find the probability that more passengers show up for the flight than there are seats available.

Exercise 1

7

A commuter airline sells 32 tickets for a flight on a plane that has 30 seats. The probability that any particular passenger will not show up for a flight is 0.10, independent of other passengers. Find the probability that more passengers show up for the flight than there are seats available.

Let $N = \#$ of passengers that show up. $N \sim \text{Binomial}(n = 32, p = 0.9)$.

$$P[N = 32] = \binom{32}{32} \cdot (0.9)^{32} = 0.9^{32}$$

$$P[N = 31] = \binom{32}{31} \cdot (0.9)^{31}(0.1)^1 = 32(0.9)^{31}(0.1)^1$$

$$\begin{aligned} P[N > 30] &= P[N = 32] + P[N = 31] \\ &= 0.9^{32} + 32(0.9)^{31}(0.1)^1 = \boxed{0.156} \end{aligned}$$

Exercise 2

8

An airline sells 32 tickets for a flight. The probability that any particular passenger will not show up for a flight is 0.10, independent of other passengers. What are the mean and variance of the number of passengers who show up?

Exercise 2

8

An airline sells 32 tickets for a flight. The probability that any particular passenger will not show up for a flight is 0.10, independent of other passengers. What are the mean and variance of the number of passengers who show up?

The number of passengers N who arrive is binomial with $n = 32$ and $p = 0.9$, so

$$E[N] = 32 \cdot 0.9 = 28.8$$

$$\text{Var}[N] = 32 \cdot 0.9 \cdot 0.1 = 2.88$$

$$\text{SD}[N] = 1.69$$

Multinomial Distribution

1

Example

Multinomial Distribution

Exercise

Example

2

Accidents are categorized into three groups: minor, moderate, and severe. These occur with probabilities 0.5 for minor, 0.4 for moderate, and 0.1 for severe.

Two accidents occur independently in one month. Find the probability that neither accident is severe and at most one is moderate.

This is not binomial because each trial / accident has more than 2 possible outcomes. But many of the ideas remain the same.

We want $P[1 \text{ minor}, 1 \text{ moderate}] + P[2 \text{ minor}]$

$$= \binom{2}{1} (0.5)(0.4) + \binom{2}{2} (0.5)^2 = 2(0.4)(0.5) + (0.5)^2 = 0.65$$

↑

Choose which accident is minor

Multinomial Distribution: 3 Outcomes

3

That example had 3 possible outcomes instead of 2.
That made it an example of the multinomial distribution.

Suppose there are n trials with 3 possible outcomes. Let the probabilities of those outcomes be p_1, p_2 and p_3 (so $\sum p_i = 1$) and let X_i be the number of trials that have outcome i . Then

$$\begin{aligned} P[X_1 = k_1, X_2 = k_2, X_3 = k_3] &= \binom{n}{k_1} \binom{n - k_1}{k_2} \binom{n - k_1 - k_2}{k_3} p_1^{k_1} p_2^{k_2} p_3^{k_3} \\ &= \frac{n!}{k_1!(n - k_1)!} \cdot \frac{(n - k_1)!}{k_2!(n - k_1 - k_2)!} \binom{k_3}{k_3} p_1^{k_1} p_2^{k_2} p_3^{k_3} \\ &= \frac{n!}{k_1!k_2!k_3!} p_1^{k_1} p_2^{k_2} p_3^{k_3} \end{aligned}$$

Multinomial Distribution: General Case

4

Suppose that there are n independent trials, each with the same r possible outcomes. Let $p_1, p_2, \dots, p_r$ be the probabilities of the outcomes, and X_i the number of trials resulting in the i -th outcome. Then

$$P[X_1 = k_1, X_2 = k_2, \dots, X_r = k_r] = \frac{n!}{k_1!k_2! \dots k_r!} p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$$

As with binomial, need:

- A fixed number of trials
- Different trials are independent
- All trials have same distribution

Difference is now can have > 2 possibilities.

Exercise

5

Accidents are categorized as minor, moderate, or severe. The probability that a given accident is minor is 0.5, that it is moderate is 0.4, and that it is severe is 0.1.

Four accidents occur independently in one month. Find the probability that there is at least one accident of each type.

Exercise

5

Accidents are categorized as minor, moderate, or severe. The probability that a given accident is minor is 0.5, that it is moderate is 0.4, and that it is severe is 0.1.

Four accidents occur independently in one month. Find the probability that there is at least one accident of each type.

There are 3 cases:

$$P[2 \text{ minor, } 1 \text{ moderate, } 1 \text{ severe}] = \frac{4!}{2!1!1!} (0.5)^2 (0.4) (0.1)$$

$$P[1 \text{ minor, } 2 \text{ moderate, } 1 \text{ severe}] = \frac{4!}{1!2!1!} (0.5) (0.4)^2 (0.1)$$

$$P[1 \text{ minor, } 1 \text{ moderate, } 2 \text{ severe}] = \frac{4!}{1!1!2!} (0.5) (0.4) (0.1)^2$$

$$\begin{aligned} P[\text{Total}] &= 0.12 + 0.096 + 0.024 \\ &= 0.24 \end{aligned}$$

Hypergeometric Distribution

1

Hypergeometric Distribution

Binomial vs Hypergeometric

Examples

Exercises

Non-Independent Draws

2

A crate of 10 electrical components has 4 defective components. If 3 components are randomly selected, find the probability that at most one of them is defective.

Two cases: none of the 3 are defective, or 1 of the three is defective. Those are mutually disjoint, so we can sum their probabilities, giving

$$\begin{aligned} & P[\text{none are defective}] + P[1 \text{ is defective}] \\ &= \frac{\binom{6}{3}}{\binom{10}{3}} + \frac{\binom{6}{2} \binom{4}{1}}{\binom{10}{3}} \\ &= \frac{20}{120} + \frac{15 \cdot 4}{120} = \frac{2}{3} \end{aligned}$$

Hypergeometric Distribution

3

Hypergeometric: have G good pieces out of N total. Choose n , and want the probability of choosing g good items.

$$\frac{\text{Number of ways to choose exactly } g \text{ good items}}{\text{Number of ways to choose } n \text{ total items}} = \frac{\binom{G}{g} \binom{N-G}{n-g}}{\binom{N}{n}}$$

This is not a binomial distribution: knowing whether or not the first item is good gives information about whether or not the second one will be good.

If $n > g$, then it isn't possible to have n successes, while that would be possible with a binomial.

Binomial vs Hypergeometric

4

Key words to tell hypergeometric vs binomial: Is sample with or without replacement?

A binomial needs each trial to be independent. This occurs when sampling *with* replacement.

A hypergeometric distribution comes up when we are sampling *without* replacement from a finite population.

Examples

5

When packing for a trip, I draw 6 socks without replacement from a drawer that contains 16 black socks and 4 white socks.

What is the probability that I will draw 6 white socks? What is the probability that I will draw 4 black socks and 2 white socks?

This is hypergeometric since the different draws are not independent.
Let W denote the number of white socks that are drawn.

$P[W = 6] = 0$ since there are only 4 white socks in the drawer.

$$P[W = 2] = \frac{\binom{4}{2} \cdot \binom{16}{4}}{\binom{20}{6}} = 0.282$$

Binomial variation

6

Suppose that I draw 6 socks *with* replacement from a drawer that contains 16 black socks and 4 white socks.

What is the probability that I will draw 6 white socks? What is the probability that I will draw 4 black socks and 2 white socks?

This is binomial distribution since the different draws are independent.
Let W denote the number of white socks that are drawn.

$$\begin{aligned} P[W = 6] &= 0.2^6 \\ P[W = 2] &= \binom{6}{2} (0.2)^2 (0.8)^4 \\ &= 0.246 \end{aligned}$$

Exercise 1

7

I draw 6 socks without replacement from a drawer that contains 10 black socks, 6 brown socks, and 4 white socks. Find the probability that I will draw 2 socks of each color.

Exercise 1

7

I draw 6 socks without replacement from a drawer that contains 10 black socks, 6 brown socks, and 4 white socks. Find the probability that I will draw 2 socks of each color.

Draws are not independent b/c sampling without replacement.

Let Bl , Br and W equal # of black, brown, and white socks drawn.

$$P[Bl = Br = W = 2] = \frac{\binom{10}{2} \binom{6}{2} \binom{4}{2}}{\binom{20}{6}} = 0.104$$

Exercise 2

8

I randomly select 6 socks from a drawer. Each sock has a 50% chance of being black, a 30% chance of being brown, and a 20% chance of being white, independently of the other socks. Find the probability that I will draw 2 socks of each color.

Exercise 2

8

I randomly select 6 socks from a drawer. Each sock has a 50% chance of being black, a 30% chance of being brown, and a 20% chance of being white, independently of the other socks. Find the probability that I will draw 2 socks of each color.

This is multinomial distribution because the draws are independent.

Let Bl , Br and W equal # of black, brown, and white socks drawn. Then

$$\begin{aligned}
 P[Bl = Br = W = 2] &= \binom{6}{2} \binom{4}{2} (0.5)^2 (0.3)^2 (0.2)^2 \\
 &= \frac{6!}{2!2!2!} (0.5)^2 (0.3)^2 (0.2)^2 \\
 &= 0.081
 \end{aligned}$$

Geometric Series

1

Overview

Geometric Series

Generalizations

Overview

2

Many discrete distributions can equal any non-negative integer.

Deriving the mean / variance of these requires using infinite series.

Exam questions will not require derivations, just the results.

The material in this lesson is optional, will only be used for theory.

Geometric Series

3

Suppose that $|r| < 1$. Then

$$\begin{aligned}
 S &= \sum_{n=0}^{\infty} ar^n \\
 S &= a + ar + ar^2 + \dots \\
 &\quad \searrow \quad \searrow \quad \searrow \\
 Sr &= \quad ar + ar^2 + ar^3 + \dots \\
 S - Sr &= a \\
 S &= \frac{a}{1-r}
 \end{aligned}$$

Geometric Series

4

In words, the sum of geometric series is

$$\begin{aligned}
 S &= \frac{\text{first term}}{1 - \text{ratio}} \\
 \sum_{n=10}^{\infty} ar^n &= \frac{ar^{10}}{1-r}
 \end{aligned}$$

For a partial sum (the final result holds for any r),

$$\begin{aligned}
 \sum_{n=0}^m ar^n &= \sum_{n=0}^{\infty} ar^n - \sum_{n=m+1}^{\infty} ar^n \\
 &= \frac{a}{1-r} - \frac{ar^{m+1}}{1-r} \\
 &= \frac{\text{first term} - \text{first missing term}}{1-r}
 \end{aligned}$$

Example

5

Suppose we want to find $\sum_{n=3}^{17} 5 \cdot \frac{e^{n+2}}{2^{3n} 3^{-n}}$.

$n = 3$ is the first term, and $n = 18$ is the first missing term.

The ratio is $e/(2^3 3^{-1}) = 3e/8$ so we get

$$\begin{aligned} \sum_{n=3}^{17} 5 \cdot \frac{e^{n+2}}{2^{3n} 3^{-n}} &= \frac{\text{first term} - \text{first missing term}}{1 - \text{ratio}} \\ &= \frac{\frac{5e^{3+2}}{2^9 3^{-3}} - \frac{5e^{18+2}}{2^{3 \cdot 18} 3^{-18}}}{1 - \frac{3e}{8}} \end{aligned}$$

 $n \cdot ar^n$

6

What if instead we wanted

$$\begin{aligned} S &= \sum_{n=0}^{\infty} n \cdot ar^n, \quad |r| < 1 \\ S &= 0 \cdot a + 1 \cdot ar + 2 \cdot ar^2 + \dots \\ &\quad \searrow \quad \quad \searrow \quad \quad \searrow \\ Sr &= 0 \cdot ar + 1 \cdot ar^2 + 2 \cdot ar^3 + \dots \\ S - Sr &= 0 \cdot a + 1 \cdot ar + 1 \cdot ar^2 + \dots \\ S(1 - r) &= \frac{ar}{1 - r} \\ S &= \frac{ar}{(1 - r)^2} \end{aligned}$$

Will use this to find mean of a geometric random variable

$$n^2 \cdot ar^n$$

7

Harder still:

$$\begin{aligned}
 S &= \sum_{n=0}^{\infty} n^2 \cdot ar^n, \quad |r| < 1 \\
 S &= 0^2 \cdot a + 1^2 \cdot ar + 2^2 \cdot ar^2 + \dots \\
 &\quad \searrow \quad \quad \searrow \quad \quad \searrow \\
 Sr &= 0^2 \cdot ar + 1^2 \cdot ar^2 + 2^2 \cdot ar^3 + \dots \\
 S - Sr &= 1 \cdot ar + 3 \cdot ar^2 + 5 \cdot ar^3 + \dots \\
 S(1 - r) &= \sum_{n=1}^{\infty} (2n - 1)ar^n = 2 \sum_{n=1}^{\infty} nar^n - \sum_{n=1}^{\infty} ar^n \\
 S(1 - r) &= 2 \cdot \frac{ar}{(1 - r)^2} - \frac{ar}{1 - r} \\
 S &= \frac{ar(1 + r)}{(1 - r)^3}
 \end{aligned}$$

Calculus Approach

8

Alternatively,

$$\begin{aligned}
 \sum_{n=0}^{\infty} ar^n &= \frac{a}{1 - r} \\
 \frac{d}{dr} \sum_{n=0}^{\infty} ar^n &= \sum_{n=0}^{\infty} \frac{d}{dr} ar^n = \frac{d}{dr} \frac{a}{1 - r} \\
 \sum_{n=0}^{\infty} nar^{n-1} &= \frac{a}{(1 - r)^2} \\
 \sum_{n=0}^{\infty} nar^n &= \frac{ar}{(1 - r)^2}
 \end{aligned}$$

Taking a second derivative leads to $\sum n^2 ar^n = \frac{ar(1 + r)}{(1 - r)^3}$

Taylor Series for e^x

1

The Taylor Series for e^x

Examples

The Taylor Series for e^x

2

When we study the Poisson distribution, we will need facts about e^x .

$$\begin{aligned}
 e^x &= 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots \\
 \text{Note : } \frac{d}{dx} \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots \right) &= \left(0 + 1 + 2 \cdot \frac{x}{2!} + 3 \cdot \frac{x^2}{3!} + \cdots \right) \\
 &= 1 + x + \frac{x^2}{2!} + \cdots = e^x \\
 e^u &= 1 + u + \frac{u^2}{2} + \frac{u^3}{3!} + \cdots \\
 e^{-x} &= 1 - x + \frac{x^2}{2} - \frac{x^3}{3!} + \cdots
 \end{aligned}$$

Examples

3

$$\begin{aligned}
 \sum_{n=0}^{\infty} \frac{5^n e^{tn}}{n!} \cdot e^{-5} &= e^{-5} \sum_{n=0}^{\infty} \frac{(5e^t)^n}{n!} \\
 &= e^{-5} \sum_{n=0}^{\infty} \frac{u^n}{n!} \quad u = 5e^t \\
 &= e^{-5} \cdot e^u = e^{-5} \cdot e^{(5e^t)} \\
 &= e^{5e^t - 5}
 \end{aligned}$$

Missing Terms

4

Suppose we want to find $\sum_{n=2}^{\infty} \frac{2^n}{n!}$

We know $\sum_{n=0}^{\infty} \frac{2^n}{n!} = e^2$

Our sum is missing the $n = 0$ and $n = 1$ terms, so

$$\begin{aligned}
 \sum_{n=2}^{\infty} \frac{2^n}{n!} &= \sum_{n=0}^{\infty} \frac{2^n}{n!} - \left(\frac{2^0}{0!} + \frac{2^1}{1!} \right) \\
 &= e^2 - \left(\frac{1}{1} + \frac{2}{1} \right) \\
 &= e^2 - 3
 \end{aligned}$$

The Geometric Distribution

1

Rolling a die until the first six

Geometric random variables

Exercise

Rolling a die until the first six

2

Suppose I roll a die until I get a 6. Let N be the total number of rolls. What is the distribution of N ?

$$\begin{array}{ll}
 P[N > 0] = 1 & P[N = 1] = \frac{1}{6} \\
 P[N > 1] = \frac{5}{6} & P[N = 2] = \left(\frac{5}{6}\right) \cdot \frac{1}{6} \\
 P[N > 2] = \left(\frac{5}{6}\right)^2 & P[N = 3] = \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} \\
 \vdots & \vdots \\
 P[N > k] = \left(\frac{5}{6}\right)^k & P[N = n] = \left(\frac{5}{6}\right)^{n-1} \cdot \frac{1}{6}
 \end{array}$$

Expected Value

3

$$\begin{aligned}
 E[N] &= \sum_{n=1}^{\infty} n \cdot P[N = n] = \sum_{k=0}^{\infty} P[N > k] \\
 &= \sum_{n=1}^{\infty} n \cdot \frac{1}{6} \cdot \left(\frac{5}{6}\right)^{n-1} = \sum_{k=0}^{\infty} \left(\frac{5}{6}\right)^k
 \end{aligned}$$

The survival approach is nicer here because it gives us a geometric series.

$$E[N] = \frac{1}{1 - \frac{5}{6}} = 6$$

Variance

4

$$\begin{aligned}
 E[N^2] &= \sum_{n=1}^{\infty} n^2 \cdot P[N = n] \\
 &= \sum_{n=0}^{\infty} n^2 \cdot \frac{1}{6} \cdot \left(\frac{5}{6}\right)^{n-1} = \sum_{n=0}^{\infty} n^2 \cdot \frac{1}{6} \cdot \frac{6}{5} \cdot \left(\frac{5}{6}\right)^n \\
 &= \frac{ar(1+r)}{(1-r)^3} = \frac{\left[\frac{1}{6} \cdot \frac{6}{5}\right] \cdot \frac{5}{6} \cdot \frac{11}{6}}{\left(1 - \frac{5}{6}\right)^3} \\
 &= \frac{\frac{1}{5} \cdot \frac{5}{6} \cdot \frac{11}{6}}{\frac{1}{6^3}} = 66 \\
 \text{Var}[N] &= 66 - 6^2 = 30
 \end{aligned}$$

Geometric starting at 1

5

X is a geometric random variable on $\{1, 2, \dots\}$ with parameter p if X is the number of trials up to, and including, the first success.

$$P[X = n] = p \cdot (1 - p)^{n-1}$$

$$E[X] = \frac{1}{p}, \text{ memorize or use survival method}$$

$$E[X^2] = \frac{2}{p^2} - \frac{1}{p} \quad (\text{don't memorize})$$

$$\text{Var}[X] = \left(\frac{2}{p^2} - \frac{1}{p} \right) - \left(\frac{1}{p} \right)^2 = \frac{1}{p^2} - \frac{1}{p}$$

$$\text{Var}[X] = \frac{1-p}{p^2} \quad (\text{memorize})$$

Geometrics Starting at 0

6

Y is a geometric random variable on $0, 1, 2, \dots$ if Y counts the number of failures before the first success.

$$P[Y = n] = p \cdot (1 - p)^n$$

$Y = X - 1$ since the # of trials until the first success, including the success, is always exactly one plus the number of failures. That means that

$$E[Y] = E[X] - 1 = \frac{1}{p} - 1$$

$$E[Y] = \frac{1-p}{p}$$

$$\text{Var}[Y] = \text{Var}[X]$$

$$\text{Var}[Y] = \frac{1-p}{p^2} = E[Y] \cdot E[X]$$

Exercise

7

Let N be the number of visits (possibly 0) that a randomly chosen insured makes to the doctor in a year. If N has a geometric distribution with mean 3, what is the probability that a randomly chosen insured makes at least 2 visits to the doctor in a year?

Exercise

7

Let N be the number of visits (possibly 0) that a randomly chosen insured makes to the doctor in a year. If N has a geometric distribution with mean 3, what is the probability that a randomly chosen insured makes at least 2 visits to the doctor in a year?

We are told that N can be 0, so N is a geometric on $\{0, 1, 2, \dots\}$

$$\begin{aligned}
 E[N] &= \frac{1-p}{p} \\
 3 &= \frac{1-p}{p}, \quad p = \frac{1}{4} \\
 P[N \geq 2] &= 1 - P[N = 0] - P[N = 1] \\
 &= 1 - p - (1-p) \cdot p \\
 &= 1 - 2p + p^2 \\
 &= (1-p)^2 = \boxed{\frac{9}{16}}
 \end{aligned}$$

Memoryless Property

1

Intuitive Example

Algebra Approach

Memoryless Property

Variance

Exercise

“Nines”

2

In each round of the dice game “Nines” I roll two fair six-sided dice. The game ends if either a 7 or 9 is rolled, and continues to the next round on any other outcome.

If I play a game of Nines, what is the expected number of rounds that I will play?

The game ends on a given round with probability $\frac{6}{36} + \frac{4}{36} = \frac{5}{18}$.

The length is therefore a geometric random variable (starting at 1) with $p = 5/18$.

Expected game length is $1/p = 18/5$ rounds.

Watching “Nines”

3

If I watch someone play Nines, find the expected number of rounds that I will watch.

There is the same as the previous example, so the answer is still $18/5$.

Now suppose I start watching a game after the 3rd round. How many more rounds will I watch?

Intuitively, this is no different than before. Each round I watch will end the game with probability $5/18$, the number of rounds I watch is a geometric starting at 1, and the answer is still $18/5$.

Algebra Approach

4

Let $N =$ game length.

Start watching after 3 rounds, so watch for $N - 3$ rounds.

The game lasts for more than 3 rounds, so we are given $N > 3$.

$$\begin{aligned} P[N - 3 = k \mid N > 3] &= \frac{P[N = k + 3, N > 3]}{P[N > 3]} \\ &= \frac{(1 - p)^{k+3-1} \cdot p}{(1 - p)^3} \\ &= (1 - p)^{k-1} \cdot p \\ &= P[N = k] \end{aligned}$$

so $(N - 3 \mid N > 3)$ and (the original) N have the same distribution.
In particular, $E[N - 3 \mid N > 3] = E[N] = 18/5$.

Memoryless Property

5

If N is a geometric with parameter p , then $(N - k \mid N > k)$ is a geometric starting at 1 with the same p .

- $(N - k \mid N > k)$ is at least 1, hence starting at 1
- Holds whether N starts at 0 or 1
- Geometrics are only discrete distribution with this property

Variance

6

A game of Nines lasts for at least 4 rounds. What are the mean and variance of the length of the game?

We are given that $N \geq 4$, so $N > 3$.

$$\begin{aligned}
 E[N \mid N > 3] &= E[N - 3 + 3 \mid N > 3] \\
 &= E[N - 3 \mid N > 3] + 3 \\
 &= E[N] + 3 = \frac{18}{5} + 3 = 6.6 \\
 \text{Var}[N \mid N > 3] &= \text{Var}[N - 3 + 3 \mid N > 3] \\
 &= \text{Var}[N - 3 \mid N > 3] \\
 &= \text{Var}[N] = \frac{1 - p}{p^2} \\
 &= 9.36
 \end{aligned}$$

Exercise

7

Suppose X satisfies $P[X = k] = 0.2(0.8)^k$ for $k = 0, 1, 2, \dots$. Find $E[X \mid X > 6]$ and $\text{Var}[X \mid X > 6]$.

Exercise

7

Suppose X satisfies $P[X = k] = 0.2(0.8)^k$ for $k = 0, 1, 2, \dots$. Find $E[X \mid X > 6]$ and $\text{Var}[X \mid X > 6]$.

X can be 0, $P[X = k]$ decays geometrically, so X is a geometric starting at 0.
 $P[X = 0] = p = 0.2(0.8)^0$, so $p = 0.2$

$$\begin{aligned}
 E[X \mid X > 6] &= E[X - 6 + 6 \mid X > 6] \\
 &= E[X - 6 \mid X > 6] + 6 \\
 &= E[\text{Geo starting at 1, } p = 0.2] + 6 \\
 &= \frac{1}{0.2} + 6 = 11 \\
 \text{Var}[X \mid X > 6] &= \text{Var}[X - 6 \mid X > 6] \\
 &= \text{Var}[\text{Geo starting at 1, } p = 0.2] \\
 &= \frac{1 - 0.2}{0.2^2} = 20
 \end{aligned}$$

Negative Binomial Distribution

1

Rolling a die until the third six

Negative binomial

Example

Exercise

Rolling a die until the third six

2

Roll a die until the third time that a 6 is rolled. Let N denote the number of non-sixes (failures) that we roll. What is the distribution of N ?

If $N = n$, then roll $n + 3$ was a 6.

The first $(n + 3) - 1$ rolls had $3 - 1 = 2$ sixes. Since there were exactly 3 sixes in the first $n + 3$ rolls, there were n non-sixes.

$$\begin{aligned} P[N = n] &= \binom{(n+3)-1}{3-1} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^n \frac{1}{6} \\ &= \binom{n+(3-1)}{3-1} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^n \\ &= \binom{n+(3-1)}{n} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^n \end{aligned}$$

Rolling a die until the third six

3

What are the mean and variance of N ?

We can use a trick to find them.

Let $N_1 = \#$ failures before first 6

$N_2 = \#$ number of failures before 2nd 6

$N = N_3 = \#$ number of failures before 3rd 6

$$N = (N_1 - 0) + (N_2 - N_1) + (N_3 - N_2)$$

$N_1 - 0$ is the number of failures before the first six.

$N_2 - N_1$ is the number of failures after the first six, but before the second six.

$N_3 - N_2$ is the number of failures after the second six, but before the third six.

Rolling a die until the third six comes up

4

Key point: $\#$ of failures between sixes is a geometric that starts at 0.

N is the sum of 3 independent geometrics on $\{0, 1, 2, \dots\}$ and

$$E[N] = E[N_1] + E[N_2 - N_1] + E[N_3 - N_2]$$

$$E[N] = 3E[N_1] = 3 \cdot (6 - 1) = \boxed{15}$$

$$\text{Var}[N] = 3 \left(\text{Var of a geometric} \left(p = \frac{1}{6} \right) \right)$$

$$= 3 \cdot \frac{1-p}{p^2}, \quad p = \frac{1}{6}$$

$$\text{Var}[N] = \frac{3 \cdot \frac{5}{6}}{\left(\frac{1}{6}\right)^2} = \boxed{90}$$

Negative binomial

5

N is a negative binomial random variable with parameters r and p if it is the sum of r independent geometric random variables starting at 0. It is the number of failures before the r -th success.

$$\begin{aligned} P[N = n] &= \binom{n + (r - 1)}{r - 1} \cdot p^r \cdot (1 - p)^n \\ &= \binom{n + (r - 1)}{n} \cdot p^r \cdot (1 - p)^n \\ E[N] &= r \cdot \frac{1 - p}{p} \\ \text{Var}[N] &= r \cdot \frac{1 - p}{p^2} = \frac{r(1 - p)}{p^2} \end{aligned}$$

Example

6

An insurance policy covers accidents at a manufacturing plant. The probability that one or more accidents will occur during any given month is $3/5$. The number of accidents that occur in any given month is independent of the number of accidents that occur in all other months.

Find the probability that June will be the fourth month in 2025 in which at least one accident occurs.

Having an accident = “success,” $p = 3/5$.

Want $r = 4$ th success in 6th try, $n = 6 - 4 = 2$ “failures”.

$$\binom{2 + (4 - 1)}{4 - 1} \cdot \left(\frac{3}{5}\right)^4 \cdot \left(\frac{2}{5}\right)^2$$

Exercise

7

Let N be the sum of r independent geometrics on $\{0, 1, 2, \dots\}$. Suppose that $E[N] = 12$ and $\text{Var}[N] = 60$. Find the probability that N is no more than 2.

Exercise

7

Let N be the sum of r independent geometrics on $\{0, 1, 2, \dots\}$. Suppose that $E[N] = 12$ and $\text{Var}[N] = 60$. Find the probability that N is no more than 2.

$$E[N] = \frac{r(1-p)}{p} = 12$$

$$\text{Var}[N] = \frac{r(1-p)}{p^2} = \frac{r(1-p)}{p} \cdot \frac{1}{p}$$

$$60 = 12 \cdot \frac{1}{p} \Rightarrow p = \frac{1}{5}, \quad r = 3$$

$$\begin{aligned} P[N \leq 2] &= P[N = 0] + P[N = 1] + P[N = 2] \\ &= \binom{0+3-1}{3-1} p^3 + \binom{1+2}{2} (1-p)p^3 + \binom{2+2}{2} (1-p)^2 p^3 \\ &= p^3 + \binom{3}{2} (1-p)p^3 + \binom{4}{2} (1-p)^2 p^3 \\ &= \boxed{0.05792} \end{aligned}$$

The Poisson distribution

1

The Poisson distribution

Mean and variance

Key Points

Exercises

The Poisson distribution

2

X is a $\text{Poisson}(\lambda)$ random variable if

$$P[X = n] = e^{-\lambda} \cdot \frac{\lambda^n}{n!}, \quad n = 0, 1, 2, \dots$$

To remember this, recall that the Taylor series for e^λ is

$$e^\lambda = 1 + \lambda + \frac{\lambda^2}{2!} + \dots + \frac{\lambda^n}{n!} + \dots$$

The $e^{-\lambda}$ term is the constant needed to make the probabilities sum to 1.

Poisson variables arise in nature by counting the number of occurrences of unusual events if the number of occurrences in disjoint time intervals are independent.

Poisson Means

3

$$\begin{aligned}
 E[N] &= \sum_{n=0}^{\infty} n \cdot P[N = n] = \sum_{n=1}^{\infty} n \cdot e^{-\lambda} \cdot \frac{\lambda^n}{n!} \\
 &= \lambda \sum_{n=1}^{\infty} e^{-\lambda} \cdot \frac{\lambda^{n-1}}{(n-1)!} \\
 &= \lambda \sum_{m=0}^{\infty} e^{-\lambda} \cdot \frac{\lambda^m}{m!} \quad \text{by letting } m = n - 1 \\
 &= \lambda \cdot 1 \text{ since the sum is } \sum_{m=0}^{\infty} P[N = m] = 1
 \end{aligned}$$

Point: The mean of a $\text{Poisson}(\lambda)$ is λ .

Variance

4

$$\begin{aligned}
 E[N^2] &= \sum_{n=0}^{\infty} n^2 \cdot e^{-\lambda} \cdot \frac{\lambda^n}{n!} = \sum_{n=1}^{\infty} n \cdot e^{-\lambda} \cdot \frac{\lambda \cdot \lambda^{n-1}}{(n-1)!} \\
 &= \lambda \sum_{m=0}^{\infty} (m+1) \cdot e^{-\lambda} \cdot \frac{\lambda^m}{m!} \quad \text{by letting } m = n - 1 \\
 &= \lambda \left(\sum_{m=0}^{\infty} m \cdot \frac{e^{-\lambda} \cdot \lambda^m}{m!} \right) + \lambda \left(\sum_{m=0}^{\infty} \frac{e^{-\lambda} \cdot \lambda^m}{m!} \right) \\
 &= \lambda \left(\sum_{m=0}^{\infty} m \cdot P[N = m] \right) + \lambda \left(\sum_{m=0}^{\infty} P[N = m] \right) \\
 &= \lambda \cdot \lambda + \lambda \cdot 1 = \lambda^2 + \lambda \\
 \text{Var}[N] &= (\lambda^2 + \lambda) - \lambda^2 = \lambda = E[N]
 \end{aligned}$$

Key Points

5

The derivations are harder than what you will be doing on the exam.

What you need to know so far:

- The probability mass function

$$P[N = k] = e^{-\lambda} \cdot \frac{\lambda^k}{k!}$$

- $E[N] = \text{Var}[N] = \lambda$

Exercise 1

6

Policyholders are three times as likely to file two claims as to file four claims. If the number of claims filed has a Poisson distribution, find the variance of the number of claims filed.

Exercise 1

6

Policyholders are three times as likely to file two claims as to file four claims. If the number of claims filed has a Poisson distribution, find the variance of the number of claims filed.

Let N be the number of claims. $\text{Var}[N] = \lambda$, so we need to find λ .

$$P[N = 2] = 3 \cdot P[N = 4]$$

$$e^{-\lambda} \cdot \frac{\lambda^2}{2} = 3e^{-\lambda} \cdot \frac{\lambda^4}{4!}$$

$$\frac{4!}{2 \cdot 3} = \frac{\lambda^4}{\lambda^2}$$

$$4 = \lambda^2$$

$$\lambda = 2 \quad \text{or} \quad -2 \text{ but } \lambda > 0$$

$$\text{so } \lambda = \text{Var}[N] = \boxed{2}$$

Exercise 2

7

The number of annual losses has a Poisson distribution with second moment equal to 12. Find the probability that the number of annual losses is at least 2.

Exercise 2

7

The number of annual losses has a Poisson distribution with second moment equal to 12. Find the probability that the number of annual losses is at least 2.

Let N be the number of annual losses.

$$E[N^2] = \text{Var}[N] + (E[N])^2$$

$$12 = \lambda + \lambda^2$$

$$0 = \lambda^2 + \lambda - 12$$

$$0 = (\lambda - 3)(\lambda + 4)$$

$$\lambda = 3 \quad \text{because } \lambda > 0$$

$$\begin{aligned} P[N \geq 2] &= 1 - P[N = 0] - P[N = 1] \\ &= 1 - e^{-3} - 3e^{-3} \\ &= 0.80 \end{aligned}$$

Sums of Independent Poissons

1

Example

Sums of Poisson

Exercise

Example

2

If X is Poisson with mean 1.7 and Y is an independent Poisson with mean 1.3, find:

- a) $E[X + Y]$
- b) $\text{Var}[X + Y]$
- c) $P[X + Y = 2]$

$$E[X + Y] = E[X] + E[Y] = 1.7 + 1.3 = 3$$

$$\begin{aligned}\text{Var}[X + Y] &= \text{Var}[X] + \text{Var}[Y] \text{ by independence} \\ &= 1.7 + 1.3 = 3\end{aligned}$$

$$\begin{aligned}P[X + Y = 2] &= P[X = 0, Y = 2] + P[X = 1, Y = 1] + P[X = 2, Y = 0] \\ &= e^{-1.7} \cdot \frac{1.3^2}{2} e^{-1.3} + 1.7 e^{-1.7} \cdot 1.3 e^{-1.3} + \frac{1.7^2}{2} e^{-1.7} \cdot e^{-1.3} \\ &= 4.5 e^{-3} = \frac{3^2}{2} e^{-3}\end{aligned}$$

Note: answers are consistent with $X + Y \sim \text{Poisson}(3)$

Sums of Poisson

3

If $N \sim \text{Poisson}(\lambda)$, $M \sim \text{Poisson}(\mu)$ and they are independent, find $P[N + M = n]$.

$$\begin{aligned}
 P[N + M = n] &= \sum_{k=0}^n P[N = k] \cdot P[M = n - k] \\
 &= \sum_{k=0}^n e^{-\lambda} \cdot \frac{\lambda^k}{k!} \cdot e^{-\mu} \cdot \frac{\mu^{n-k}}{(n-k)!} \\
 &= e^{-(\lambda+\mu)} \left(\sum_{k=0}^n \frac{\lambda^k \cdot \mu^{n-k}}{k!(n-k)!} \cdot n! \right) \cdot \frac{1}{n!} \\
 &= \frac{e^{-(\lambda+\mu)}}{n!} \sum_{k=0}^n \lambda^k \cdot \mu^{n-k} \cdot \binom{n}{k} \\
 &= e^{-(\lambda+\mu)} \cdot \frac{(\lambda + \mu)^n}{n!} \\
 &= P[\text{Poisson}(\lambda + \mu) = n]
 \end{aligned}$$

Key Point

4

If $N_1, \dots, N_k$ are *independent* Poissons, then their sum is Poisson.

$$E[\sum N_i] = \sum E[N_i].$$

Revisiting our first example: If X is Poisson with mean 1.7 and Y is an independent Poisson with mean 1.3, find $P[X + Y = 2]$

$$X + Y \sim \text{Poisson}(\lambda = 1.3 + 1.7 = 3)$$

$$P[X + Y = 2] = \frac{3^2}{2} e^{-3}$$

Exercise

5

The number of accidents per day at a busy intersection has a Poisson distribution with mean 0.5 during a workday and 0.3 during a weekend day. If the number of accidents on different days is independent, what is the probability that there will be exactly three accidents at the intersection during a week?

Exercise

5

The number of accidents per day at a busy intersection has a Poisson distribution with mean 0.5 during a workday and 0.3 during a weekend day. If the number of accidents on different days is independent, what is the probability that there will be exactly three accidents at the intersection during a week?

The sum of independent Poissons is Poisson, so the number of accidents per week is a Poisson with mean

$$\begin{aligned}\lambda &= 5 \cdot 0.5 + 2 \cdot 0.3 = 3.1 \\ P[N = 3] &= e^{-\lambda} \cdot \frac{\lambda^3}{3!} \\ &= e^{-3.1} \cdot \frac{3.1^3}{6} \\ &= \boxed{0.224}\end{aligned}$$

Deductibles

1

Deductibles

Finding Expected Payments

Exercises

Deductibles

2

Suppose that X represents the amount of a loss.

If there is a deductible of d , then the resulting payment is

$$\text{Payment} = (X - d)_+ = \begin{cases} 0 & X \leq d \\ X - d & X > d \end{cases}$$

The uncovered cost to the insured is

$$\text{Uncovered Cost} = \min\{X, d\} = X \wedge d = \begin{cases} X & X \leq d \\ d & X > d \end{cases}$$

In words, note that

Total Loss = Insurance Payment + Uncovered Cost to Insured

$$X = (X - d)_+ + (X \wedge d)$$

Example

3

Suppose that loss amounts are uniform on $\{1, 2, 3, 4, 5\}$ and that there is a deductible of 2. What is the expected payment? What is the probability that the uncovered loss will be 2?

x	$P[X = x]$	Payment	Uncovered Loss
1	$1/5$	0	1
2	$1/5$	0	2
3	$1/5$	1	2
4	$1/5$	2	2
5	$1/5$	3	2

$$E[\text{Payment}] = \frac{1}{5} \cdot 0 + \frac{1}{5} \cdot 0 + \frac{1}{5} \cdot 1 + \frac{1}{5} \cdot 2 + \frac{1}{5} \cdot 3$$

$$= 1.2$$

$$P[\text{Uncovered Loss} = 2] = \frac{4}{5}$$

Finding the Expected Payment

4

There are often fewer possible values for the uncovered loss than for the payment, which means that it is often easier to find $E[X \wedge d]$ than $E[(X - d)_+]$. We can take advantage of this as follows:

$$X = (X - d)_+ + (X \wedge d)$$

$$E[X] = E[(X - d)_+] + E[X \wedge d]$$

$$E[(X - d)_+] = E[X] - E[X \wedge d]$$

Warning: This only works for first moments:

$$X^2 \neq (X - d)_+^2 + (X \wedge d)^2$$

$$E[X^2] \neq E[(X - d)_+^2] + E[(X \wedge d)^2]$$

Exam questions about second moments are very rare.

Exercise 1

5

A farm is insured against tornado damage. During tornado season, each week has either 0 or 1 tornados, with a probability of 0.3 of having a tornado. The policy pays \$100 per tornado, with an annual deductible of \$50.

Tornado season is 8 weeks long and the number of tornados in different weeks are independent. Find the expected annual insurance payment.

Exercise 1

5

A farm is insured against tornado damage. During tornado season, each week has either 0 or 1 tornados, with a probability of 0.3 of having a tornado. The policy pays \$100 per tornado, with an annual deductible of \$50.

Tornado season is 8 weeks long and the number of tornados in different weeks are independent. Find the expected annual insurance payment.

Let N be the number of storms, and $X = 100N$ the total loss. The uncovered loss is either 0 (if there are no tornados) or 50 (if there is at least 1 tornado). So

$$\begin{aligned} E[X \wedge 50] &= 0 \cdot P[N = 0] + 50 \cdot P[N \geq 1] \\ &= 0 + 50 \cdot (1 - 0.7^8) \\ &= 47.12 \end{aligned}$$

$$E[X] = 100 E[N] = 100 \cdot 8 \cdot 0.3 = 240$$

$$E[\text{Payment}] = 240 - 47.12 = \boxed{192.88}$$

Exercise 2

6

The number of annual losses N is a geometric on $\{0, 1, 2, \dots\}$ with mean 2. Losses are insured for \$100 each, with an annual deductible of \$150. Find the expected annual payment.

Exercise 2

6

The number of annual losses N is a geometric on $\{0, 1, 2, \dots\}$ with mean 2. Losses are insured for \$100 each, with an annual deductible of \$150. Find the expected annual payment.

$$E[N] = 2 = \frac{1-p}{p}$$

$$p = 1/3$$

$$\begin{aligned} E[\text{Uncovered Loss}] &= 0 \cdot P[N = 0] + 100 \cdot P[N = 1] + 150 \cdot P[N \geq 2] \\ &= 0 \cdot p + 100 \cdot (1-p)p + 150 \cdot (1-p)^2 \\ &= 88.9 \end{aligned}$$

$$\begin{aligned} E[\text{Payment}] &= E[\text{Total Loss}] - E[\text{Uncovered Loss}] \\ &= 2 \cdot 100 - 88.9 \\ &= \boxed{111.1} \end{aligned}$$

Policy Limits

1

Policy Limits

Exercises

Policy Limits

2

Another way for the payment to be less than the total loss is to have a policy limit.

Let X be the loss amount, and u the policy limit. With no deductible,

$$\text{Payment} = \begin{cases} X & X \leq u \\ u & u < X \end{cases}$$

In this case, $\text{Payment} = \min\{X, u\} = X \wedge u$.

Limits and Deductibles

3

With both a deductible of d and a limit of u , then there are different types of limits.

Exam questions will be explicit about how the limit works.

Examples in the study note all have a payment limit, meaning that u is the maximum payment allowed. In that case:

$$\text{Payment} = \begin{cases} 0 & X \leq d \\ X - d & d < X \leq d + u \\ u & d + u < X \end{cases}$$

The expected payment is also called the *net premium* or the *benefit premium*.

Exercise 1

4

The number of annual losses is Poisson with mean 2.4. Each loss results in 50 in damages. Total annual claims are insured with a payment limit of 75. Find the expected annual payment.

Exercise 1

4

The number of annual losses is Poisson with mean 2.4. Each loss results in 50 in damages. Total annual claims are insured with a payment limit of 75. Find the expected annual payment.

Let N be the number of losses. The payment is 0 when $N = 0$, 50 when $N = 1$, and 75 when $N \geq 2$, so

$$\begin{aligned} E[\text{Payment}] &= 0 \cdot P[N = 0] + 50 \cdot P[N = 1] + 75 \cdot P[N \geq 2] \\ &= 0 + 50 \cdot 2.4e^{-2.4} + 75(1 - e^{-2.4} - 2.4e^{-2.4}) \\ &= 62.75 \end{aligned}$$

Exercise 2

5

Loss amounts X have a binomial distribution with $n = 5$ and $p = 0.4$. If there is a deductible of 1 and a payment limit of 3, find the expected payment for a randomly selected loss.

Exercise 2

5

Loss amounts X have a binomial distribution with $n = 5$ and $p = 0.4$. If there is a deductible of 1 and a payment limit of 3, find the expected payment for a randomly selected loss.

x	0	1	2	3	4	5
$P[X = x]$	0.6^5	$5 \cdot 0.4 \cdot 0.6^4$	$10 \cdot 0.4^2 0.6^3$	$10 \cdot 0.4^3 0.6^2$	$5 \cdot 0.4^4 0.6$	0.4^5
Payment	0	0	1	2	3	3
Uncovered Loss	0	1	1	1	1	2

$$\begin{aligned} E[\text{Payment}] &= 0.3456 \cdot 1 + 0.2304 \cdot 2 + (0.0768 + 0.0102) \cdot 3 \\ &= 1.06752 \end{aligned}$$

$$E[\text{Uncovered Loss}] = 1 \cdot (1 - 0.6^5 - 0.4^5) + 2 \cdot 0.4^5 = 0.93248$$

$$\begin{aligned} E[\text{Payment}] &= E[X] - E[\text{Uncovered Loss}] \\ &= 5 \cdot 0.4 - 0.93248 = 1.06752 \end{aligned}$$

Calculator Approach to Deductibles

1

Examples Revisited

Example

2

Suppose that loss amounts are uniform on $\{1, 2, 3, 4, 5\}$ and that there is a deductible of 2. What is the expected payment?

Let X be a random loss amount.

When X exceeds the deductible, the payment is $X - 2$.

When X is below the deductible, the payment is 0.

Put x in L_1 , $P[X = x]$ in L_2 and payment in L_3 . For the payment, we will start with a formula ($L_3 = L_1 - 2$) and then edit to take into account the deductible.

$$E[\text{Payment}] = 1.2$$

Example

3

A farm is insured against damage from tornados. Each week during tornado season has either 0 or 1 tornados, with a probability of 0.3 of having a tornado. The insurance policy pays \$100 per tornado, with an annual deductible of \$50.

If tornado season is 8 weeks long and the number of tornados in different weeks are independent, what is the expected annual insurance payment?

Let N be the number of storms. $N \sim \text{Binomial}(8, 0.3)$

Put N in L_1 , probabilities in L_2 .

The payment (L_3) is $100N - 50$ when that is positive, and is 0 otherwise.

$$E[\text{Payment}] = 193$$

Example

4

The number of annual losses N is geometric on $\{0, 1, 2, \dots\}$ with mean 2. Each loss is insured for \$100, with annual deductible of \$150. Find the expected annual payment.

Can't enter all possible values, but $E[N]$ is only 2. Turns out going up to 10 isn't enough, 15 gets pretty close.

$$E[N] = 2 = \frac{1-p}{p}$$

$$2p = 1 - p \quad \Rightarrow \quad p = \frac{1}{3}$$

$$P[N = k] = p(1-p)^k = \frac{1}{3} \cdot \left(\frac{2}{3}\right)^k$$

Payment is $100N - 150$ when positive, 0 otherwise.

Again, k in L_1 , $P[N = k]$ in L_2 , and payment in L_3 . $E[\text{Payment}] = 111.1$

Only get 99 if stopping at $k = 10$. Need n to be closer to 1.

Example

5

Loss amounts X have a binomial distribution with $n = 5$ and $p = 0.4$. Suppose that there is a deductible of 1 and a payment limit of 3. What is the resulting expected payment?

When X is above the deductible and the limit hasn't yet been reached, the payment is $X - 1$. Below the deductible the payment is 0 and above the limit it is 3.

This time we will edit L_3 for both the deductible and limit.

$$E[\text{Payment}] = 1.07$$

Discrete Review

1

Basic Formulas

Moments

Combinatorics

Key Distributions

General Probability Rules

2

- $0 \leq P[A] \leq 1$
- $P[\text{State Space}] = 1, P[\emptyset] = 0$
- $P[A \cup B] = P[A] + P[B] - P[A \cap B]$
If A and B are mutually exclusive, then $P[A \cap B] = 0$ and $P[A \cup B] = P[A] + P[B]$
- $A \cup A' = \text{State Space}, A \cap A' = \emptyset$
- $P[A^c] = P[A'] = 1 - P[A], P[A] = 1 - P[A']$
- $P[A | B] = \frac{P[A \cap B]}{P[B]}$
- $P[A \cap B] = P[AB] = P[A] \cdot P[B | A]$
- A and B are independent if and only if $P[AB] = P[A] \cdot P[B]$.
In that case, $P[A | B] = P[A]$

Examples

3

If A and B are independent with $P[A \cup B] = 0.58$ and $P[A] = 0.3$, find $P[A' \cup B']$.

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$

$$P[A \cup B] = P[A] + P[B] - P[A] \cdot P[B]$$

$$0.58 = 0.3 + P[B] - 0.3P[B]$$

$$P[B] = 0.4$$

$$P[A' \cup B'] = P[A'] + P[B'] - P[A' \cap B']$$

$$= 0.7 + 0.6 - 0.7 \cdot 0.6$$

$$= \boxed{0.88}$$

$$\text{or: } A' \cup B' = (A \cap B)'$$

$$P[A' \cup B'] = 1 - P[AB]$$

$$= 1 - 0.3 \cdot 0.4$$

$$= 0.88$$

Examples

4

If $P[A | B] = 3P[B | A]$, $P[A \cup B] = 7P[AB]$ and $P[B] = 0.1$, then what is $P[AB]$?

$$P[A | B] = 3P[B | A]$$

$$\frac{P[AB]}{P[B]} = 3 \cdot \frac{P[AB]}{P[A]}$$

$$P[A] = 3 \cdot P[B] = 0.3$$

$$P[A \cup B] = P[A] + P[B] - P[AB]$$

$$7P[AB] = 0.3 + 0.1 - P[AB]$$

$$P[AB] = \frac{0.4}{8} = \boxed{0.05}$$

Moments, etc.

5

Let X be a random variable.

- y is the mode of X if $P[X = y] \geq P[X = x]$ for all x
- m is the median of X if m is the smallest value such that $P[X \leq m] = F(m) \geq 1/2$
- $E[X] = \sum_x x \cdot P[X = x]$
- $E[X^2] = \sum_x x^2 \cdot P[X = x]$
- $E[g(X)] = \sum_x g(x) \cdot P[X = x]$
- $\text{Var}[X] = E[X^2] - (E[X])^2 = E[(X - \mu)^2]$
- $E[X^2] = \text{Var}[X] + (E[X])^2$

Example

6

Suppose that $P[X = x] = cx$ for $x = 1, 2, 3, 4$ or 5 . Find the median and mode of X .

x	$P[X = x]$	$P[X \leq x]$
1	c	$1/15$
2	$2c$	$3/15$
3	$3c$	$6/15$
4	$4c$	$10/15$
5	$5c$	$15/15$

$$1 = \sum_x P[X = x] = c + 2c + 3c + 4c + 5c, \quad c = \frac{1}{15}$$

The mode is 5 since that maximizes $P[X = x]$ and the median is 4 since that is the first time $F(x) = P[X \leq x]$ exceeds $1/2$.

Example

7

$P[X = x] = x/15$ for $x = 1, 2, 3, 4$ or 5 . Find the variance of X .

$$E[X] = 1 \cdot \frac{1}{15} + 2 \cdot \frac{2}{15} + \cdots + 5 \cdot \frac{5}{15}$$

$$E[X] = \frac{55}{15} = \frac{11}{3}$$

$$E[X^2] = 1^2 \cdot \frac{1}{15} + 2^2 \cdot \frac{2}{15} + \cdots + 5^2 \cdot \frac{5}{15}$$

$$E[X^2] = \frac{225}{15} = 15$$

$$\text{Var}[X] = 15 - \left(\frac{11}{3}\right)^2 = \boxed{\frac{14}{9}}$$

Breaking Down Into Cases

8

Let $A_1, A_2, \dots, A_n$ be a partition of the sample space, i.e., $A_i \cap A_j = \emptyset$ for $i \neq j$ and $\bigcup_i A_i = S$.

Then

- $P[B] = \sum_i P[A_i] \cdot P[B | A_i]$
- $E[X] = \sum_i P[A_i] \cdot E[X | A_i]$
- $P[A_i | B] = \frac{P[A_i B]}{P[B]} = \frac{P[A_i] \cdot P[B | A_i]}{\sum_j P[A_j] \cdot P[B | A_j]}$

Combinatorics

9

$$n! = n(n-1)(n-2)\dots 2 \cdot 1$$

= # of ways to order n objects

$$1! = 1$$

$$0! = 1$$

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

$$= \binom{n}{n-k}$$

= # of ways to choose k objects from a set of n

Example

10

I randomly select two socks from a drawer that has 4 black socks and 6 brown socks, and put them into a bag that initially had 3 black and 3 brown socks. I then randomly select a sock from the bag. Find the probability that both socks from the drawer were brown given that the sock taken from the bag was brown.

Let X be the number of brown socks taken from the drawer and Y the number of brown socks drawn from the bag.

$$\begin{aligned} P[X = 2 \mid Y = 1] &= \frac{P[X = 2, Y = 1]}{P[Y = 1]} \\ &= \frac{\frac{\binom{6}{2}}{\binom{10}{2}} \cdot \frac{5}{8}}{\frac{\binom{6}{2}}{\binom{10}{2}} \cdot \frac{5}{8} + \frac{\binom{6}{1}\binom{4}{1}}{\binom{10}{2}} \cdot \frac{4}{8} + \frac{\binom{4}{2}}{\binom{10}{2}} \cdot \frac{3}{8}} \\ &= \frac{15 \cdot 5}{15 \cdot 5 + 24 \cdot 4 + 6 \cdot 3} = \boxed{\frac{25}{63}} \end{aligned}$$

Key Distributions

11

Type of Variable	Key Properties
Uniform	All possibilities are equally likely
Bernoulli	0 or 1
Binomial	Sum of Bernoullis Number of successes in n independent trials
Geometric on $\{0, 1, 2, \dots\}$	Number of <i>failures</i> before first success
Geometric on $\{1, 2, 3, \dots\}$	Number of <i>trials</i> including first success
Negative Binomial	Number of failures before r -th success Sum of geometrics
Poisson	Sum of independent Poissons is Poisson

Key Distributions

12

Variable	$P[X = x]$	$E[X]$	$\text{Var}[X]$
Uniform on $\{1, 2, \dots, n\}$	$\frac{1}{n}$ $1 \leq x \leq n$	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$
Bernoulli(p)	$p, x = 1$ $1 - p, x = 0$	p	$p(1-p)$
Binomial(n, p)	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Geometric(p) on $\{0, 1, 2, \dots\}$	$p(1-p)^x$	$\frac{1}{p} - 1 = \frac{1-p}{p}$	$\frac{1-p}{p^2}$
Geometric(p) on $\{1, 2, 3, \dots\}$	$p(1-p)^{x-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Binomial	$\binom{x+(r-1)}{r-1} \cdot p^r \cdot (1-p)^x$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$
Poisson	$e^{-\lambda} \cdot \frac{\lambda^x}{x!}$	λ	λ



B.0 One Dimensional Calculus - Outline

B.0.1 One-Dimensional Derivatives

Definition of Derivative

Basic Formulas

Chain Rule

Product Rule

Absolute Values

Sines and Cosines

Further Examples

B.0.2 1-Dimensional integrals

B.0.3 Integration By Parts

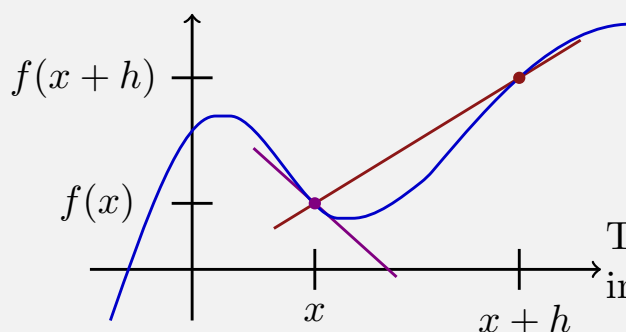
Definition of Derivative (Background Only)



The average rate of change
from x to $x + h$ is

$$= \frac{\text{total change}}{\text{length of interval}}$$

$$= \frac{f(x + h) - f(x)}{h}$$



The derivative is the
instantaneous rate of change

$$= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

$$= \frac{f(x + dx) - f(x)}{dx} = \frac{df}{dx}$$



Example

The definition is typically cumbersome to use.
For example,

$$\begin{aligned}
 \frac{d}{dx}x^2 &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2) - x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\
 &= \lim_{h \rightarrow 0} (2x + h) \\
 &= 2x
 \end{aligned}$$

Instead of always doing this, in practice people use a smaller number of key formulas.

Basic Formulas



$$\begin{aligned}
 \frac{d}{dx}f(x) &= f'(x) \\
 \frac{d}{dx}a &= 0 \text{ for any constant } a \\
 \frac{d}{dx}x^n &= nx^{n-1} \\
 \frac{d}{dx}[\ln(x)] &= \frac{1}{x} \\
 \frac{d}{dx}e^x &= e^x \\
 \frac{d}{dx}[f(x) + g(x)] &= f'(x) + g'(x) \\
 \frac{d}{dx}[cf(x)] &= c\frac{d}{dx}f(x) = cf'(x)
 \end{aligned}$$



Basic Formulas: Examples

$$\frac{d}{dx}x^3 = 3x^2$$

$$\frac{d}{dy}3y^2 = 3 \cdot 2y^1 = 6y$$

$$\frac{d}{dt}(5e^t + 3t^4) = 5e^t + 3 \cdot 4t^3$$

$$\begin{aligned}\frac{d}{ds}\frac{2}{s^3} &= \frac{d}{ds}2s^{-3} \\ &= -6s^{-4} \\ &= \frac{-6}{s^4}\end{aligned}$$

Chain Rule



Theorem (Chain Rule)

If f and u are differentiable functions,

$$\frac{d}{dx}[f(u)] = f'(u) \cdot \frac{du}{dx}$$

Examples

$$\frac{d}{dx}u^n = nu^{n-1} \frac{du}{dx}$$

$$\frac{d}{dx}e^u = e^u \cdot \frac{du}{dx}$$



Chain Rule: Examples

Suppose we want to find $\frac{d}{dt} \exp[-2t + t^2]$. Let $u = -2t + t^2$.

$$\begin{aligned} \frac{d}{dt} \exp[-2t + t^2] &= \frac{d}{dt} \exp[u] \\ &= \exp[u] \cdot \frac{du}{dt} \\ &= \exp[-2t + t^2] \cdot (-2 + 2t) \end{aligned}$$

$$\frac{d}{dx} (2x^2 + 5x + 3)^5 = 5 (2x^2 + 5x + 3)^4 \cdot (4x + 5)$$

$$\frac{d}{dt} \exp[5e^t - 5 + 3t] = (\exp[5e^t - 5 + 3t]) \cdot (5e^t + 3)$$

Chain Rule: Examples



Suppose we want to find $\frac{d}{dx} 2^x$. We know how to find $\frac{d}{dx} e^x$, so let's rewrite 2^x in terms of e^x .

$$2 = e^{\ln(2)}, 2^x = \left(e^{\ln(2)}\right)^x = e^{x \ln(2)}$$

$$\frac{d}{dx} 2^x = \frac{d}{dx} e^{x \ln(2)} = \ln(2) \cdot e^{x \ln(2)} = (\ln 2) \cdot 2^x$$

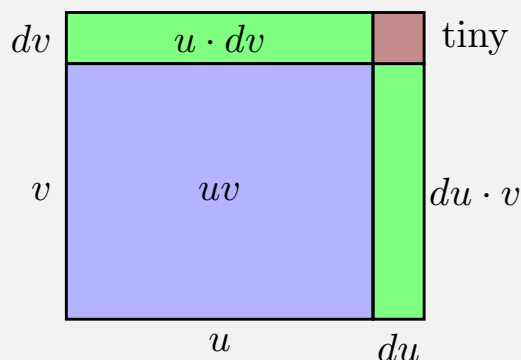
More generally, for any a ,

$$\begin{aligned} \frac{d}{dx} a^x &= (\ln a) \cdot a^x \\ \frac{d}{dx} 2^{x^3-3x} &= \frac{d}{dx} 2^u \quad u = x^3 - 3x \\ &= 2^u \cdot \ln(2) \cdot \frac{du}{dx} \\ &= 2^{(x^3-3x)} \cdot \ln(2) \cdot [3x^2 - 3] \end{aligned}$$



Product Rule

What about the derivatives of products or quotients of two functions?



$$\frac{d}{dx} [u \cdot v] = u \cdot \frac{dv}{dx} + \frac{du}{dx} \cdot v$$

$$(uv)' = u \cdot v' + u' \cdot v$$

Quotients can be done by rewriting them as products

$$\begin{aligned} \frac{d}{dx} \left[u \cdot \frac{1}{v} \right] &= u \cdot \frac{d}{dx} \frac{1}{v} + u' \cdot \frac{1}{v} \\ &= \frac{-u v'}{v^2} + \frac{u'}{v} \\ &= \frac{-uv' + u'v}{v^2} \end{aligned}$$

Product Rule Examples



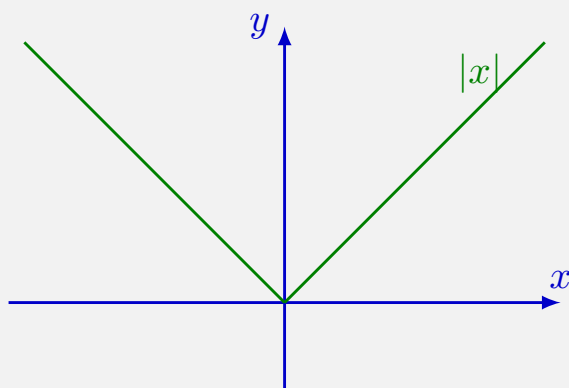
$$\frac{d}{dx} [x^2 e^{3x}] = x^2 \cdot 3e^{3x} + 2x \cdot e^{3x}$$

$$\begin{aligned} \frac{d}{dx} \frac{e^{-x}}{x^3} &= \frac{d}{dx} [e^{-x} \cdot x^{-3}] \\ &= e^{-x} \cdot \frac{-3}{x^4} + (-e^{-x}) \cdot \frac{1}{x^3} \end{aligned}$$

$$\begin{aligned} \frac{d}{dx} [(x^2 + 3x + 5)^3 \cdot e^{4x}] &= (x^2 + 3x + 5)^3 \cdot \frac{d}{dx} e^{4x} \\ &\quad + \frac{d}{dx} [(x^2 + 3x + 5)^3] \cdot e^{4x} \\ &= (x^2 + 3x + 5)^3 \cdot 4e^{4x} \\ &\quad + 3 \cdot (x^2 + 3x + 5)^2 \cdot (2x + 3) \cdot e^{4x} \end{aligned}$$



Absolute Values



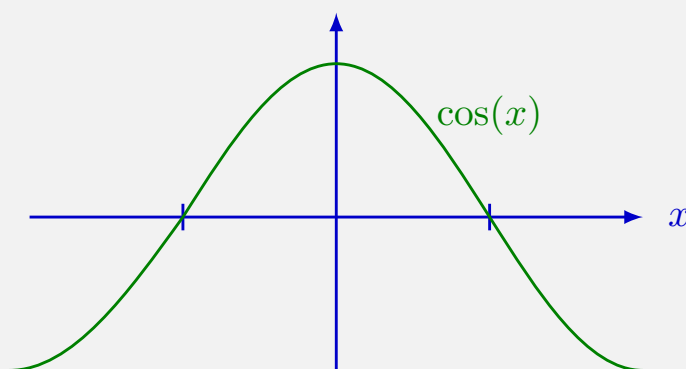
$$|x| = x \quad \text{if } x \geq 0$$

$$|x| = -x \quad \text{if } x < 0$$

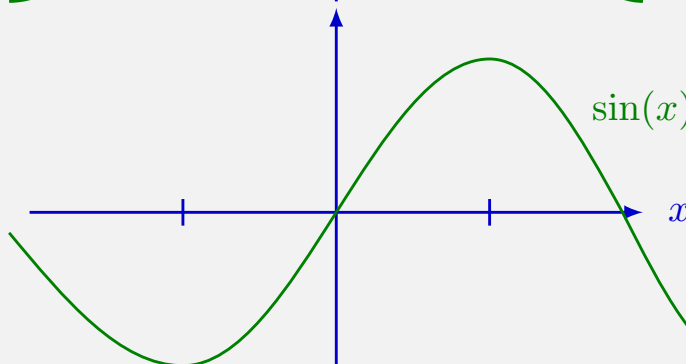
$$\frac{d}{dx}|x| = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$

Note that $\frac{d|x|}{dx}$ is undefined if $x = 0$.

Sines and Cosines



$$\frac{d}{dx} \cos(x) = -\sin(x)$$



$$\frac{d}{dx} \sin(x) = \cos(x)$$



Further Examples

$$\frac{d}{dx} \frac{2x+5}{x^2-3x+4} = (2x+5) \frac{(-1)(2x-3)}{(x^2-3x+4)^2} + \frac{2}{x^2-3x+4}$$

$$\frac{d}{dx} (x+2)e^{x^2-5x} = (x+2)(2x-5)e^{x^2-5x} + 1 \cdot e^{x^2-5x}$$

$$\frac{d}{dx} x^2 e^{-3x^2} = x^2(-6x)e^{-3x^2} + 2x \cdot e^{-3x^2}$$



Further Examples

$$\begin{aligned} \frac{d}{dx} \sin |x+2| &= \frac{d}{dx} \sin(x+2) \quad \text{if } x+2 > 0 \\ &= \cos(x+2) \quad x > -2 \end{aligned}$$

$$\begin{aligned} \frac{d}{dx} \sin |x+2| &= \frac{d}{dx} \sin(-x-2) \quad \text{if } x+2 < 0 \\ &= -\cos(-x-2) \quad x < -2 \end{aligned}$$

Key point: We get two cases based on whether or not what is inside the absolute value is positive. If $x+2 > 0$ then what is inside the absolute value is positive, so $|x+2| = x+2$, while if $x+2 < 0$ then what is inside the absolute value is negative so $|x+2| = -x-2$.



B.0 One Dimensional Calculus - Outline

B.0.1 One-Dimensional Derivatives

B.0.2 1-Dimensional integrals

What is an Integral?

The Fundamental Theorem of Calculus

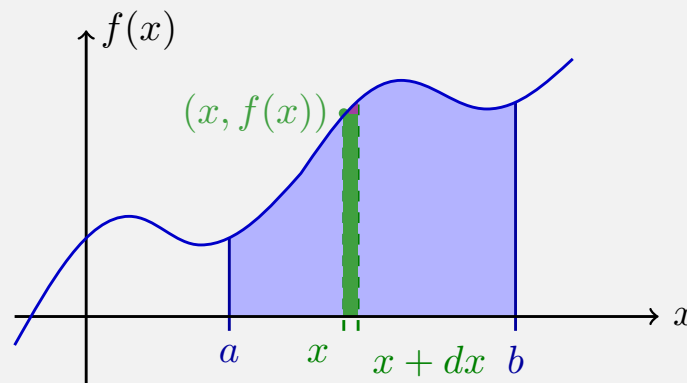
Common Formulas

Substitution

Other Formulas

B.0.3 Integration By Parts

Definition of an Integral



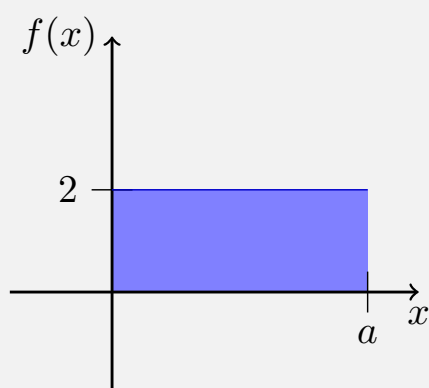
$$\int_a^b f(x) dx = \text{area under curve}$$

In some sense, $f(x) dx$ is the area of an infinitely thin rectangle and the integral says that the area under the curve is the sum of the areas of infinitely many of these thin rectangles.

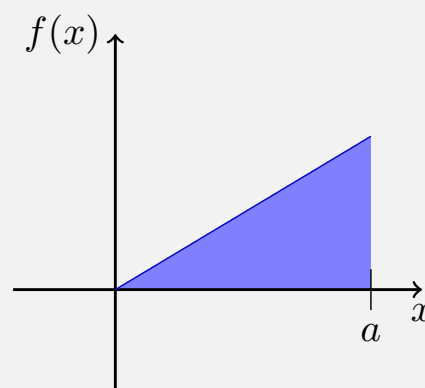


Geometric Examples

Often we can use geometry to find the integral/area under the curve.



$$\int_0^a 2dx = 2a$$



$$\int_0^a xdx = \frac{a^2}{2}$$

Geometric Examples



In that example,

1. The integral of a constant was a linear function.
2. The integral of a line was a quadratic function.

So in these two examples, when we integrated the power of a polynomial increased by 1.

When we differentiate,

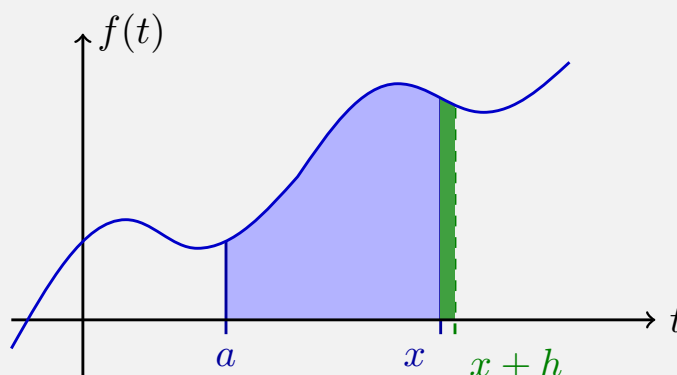
1. The derivative of a linear function is a constant.
2. The derivative of a quadratic function is linear.

More generally, when we differentiate the power of a polynomial decreases by 1. That is the opposite of when we integrate.

Hmmm.....isn't that an interesting coincidence?



The Fundamental Theorem of Calculus



$$\begin{aligned}\frac{d}{dx}F(x) &= \frac{d}{dx} \int_a^x f(t) dt = \lim_{h \rightarrow 0} \frac{\int_a^{x+h} f(t) dt - \int_a^x f(t) dt}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x) \cdot h}{h} = f(x)\end{aligned}$$

The Fundamental Theorem of Calculus



Theorem (Fundamental Theorem of Calculus)

$$\frac{d}{dx} \int_a^x f(t) dx = f(x)$$

Generalization: If v and u are functions,

$$\frac{d}{dx} \int_a^v f(t) dt = f(v) \frac{dv}{dx} - f(u) \frac{du}{dx}$$

In words, the Fundamental Theorem of Calculus says that derivatives and integrals are inverse operations.



Evaluating Integrals

The Fundamental Theorem of Calculus says that derivatives and integrals are inverse operations. To find the integral of $f(x)$, we need to find a function whose derivative is $f(x)$.

Examples

$$\int_0^5 x \, dx = \left. \frac{1}{2}x^2 \right|_0^5 = \frac{5^2}{2} - \frac{0^2}{2} = \frac{25}{2}$$

$$\begin{aligned} \int_a^b x^n \, dx &= \left. \frac{1}{n+1} \cdot x^{n+1} \right|_a^b \\ &= \frac{b^{n+1}}{n+1} - \frac{a^{n+1}}{n+1} \end{aligned}$$

Common Formulas



$$\int a \, dx = ax + C$$

$$\int x \, dx = \frac{x^2}{2} + C$$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad \text{for } n \neq -1$$

$$\int e^{bx} \, dx = \frac{1}{b}e^{bx} + C$$

$$\begin{aligned} \int a^x \, dx &= \int e^{x \ln a} \, dx \\ &= \frac{1}{\ln a} a^x + C \end{aligned}$$



Examples

$$\begin{aligned}
 \int_{-2}^5 3x^4 dx &= 3 \cdot \frac{x^5}{5} \Big|_{-2}^5 \\
 &= 3 \cdot \frac{5^5}{5} - 3 \cdot \frac{(-2)^5}{5} \\
 \int_2^\infty \frac{3}{x^4} dx &= \frac{3}{x^3} \cdot \frac{1}{-3} \Big|_2^\infty \\
 &= \frac{-1}{\infty^3} - \frac{-1}{2^3} \\
 &= 0 + \frac{1}{8} = \frac{1}{8}
 \end{aligned}$$

Substitution



When we differentiate, we often have nested functions and need to use the chain rule. For example,

$$\begin{aligned}
 \frac{d}{dx} e^{x^2} &= e^{x^2} \cdot 2x \\
 \frac{d}{dx} (x^2 + 3)^5 &= 5(x^2 + 3)^4 \cdot 2x
 \end{aligned}$$

In both those examples, the $2x$ factor comes from the chain rule. Often when we are doing integration, we will have a term that we need to somehow recognize as a chain rule factor. If we can do that, we can do a substitution to do the chain rule “backwards.”



Substitution

Suppose we want to integrate $2x \cdot e^{(x^2)}$. Let $u = x^2$. Then $\frac{du}{dx} = 2x$ so $du = 2x dx$ and we get

$$\begin{aligned} \int_{x=a}^{x=b} 2x e^{(x^2)} dx &= \int_{u=a^2}^{u=b^2} e^u du \\ &= e^u \Big|_{u=a^2}^{u=b^2} = e^{(x^2)} \Big|_{x=a}^{x=b} \\ &= e^{b^2} - e^{a^2} \end{aligned}$$

Note the limits! Either we convert back to x at the end, or we change the limits to be in terms of u .

Substitution Examples



$$\begin{aligned} \int_2^\infty x e^{-2x^2} dx & \quad u = 2x^2 \quad du = 4x dx \\ x = 2 \quad u &= 2 \cdot 2^2 = 8 \\ x = \infty \quad u &= 2 \cdot \infty^2 = \infty \\ &= \int_8^\infty e^{-u} \cdot \frac{du}{4} \\ &= \frac{1}{4} \cdot (-1) \cdot e^{-u} \Big|_8^\infty = 0 - \frac{-1}{4} e^{-8} \\ &= \frac{1}{4} e^{-8} \end{aligned}$$



Other Formulas

$$\int \frac{dx}{x} = \ln x + C$$

$$\int cf(x) dx = c \int f(x) dx$$

$$\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$$

$$\int \cos x dx = \sin x + C \quad \text{because } \frac{d}{dx} \sin x = \cos x$$

$$\int \sin x dx = -\cos x + C \quad \text{because } \frac{d}{dx} \cos x = -\sin x$$

Examples



$$\begin{aligned} u &= x + 5 & x &= 2, u = 2 + 5 = 7 \\ du &= dx & x &= 5, u = 5 + 5 = 10 \end{aligned}$$

$$\begin{aligned} \int_2^5 \frac{3x}{(x+5)^2} dx &= \int_7^{10} \frac{3(u-5)}{u^2} du \\ &= \int_7^{10} \frac{3}{u} - \frac{15}{u^2} du \\ &= \left(3 \ln u + \frac{15}{u} \right) \Big|_7^{10} \\ &= \left(3 \ln 10 + \frac{15}{10} \right) - \left(3 \ln 7 + \frac{15}{7} \right) \end{aligned}$$



Examples

$$\begin{aligned}
 \int_0^\pi (1 + \cos t) dt &= t + \sin t \Big|_0^\pi \\
 &= (\pi + 0) - (0 + 0) = \pi \\
 \int_{-2}^5 |x| dx &= \int_{-2}^0 -x dx + \int_0^5 x dx \\
 &= \left. \frac{-x^2}{2} \right|_{-2}^0 + \left. \frac{x^2}{2} \right|_0^5 = \frac{(-2)^2}{2} + \frac{25}{2} = \frac{29}{2} \\
 \frac{d}{dx} \int_{-2x}^{x^3} e^{5t-5} dt &= e^{5x^3-5} \cdot 3x^2 - e^{5(-2x)-5} (-2)
 \end{aligned}$$

B.0 One Dimensional Calculus - Outline



B.0.1 One-Dimensional Derivatives

B.0.2 1-Dimensional integrals

B.0.3 Integration By Parts

Integration By Parts

Tabular integration

The Gamma trick



Integration By Parts

Integration by parts is doing the product rule backwards.

$$\begin{aligned}\frac{d}{dx} uv &= u \cdot \frac{dv}{dx} + \frac{du}{dx} \cdot v \\ \int d(uv) &= uv = \int u dv + \int v du \\ \int u \cdot dv &= uv - \int v du\end{aligned}$$

Integration By Parts



Example

$$\begin{aligned}\int x e^x dx &= \int u dv \\ u &= x & dv &= e^x dx \\ du &= dx & v &= e^x \\ \text{so} \\ \int x e^x dx &= uv - \int v du \\ &= x e^x - \int e^x dx \\ &= x e^x - e^x + C\end{aligned}$$



Logarithms

You can use integration by parts to handle functions whose derivatives are easier to find than their integrals.

$$\begin{aligned}\int \ln x \, dx &= \int u \, dv \\ u &= \ln x & dv &= dx \\ du &= \frac{dx}{x} & v &= x\end{aligned}$$

so

$$\begin{aligned}\int \ln x \, dx &= uv - \int v \, du \\ &= (\ln x)x - \int x \frac{dx}{x} \\ &= x \ln x - x + C\end{aligned}$$

Logarithms



$$\begin{aligned}\int x \ln x \, dx &= \int u \, dv \\ u &= \ln x & dv &= x \, dx \\ du &= \frac{dx}{x} & v &= \frac{x^2}{2}\end{aligned}$$

so

$$\begin{aligned}\int x \ln x \, dx &= uv - \int v \, du \\ &= \frac{x^2}{2} \ln x - \int \frac{x^2}{2x} \, dx \\ &= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C\end{aligned}$$



Which Part to Use

How do we choose u and dv ? We need dv to be something that is easy to integrate and we need u to be something that is easy to differentiate.

Ideally we also want u to become simpler when you differentiate.

Easy to Differentiate		Easy to Integrate
Logs	Polynomials	Exponentials

Integration by parts



Iterated Parts

$$\begin{aligned}
 \int x^2 e^{2x} dx & \quad \text{Let } u = x^2 \quad dv = e^{2x} dx \\
 & \quad du = 2x dx \quad v = \frac{1}{2} e^{2x} \\
 & = x^2 \cdot \frac{1}{2} e^{2x} - \int \frac{1}{2} e^{2x} 2x dx \\
 & = \frac{1}{2} x^2 e^{2x} - \int x e^{2x} dx
 \end{aligned}$$

And to find this, we have to repeat integration by parts.



Tabular integration

Tabular integration is a way to organize our work when doing repeated integration by parts. To integrate $x^2 e^{2x}$,

<u>Derivative column</u>		<u>Integral column</u>
x^2	$+$	e^{2x}
$2x$	$-$	$\frac{1}{2} e^{2x}$
2	$+$	$\frac{1}{4} e^{2x}$
0	$-$	$\frac{1}{8} e^{2x}$

$$\int x^2 e^{2x} dx = x^2 \cdot \frac{1}{2} e^{2x} - 2x \cdot \frac{1}{4} e^{2x} + 2 \cdot \frac{1}{8} e^{2x} - 0$$

The Gamma trick



If we have a definite integral from 0 to infinity, we often can skip using integration by parts.

If $b > 0$ and a is an integer, then

$$\int_0^{\infty} x^a \cdot e^{-bx} dx = \frac{a!}{b^{a+1}}$$

Example

$$\int_0^{\infty} x^2 e^{-2x} dx = \frac{2!}{2^{2+1}} = \frac{1}{4}$$

$$\begin{aligned} -b &= -2 & b &= 2 \\ a &= 2 \end{aligned}$$



The Gamma trick

An example both ways: $\int_0^\infty 4x^2 e^{-x/3} dx$

Derivative column

Integral column

$4x^2$	$+$	$e^{-x/3}$
$8x$	$-$	$-3e^{-x/3}$
8	$+$	$9e^{-x/3}$
0		$-27e^{-x/3}$

So $\int 4x^2 e^{-x/3} dx$ is

$$\begin{aligned}
 & (4x^2)(-3e^{-x/3}) - (8x)(9e^{-x/3}) + 8 \cdot (-27e^{-x/3}) \\
 &= (-12x^2 - 72x - 8 \cdot 27)e^{-x/3}
 \end{aligned}$$

The Gamma trick



$$\begin{aligned}
 \int_0^\infty 4x^2 e^{-x/3} &= (-12x^2 - 72x - 8 \cdot 27)e^{-x/3} \Big|_0^\infty \\
 &= 8 \cdot 27
 \end{aligned}$$

or we can let $b = 1/3$ and $a = 2$ in our formula to get

$$\begin{aligned}
 \int_0^\infty x^a e^{-bx} &= \frac{a!}{b^{a+1}} \\
 \int_0^\infty x^2 e^{-x/3} &= \frac{2!}{(1/3)^{2+1}} = 2 \cdot 27 \\
 \int_0^\infty 4x^2 e^{-x/3} &= 4 \cdot \frac{2!}{\left(\frac{1}{3}\right)^{2+1}} = 8 \cdot 27
 \end{aligned}$$



Other Definite Integrals

$$\int_1^{\infty} 4x^2 e^{-x/3} \quad \begin{array}{l} \text{We want } u = 0 \text{ when } x = 1 \\ u = x - 1, \quad x = u + 1 \end{array}$$

$$= \int_0^{\infty} 4(u+1)^2 e^{(-u-1)/3} du$$

$$= 4e^{-1/3} \int_0^{\infty} (u^2 + 2u + 1) e^{-u/3} du$$

and now we plug into $\int_0^{\infty} x^a e^{-bx} = \frac{a!}{b^{a+1}}$

$$= 4e^{-1/3} \left[\frac{2!}{\left(\frac{1}{3}\right)^3} + 2 \cdot \frac{1}{\left(\frac{1}{3}\right)^2} + \frac{1}{\frac{1}{3}} \right]$$

The Gamma trick



Idea of Proof:

Let $u = x^a$ and $dv = e^{-bx} dx$

$$\begin{aligned} \int_0^{\infty} x^a e^{-bx} dx &= x^a \cdot \frac{-1}{b} e^{-bx} \Big|_0^{\infty} - \int_0^{\infty} ax^{a-1} \left(\frac{-1}{b} \right) e^{-bx} dx \\ &= 0 - 0 + \frac{a}{b} \int_0^{\infty} x^{a-1} e^{-bx} dx \\ &= \frac{a}{b} \cdot \frac{(a-1)!}{b^{a-1+1}} \\ &= \frac{a!}{b^{a+1}} \end{aligned}$$

It also is related to $E[X^a]$ when X is an exponential random variable as well as the density of a Gamma random variable.



B.0 One Dimensional Calculus - Outline

B.0.1 One-Dimensional Derivatives

Definition of Derivative

Basic Formulas

Chain Rule

Product Rule

Absolute Values

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Further Examples

B.0.2 1-Dimensional integrals

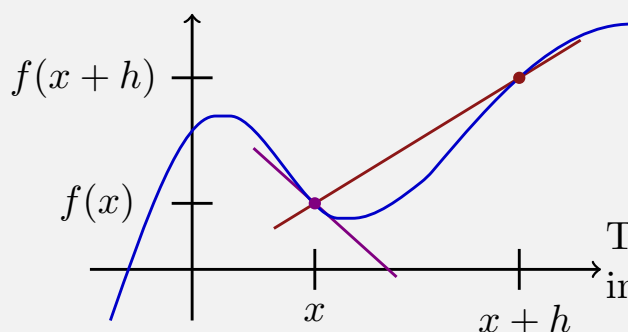
B.0.3 Integration By Parts

Definition of Derivative (Background Only)



The average rate of change
from x to $x + h$ is

$$\begin{aligned}
 &= \frac{\text{total change}}{\text{length of interval}} \\
 &= \frac{f(x + h) - f(x)}{h}
 \end{aligned}$$



The derivative is the
instantaneous rate of change

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} \\
 &= \frac{f(x + dx) - f(x)}{dx} = \frac{df}{dx}
 \end{aligned}$$



Example

The definition is typically cumbersome to use.
For example,

$$\begin{aligned}
 \frac{d}{dx}x^2 &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2) - x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\
 &= \lim_{h \rightarrow 0} (2x + h) \\
 &= 2x
 \end{aligned}$$

Instead of always doing this, in practice people use a smaller number of key formulas.

Basic Formulas



$$\begin{aligned}
 \frac{d}{dx}f(x) &= f'(x) \\
 \frac{d}{dx}a &= 0 \text{ for any constant } a \\
 \frac{d}{dx}x^n &= nx^{n-1} \\
 \frac{d}{dx}[\ln(x)] &= \frac{1}{x} \\
 \frac{d}{dx}e^x &= e^x \\
 \frac{d}{dx}[f(x) + g(x)] &= f'(x) + g'(x) \\
 \frac{d}{dx}[cf(x)] &= c\frac{d}{dx}f(x) = cf'(x)
 \end{aligned}$$



Basic Formulas: Examples

$$\frac{d}{dx}x^3 = 3x^2$$

$$\frac{d}{dy}3y^2 = 3 \cdot 2y^1 = 6y$$

$$\frac{d}{dt}(5e^t + 3t^4) = 5e^t + 3 \cdot 4t^3$$

$$\begin{aligned}\frac{d}{ds}\frac{2}{s^3} &= \frac{d}{ds}2s^{-3} \\ &= -6s^{-4} \\ &= \frac{-6}{s^4}\end{aligned}$$

Chain Rule



Theorem (Chain Rule)

If f and u are differentiable functions,

$$\frac{d}{dx}[f(u)] = f'(u) \cdot \frac{du}{dx}$$

Examples

$$\frac{d}{dx}u^n = nu^{n-1} \frac{du}{dx}$$

$$\frac{d}{dx}e^u = e^u \cdot \frac{du}{dx}$$



Chain Rule: Examples

Suppose we want to find $\frac{d}{dt} \exp[-2t + t^2]$. Let $u = -2t + t^2$.

$$\begin{aligned} \frac{d}{dt} \exp[-2t + t^2] &= \frac{d}{dt} \exp[u] \\ &= \exp[u] \cdot \frac{du}{dt} \\ &= \exp[-2t + t^2] \cdot (-2 + 2t) \end{aligned}$$

$$\frac{d}{dx} (2x^2 + 5x + 3)^5 = 5 (2x^2 + 5x + 3)^4 \cdot (4x + 5)$$

$$\frac{d}{dt} \exp[5e^t - 5 + 3t] = (\exp[5e^t - 5 + 3t]) \cdot (5e^t + 3)$$

Chain Rule: Examples



Suppose we want to find $\frac{d}{dx} 2^x$. We know how to find $\frac{d}{dx} e^x$, so let's rewrite 2^x in terms of e^x .

$$2 = e^{\ln(2)}, 2^x = \left(e^{\ln(2)}\right)^x = e^{x \ln(2)}$$

$$\frac{d}{dx} 2^x = \frac{d}{dx} e^{x \ln(2)} = \ln(2) \cdot e^{x \ln(2)} = (\ln 2) \cdot 2^x$$

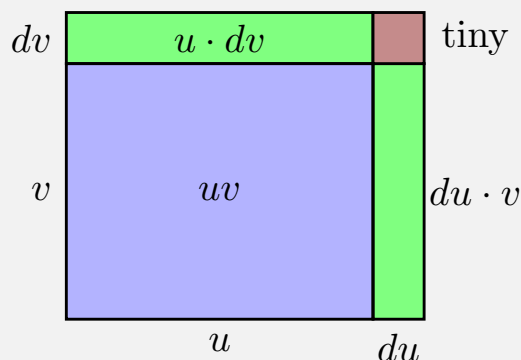
More generally, for any a ,

$$\begin{aligned} \frac{d}{dx} a^x &= (\ln a) \cdot a^x \\ \frac{d}{dx} 2^{x^3-3x} &= \frac{d}{dx} 2^u \quad u = x^3 - 3x \\ &= 2^u \cdot \ln(2) \cdot \frac{du}{dx} \\ &= 2^{(x^3-3x)} \cdot \ln(2) \cdot [3x^2 - 3] \end{aligned}$$



Product Rule

What about the derivatives of products or quotients of two functions?



$$\frac{d}{dx} [u \cdot v] = u \cdot \frac{dv}{dx} + \frac{du}{dx} \cdot v$$

$$(uv)' = u \cdot v' + u' \cdot v$$

Quotients can be done by rewriting them as products

$$\begin{aligned} \frac{d}{dx} \left[u \cdot \frac{1}{v} \right] &= u \cdot \frac{d}{dx} \frac{1}{v} + u' \cdot \frac{1}{v} \\ &= \frac{-u v'}{v^2} + \frac{u'}{v} \\ &= \frac{-uv' + u'v}{v^2} \end{aligned}$$

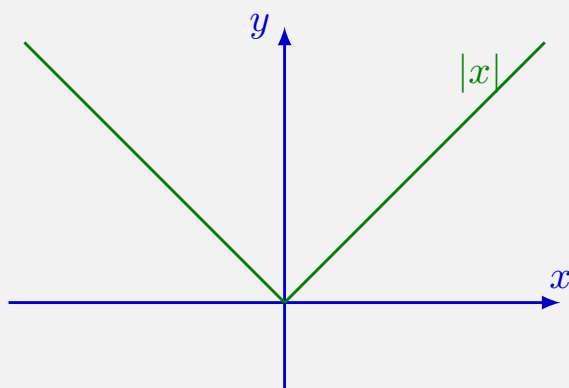
Product Rule Examples



$$\begin{aligned} \frac{d}{dx} [x^2 e^{3x}] &= x^2 \cdot 3e^{3x} + 2x \cdot e^{3x} \\ \frac{d}{dx} \frac{e^{-x}}{x^3} &= \frac{d}{dx} [e^{-x} \cdot x^{-3}] \\ &= e^{-x} \cdot \frac{-3}{x^4} + (-e^{-x}) \cdot \frac{1}{x^3} \\ \frac{d}{dx} [(x^2 + 3x + 5)^3 \cdot e^{4x}] &= (x^2 + 3x + 5)^3 \cdot \frac{d}{dx} e^{4x} \\ &\quad + \frac{d}{dx} [(x^2 + 3x + 5)^3] \cdot e^{4x} \\ &= (x^2 + 3x + 5)^3 \cdot 4e^{4x} \\ &\quad + 3 \cdot (x^2 + 3x + 5)^2 \cdot (2x + 3) \cdot e^{4x} \end{aligned}$$



Absolute Values



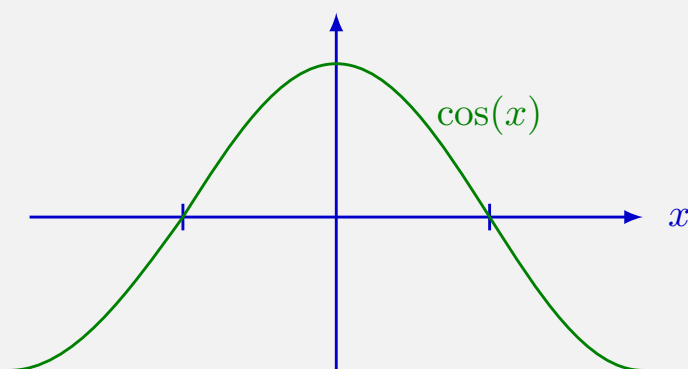
$$|x| = x \quad \text{if } x \geq 0$$

$$|x| = -x \quad \text{if } x < 0$$

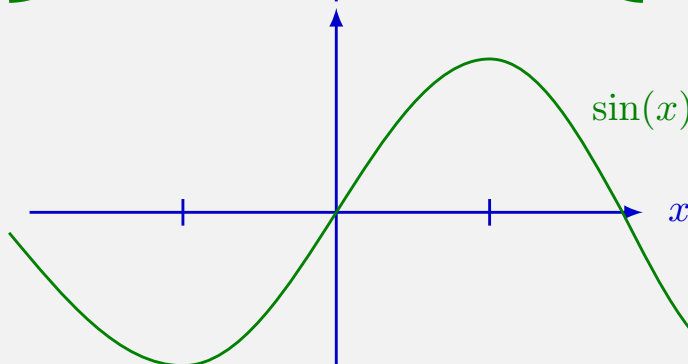
$$\frac{d}{dx}|x| = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$

Note that $\frac{d|x|}{dx}$ is undefined if $x = 0$.

Sines and Cosines



$$\frac{d}{dx} \cos(x) = -\sin(x)$$



$$\frac{d}{dx} \sin(x) = \cos(x)$$



Further Examples

$$\frac{d}{dx} \frac{2x+5}{x^2-3x+4} = (2x+5) \frac{(-1)(2x-3)}{(x^2-3x+4)^2} + \frac{2}{x^2-3x+4}$$

$$\frac{d}{dx} (x+2)e^{x^2-5x} = (x+2)(2x-5)e^{x^2-5x} + 1 \cdot e^{x^2-5x}$$

$$\frac{d}{dx} x^2 e^{-3x^2} = x^2(-6x)e^{-3x^2} + 2x \cdot e^{-3x^2}$$



Further Examples

$$\begin{aligned} \frac{d}{dx} \sin |x+2| &= \frac{d}{dx} \sin(x+2) \quad \text{if } x+2 > 0 \\ &= \cos(x+2) \quad x > -2 \end{aligned}$$

$$\begin{aligned} \frac{d}{dx} \sin |x+2| &= \frac{d}{dx} \sin(-x-2) \quad \text{if } x+2 < 0 \\ &= -\cos(-x-2) \quad x < -2 \end{aligned}$$

Key point: We get two cases based on whether or not what is inside the absolute value is positive. If $x+2 > 0$ then what is inside the absolute value is positive, so $|x+2| = x+2$, while if $x+2 < 0$ then what is inside the absolute value is negative so $|x+2| = -x-2$.



B.0 One Dimensional Calculus - Outline

B.0.1 One-Dimensional Derivatives

B.0.2 1-Dimensional integrals

What is an Integral?

The Fundamental Theorem of Calculus

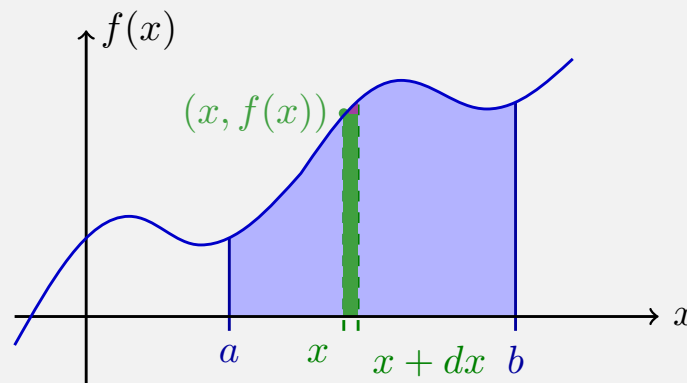
Common Formulas

Substitution

Other Formulas

B.0.3 Integration By Parts

Definition of an Integral



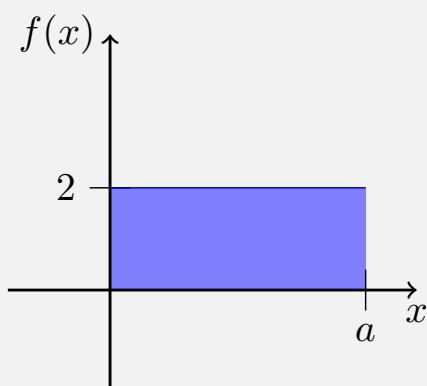
$$\int_a^b f(x) dx = \text{area under curve}$$

In some sense, $f(x) dx$ is the area of an infinitely thin rectangle and the integral says that the area under the curve is the sum of the areas of infinitely many of these thin rectangles.

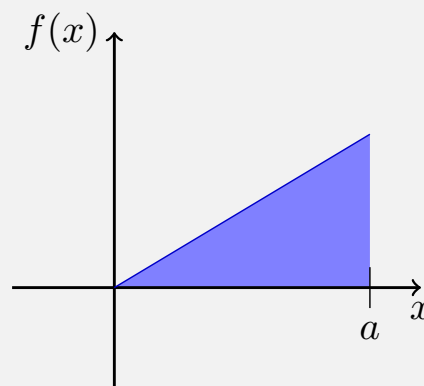


Geometric Examples

Often we can use geometry to find the integral/area under the curve.



$$\int_0^a 2dx = 2a$$



$$\int_0^a xdx = \frac{a^2}{2}$$

Geometric Examples



In that example,

1. The integral of a constant was a linear function.
2. The integral of a line was a quadratic function.

So in these two examples, when we integrated the power of a polynomial increased by 1.

When we differentiate,

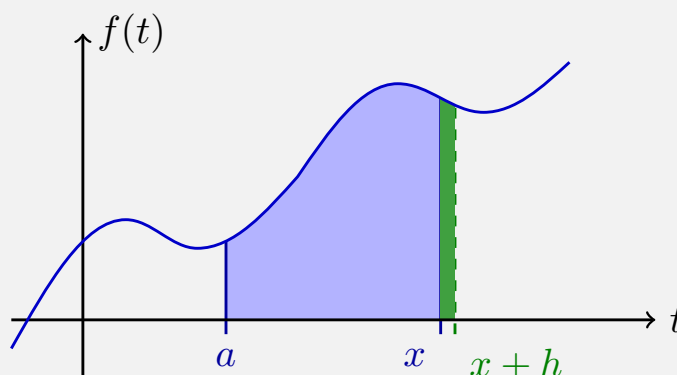
1. The derivative of a linear function is a constant.
2. The derivative of a quadratic function is linear.

More generally, when we differentiate the power of a polynomial decreases by 1. That is the opposite of when we integrate.

Hmmm.....isn't that an interesting coincidence?



The Fundamental Theorem of Calculus



$$\begin{aligned}\frac{d}{dx}F(x) &= \frac{d}{dx} \int_a^x f(t) dt = \lim_{h \rightarrow 0} \frac{\int_a^{x+h} f(t) dt - \int_a^x f(t) dt}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x) \cdot h}{h} = f(x)\end{aligned}$$

The Fundamental Theorem of Calculus



Theorem (Fundamental Theorem of Calculus)

$$\frac{d}{dx} \int_a^x f(t) dx = f(x)$$

Generalization: If v and u are functions,

$$\frac{d}{dx} \int_a^v f(t) dt = f(v) \frac{dv}{dx} - f(u) \frac{du}{dx}$$

In words, the Fundamental Theorem of Calculus says that derivatives and integrals are inverse operations.



Evaluating Integrals

The Fundamental Theorem of Calculus says that derivatives and integrals are inverse operations. To find the integral of $f(x)$, we need to find a function whose derivative is $f(x)$.

Examples

$$\int_0^5 x \, dx = \left. \frac{1}{2}x^2 \right|_0^5 = \frac{5^2}{2} - \frac{0^2}{2} = \frac{25}{2}$$

$$\begin{aligned} \int_a^b x^n \, dx &= \left. \frac{1}{n+1} \cdot x^{n+1} \right|_a^b \\ &= \frac{b^{n+1}}{n+1} - \frac{a^{n+1}}{n+1} \end{aligned}$$

Common Formulas



$$\int a \, dx = ax + C$$

$$\int x \, dx = \frac{x^2}{2} + C$$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad \text{for } n \neq -1$$

$$\int e^{bx} \, dx = \frac{1}{b}e^{bx} + C$$

$$\begin{aligned} \int a^x \, dx &= \int e^{x \ln a} \, dx \\ &= \frac{1}{\ln a} a^x + C \end{aligned}$$



Examples

$$\begin{aligned}
 \int_{-2}^5 3x^4 dx &= 3 \cdot \frac{x^5}{5} \Big|_{-2}^5 \\
 &= 3 \cdot \frac{5^5}{5} - 3 \cdot \frac{(-2)^5}{5} \\
 \int_2^\infty \frac{3}{x^4} dx &= \frac{3}{x^3} \cdot \frac{1}{-3} \Big|_2^\infty \\
 &= \frac{-1}{\infty^3} - \frac{-1}{2^3} \\
 &= 0 + \frac{1}{8} = \frac{1}{8}
 \end{aligned}$$

Substitution



When we differentiate, we often have nested functions and need to use the chain rule. For example,

$$\begin{aligned}
 \frac{d}{dx} e^{x^2} &= e^{x^2} \cdot 2x \\
 \frac{d}{dx} (x^2 + 3)^5 &= 5(x^2 + 3)^4 \cdot 2x
 \end{aligned}$$

In both those examples, the $2x$ factor comes from the chain rule. Often when we are doing integration, we will have a term that we need to somehow recognize as a chain rule factor. If we can do that, we can do a substitution to do the chain rule “backwards.”



Substitution

Suppose we want to integrate $2x \cdot e^{(x^2)}$. Let $u = x^2$. Then $\frac{du}{dx} = 2x$ so $du = 2x dx$ and we get

$$\begin{aligned} \int_{x=a}^{x=b} 2x e^{(x^2)} dx &= \int_{u=a^2}^{u=b^2} e^u du \\ &= e^u \Big|_{u=a^2}^{u=b^2} = e^{(x^2)} \Big|_{x=a}^{x=b} \\ &= e^{b^2} - e^{a^2} \end{aligned}$$

Note the limits! Either we convert back to x at the end, or we change the limits to be in terms of u .

Substitution Examples



$$\begin{aligned} \int_2^\infty x e^{-2x^2} dx & \quad u = 2x^2 \quad du = 4x dx \\ x = 2 \quad u &= 2 \cdot 2^2 = 8 \\ x = \infty \quad u &= 2 \cdot \infty^2 = \infty \\ &= \int_8^\infty e^{-u} \cdot \frac{du}{4} \\ &= \frac{1}{4} \cdot (-1) \cdot e^{-u} \Big|_8^\infty = 0 - \frac{-1}{4} e^{-8} \\ &= \frac{1}{4} e^{-8} \end{aligned}$$



Other Formulas

$$\int \frac{dx}{x} = \ln x + C$$

$$\int cf(x) dx = c \int f(x) dx$$

$$\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$$

$$\int \cos x dx = \sin x + C \quad \text{because } \frac{d}{dx} \sin x = \cos x$$

$$\int \sin x dx = -\cos x + C \quad \text{because } \frac{d}{dx} \cos x = -\sin x$$

Examples



$$\begin{aligned} u &= x + 5 & x = 2, u &= 2 + 5 = 7 \\ du &= dx & x = 5, u &= 5 + 5 = 10 \end{aligned}$$

$$\begin{aligned} \int_2^5 \frac{3x}{(x+5)^2} dx &= \int_7^{10} \frac{3(u-5)}{u^2} du \\ &= \int_7^{10} \frac{3}{u} - \frac{15}{u^2} du \\ &= \left(3 \ln u + \frac{15}{u} \right) \Big|_7^{10} \\ &= \left(3 \ln 10 + \frac{15}{10} \right) - \left(3 \ln 7 + \frac{15}{7} \right) \end{aligned}$$



Examples

$$\begin{aligned}
 \int_0^\pi (1 + \cos t) dt &= t + \sin t \Big|_0^\pi \\
 &= (\pi + 0) - (0 + 0) = \pi \\
 \int_{-2}^5 |x| dx &= \int_{-2}^0 -x dx + \int_0^5 x dx \\
 &= \left. \frac{-x^2}{2} \right|_{-2}^0 + \left. \frac{x^2}{2} \right|_0^5 = \frac{(-2)^2}{2} + \frac{25}{2} = \frac{29}{2} \\
 \frac{d}{dx} \int_{-2x}^{x^3} e^{5t-5} dt &= e^{5x^3-5} \cdot 3x^2 - e^{5(-2x)-5}(-2)
 \end{aligned}$$

B.0 One Dimensional Calculus - Outline



B.0.1 One-Dimensional Derivatives

B.0.2 1-Dimensional integrals

B.0.3 Integration By Parts

Integration By Parts

Tabular integration

The Gamma trick



Integration By Parts

Integration by parts is doing the product rule backwards.

$$\begin{aligned}\frac{d}{dx} uv &= u \cdot \frac{dv}{dx} + \frac{du}{dx} \cdot v \\ \int d(uv) &= uv = \int u dv + \int v du \\ \int u \cdot dv &= uv - \int v du\end{aligned}$$

Integration By Parts



Example

$$\begin{aligned}\int x e^x dx &= \int u dv \\ u &= x & dv &= e^x dx \\ du &= dx & v &= e^x \\ \text{so} \\ \int x e^x dx &= uv - \int v du \\ &= x e^x - \int e^x dx \\ &= x e^x - e^x + C\end{aligned}$$



Logarithms

You can use integration by parts to handle functions whose derivatives are easier to find than their integrals.

$$\int \ln x \, dx = \int u \, dv$$

$$u = \ln x \quad dv = dx$$

$$du = \frac{dx}{x} \quad v = x$$

so

$$\begin{aligned} \int \ln x \, dx &= uv - \int v \, du \\ &= (\ln x)x - \int x \frac{dx}{x} \\ &= x \ln x - x + C \end{aligned}$$

Logarithms



$$\int x \ln x \, dx = \int u \, dv$$

$$u = \ln x \quad dv = x \, dx$$

$$du = \frac{dx}{x} \quad v = \frac{x^2}{2}$$

so

$$\begin{aligned} \int x \ln x \, dx &= uv - \int v \, du \\ &= \frac{x^2}{2} \ln x - \int \frac{x^2}{2x} \, dx \\ &= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C \end{aligned}$$



Which Part to Use

How do we choose u and dv ? We need dv to be something that is easy to integrate and we need u to be something that is easy to differentiate.

Ideally we also want u to become simpler when you differentiate.

Easy to Differentiate		Easy to Integrate
Logs	Polynomials	Exponentials

Integration by parts



Iterated Parts

$$\begin{aligned}
 \int x^2 e^{2x} dx & \quad \text{Let } u = x^2 \quad dv = e^{2x} dx \\
 & \quad du = 2x dx \quad v = \frac{1}{2} e^{2x} \\
 & = x^2 \cdot \frac{1}{2} e^{2x} - \int \frac{1}{2} e^{2x} 2x dx \\
 & = \frac{1}{2} x^2 e^{2x} - \int x e^{2x} dx
 \end{aligned}$$

And to find this, we have to repeat integration by parts.



Tabular integration

Tabular integration is a way to organize our work when doing repeated integration by parts. To integrate $x^2 e^{2x}$,

<u>Derivative column</u>		<u>Integral column</u>
x^2	$+$	e^{2x}
$2x$	$-$	$\frac{1}{2} e^{2x}$
2	$+$	$\frac{1}{4} e^{2x}$
0	$-$	$\frac{1}{8} e^{2x}$

$$\int x^2 e^{2x} dx = x^2 \cdot \frac{1}{2} e^{2x} - 2x \cdot \frac{1}{4} e^{2x} + 2 \cdot \frac{1}{8} e^{2x} - 0$$

The Gamma trick



If we have a definite integral from 0 to infinity, we often can skip using integration by parts.

If $b > 0$ and a is an integer, then

$$\int_0^{\infty} x^a \cdot e^{-bx} dx = \frac{a!}{b^{a+1}}$$

Example

$$\int_0^{\infty} x^2 e^{-2x} dx = \frac{2!}{2^{2+1}} = \frac{1}{4}$$

$$\begin{aligned} -b &= -2 & b &= 2 \\ a &= 2 \end{aligned}$$



The Gamma trick

An example both ways: $\int_0^\infty 4x^2 e^{-x/3} dx$

Derivative column

Integral column

$4x^2$	$+$	$e^{-x/3}$
$8x$	$-$	$-3e^{-x/3}$
8	$+$	$9e^{-x/3}$
0		$-27e^{-x/3}$

So $\int 4x^2 e^{-x/3} dx$ is

$$\begin{aligned}
 & (4x^2)(-3e^{-x/3}) - (8x)(9e^{-x/3}) + 8 \cdot (-27e^{-x/3}) \\
 &= (-12x^2 - 72x - 8 \cdot 27)e^{-x/3}
 \end{aligned}$$

The Gamma trick



$$\begin{aligned}
 \int_0^\infty 4x^2 e^{-x/3} &= (-12x^2 - 72x - 8 \cdot 27)e^{-x/3} \Big|_0^\infty \\
 &= 8 \cdot 27
 \end{aligned}$$

or we can let $b = 1/3$ and $a = 2$ in our formula to get

$$\begin{aligned}
 \int_0^\infty x^a e^{-bx} &= \frac{a!}{b^{a+1}} \\
 \int_0^\infty x^2 e^{-x/3} &= \frac{2!}{(1/3)^{2+1}} = 2 \cdot 27 \\
 \int_0^\infty 4x^2 e^{-x/3} &= 4 \cdot \frac{2!}{\left(\frac{1}{3}\right)^{2+1}} = 8 \cdot 27
 \end{aligned}$$



Other Definite Integrals

$$\int_1^{\infty} 4x^2 e^{-x/3} \quad \begin{array}{l} \text{We want } u = 0 \text{ when } x = 1 \\ u = x - 1, \quad x = u + 1 \end{array}$$

$$= \int_0^{\infty} 4(u+1)^2 e^{(-u-1)/3} du$$

$$= 4e^{-1/3} \int_0^{\infty} (u^2 + 2u + 1) e^{-u/3} du$$

and now we plug into $\int_0^{\infty} x^a e^{-bx} = \frac{a!}{b^{a+1}}$

$$= 4e^{-1/3} \left[\frac{2!}{\left(\frac{1}{3}\right)^3} + 2 \cdot \frac{1}{\left(\frac{1}{3}\right)^2} + \frac{1}{\frac{1}{3}} \right]$$

The Gamma trick



Idea of Proof:

Let $u = x^a$ and $dv = e^{-bx} dx$

$$\begin{aligned} \int_0^{\infty} x^a e^{-bx} dx &= x^a \cdot \frac{-1}{b} e^{-bx} \Big|_0^{\infty} - \int_0^{\infty} ax^{a-1} \left(\frac{-1}{b} \right) e^{-bx} dx \\ &= 0 - 0 + \frac{a}{b} \int_0^{\infty} x^{a-1} e^{-bx} dx \\ &= \frac{a}{b} \cdot \frac{(a-1)!}{b^{a-1+1}} \\ &= \frac{a!}{b^{a+1}} \end{aligned}$$

It also is related to $E[X^a]$ when X is an exponential random variable as well as the density of a Gamma random variable.



B.0 One Dimensional Calculus - Outline

B.0.1 One-Dimensional Derivatives

Definition of Derivative

Basic Formulas

Chain Rule

Product Rule

Absolute Values

Sines and Cosines

Further Examples

B.0.2 1-Dimensional integrals

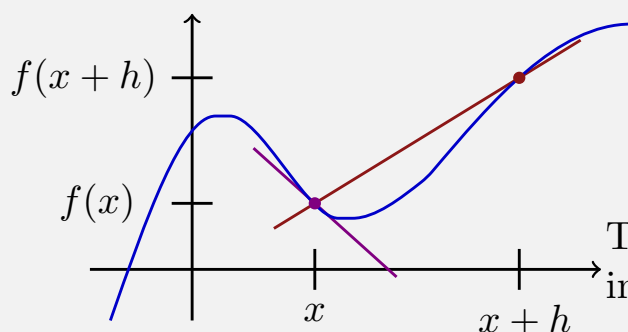
B.0.3 Integration By Parts

Definition of Derivative (Background Only)



The average rate of change
from x to $x + h$ is

$$\begin{aligned}
 &= \frac{\text{total change}}{\text{length of interval}} \\
 &= \frac{f(x + h) - f(x)}{h}
 \end{aligned}$$



The derivative is the
instantaneous rate of change

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} \\
 &= \frac{f(x + dx) - f(x)}{dx} = \frac{df}{dx}
 \end{aligned}$$



Example

The definition is typically cumbersome to use.
For example,

$$\begin{aligned}
 \frac{d}{dx}x^2 &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2) - x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\
 &= \lim_{h \rightarrow 0} (2x + h) \\
 &= 2x
 \end{aligned}$$

Instead of always doing this, in practice people use a smaller number of key formulas.

Basic Formulas



$$\begin{aligned}
 \frac{d}{dx}f(x) &= f'(x) \\
 \frac{d}{dx}a &= 0 \text{ for any constant } a \\
 \frac{d}{dx}x^n &= nx^{n-1} \\
 \frac{d}{dx}[\ln(x)] &= \frac{1}{x} \\
 \frac{d}{dx}e^x &= e^x \\
 \frac{d}{dx}[f(x) + g(x)] &= f'(x) + g'(x) \\
 \frac{d}{dx}[cf(x)] &= c \frac{d}{dx}f(x) = cf'(x)
 \end{aligned}$$



Basic Formulas: Examples

$$\frac{d}{dx}x^3 = 3x^2$$

$$\frac{d}{dy}3y^2 = 3 \cdot 2y^1 = 6y$$

$$\frac{d}{dt}(5e^t + 3t^4) = 5e^t + 3 \cdot 4t^3$$

$$\begin{aligned}\frac{d}{ds}\frac{2}{s^3} &= \frac{d}{ds}2s^{-3} \\ &= -6s^{-4} \\ &= \frac{-6}{s^4}\end{aligned}$$

Chain Rule



Theorem (Chain Rule)

If f and u are differentiable functions,

$$\frac{d}{dx}[f(u)] = f'(u) \cdot \frac{du}{dx}$$

Examples

$$\frac{d}{dx}u^n = nu^{n-1} \frac{du}{dx}$$

$$\frac{d}{dx}e^u = e^u \cdot \frac{du}{dx}$$



Chain Rule: Examples

Suppose we want to find $\frac{d}{dt} \exp[-2t + t^2]$. Let $u = -2t + t^2$.

$$\begin{aligned} \frac{d}{dt} \exp[-2t + t^2] &= \frac{d}{dt} \exp[u] \\ &= \exp[u] \cdot \frac{du}{dt} \\ &= \exp[-2t + t^2] \cdot (-2 + 2t) \end{aligned}$$

$$\frac{d}{dx} (2x^2 + 5x + 3)^5 = 5 (2x^2 + 5x + 3)^4 \cdot (4x + 5)$$

$$\frac{d}{dt} \exp[5e^t - 5 + 3t] = (\exp[5e^t - 5 + 3t]) \cdot (5e^t + 3)$$

Chain Rule: Examples



Suppose we want to find $\frac{d}{dx} 2^x$. We know how to find $\frac{d}{dx} e^x$, so let's rewrite 2^x in terms of e^x .

$$2 = e^{\ln(2)}, 2^x = \left(e^{\ln(2)}\right)^x = e^{x \ln(2)}$$

$$\frac{d}{dx} 2^x = \frac{d}{dx} e^{x \ln(2)} = \ln(2) \cdot e^{x \ln(2)} = (\ln 2) \cdot 2^x$$

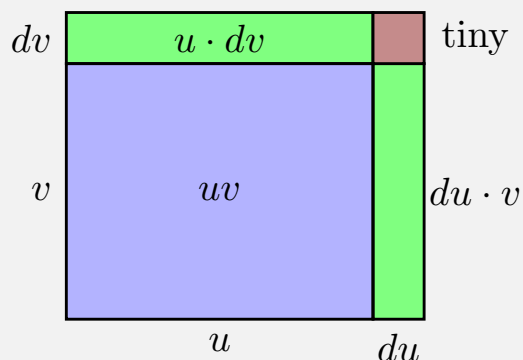
More generally, for any a ,

$$\begin{aligned} \frac{d}{dx} a^x &= (\ln a) \cdot a^x \\ \frac{d}{dx} 2^{x^3-3x} &= \frac{d}{dx} 2^u \quad u = x^3 - 3x \\ &= 2^u \cdot \ln(2) \cdot \frac{du}{dx} \\ &= 2^{(x^3-3x)} \cdot \ln(2) \cdot [3x^2 - 3] \end{aligned}$$



Product Rule

What about the derivatives of products or quotients of two functions?



$$\frac{d}{dx} [u \cdot v] = u \cdot \frac{dv}{dx} + \frac{du}{dx} \cdot v$$

$$(uv)' = u \cdot v' + u' \cdot v$$

Quotients can be done by rewriting them as products

$$\begin{aligned} \frac{d}{dx} \left[u \cdot \frac{1}{v} \right] &= u \cdot \frac{d}{dx} \frac{1}{v} + u' \cdot \frac{1}{v} \\ &= \frac{-u v'}{v^2} + \frac{u'}{v} \\ &= \frac{-uv' + u'v}{v^2} \end{aligned}$$

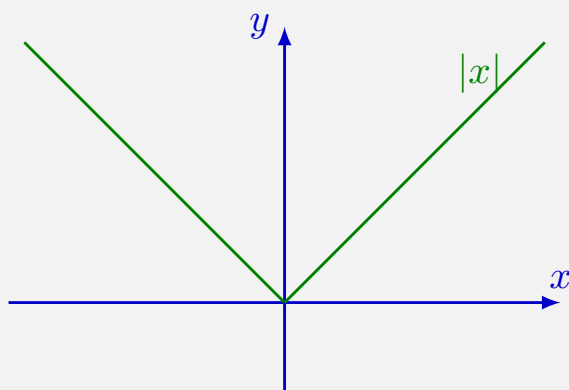
Product Rule Examples



$$\begin{aligned} \frac{d}{dx} [x^2 e^{3x}] &= x^2 \cdot 3e^{3x} + 2x \cdot e^{3x} \\ \frac{d}{dx} \frac{e^{-x}}{x^3} &= \frac{d}{dx} [e^{-x} \cdot x^{-3}] \\ &= e^{-x} \cdot \frac{-3}{x^4} + (-e^{-x}) \cdot \frac{1}{x^3} \\ \frac{d}{dx} [(x^2 + 3x + 5)^3 \cdot e^{4x}] &= (x^2 + 3x + 5)^3 \cdot \frac{d}{dx} e^{4x} \\ &\quad + \frac{d}{dx} [(x^2 + 3x + 5)^3] \cdot e^{4x} \\ &= (x^2 + 3x + 5)^3 \cdot 4e^{4x} \\ &\quad + 3 \cdot (x^2 + 3x + 5)^2 \cdot (2x + 3) \cdot e^{4x} \end{aligned}$$



Absolute Values



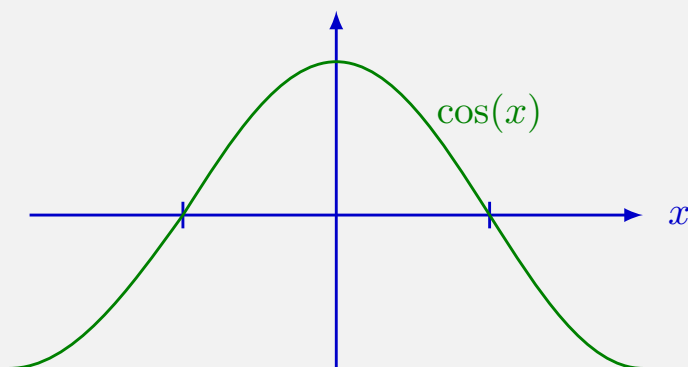
$$|x| = x \quad \text{if } x \geq 0$$

$$|x| = -x \quad \text{if } x < 0$$

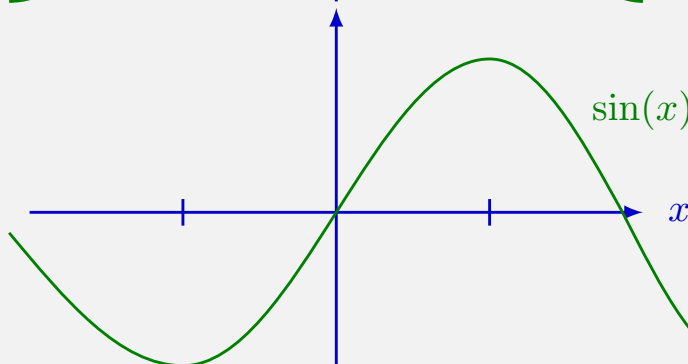
$$\frac{d}{dx}|x| = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$

Note that $\frac{d|x|}{dx}$ is undefined if $x = 0$.

Sines and Cosines



$$\frac{d}{dx} \cos(x) = -\sin(x)$$



$$\frac{d}{dx} \sin(x) = \cos(x)$$



Further Examples

$$\frac{d}{dx} \frac{2x+5}{x^2-3x+4} = (2x+5) \frac{(-1)(2x-3)}{(x^2-3x+4)^2} + \frac{2}{x^2-3x+4}$$

$$\frac{d}{dx} (x+2)e^{x^2-5x} = (x+2)(2x-5)e^{x^2-5x} + 1 \cdot e^{x^2-5x}$$

$$\frac{d}{dx} x^2 e^{-3x^2} = x^2(-6x)e^{-3x^2} + 2x \cdot e^{-3x^2}$$



Further Examples

$$\begin{aligned} \frac{d}{dx} \sin |x+2| &= \frac{d}{dx} \sin(x+2) \quad \text{if } x+2 > 0 \\ &= \cos(x+2) \quad x > -2 \end{aligned}$$

$$\begin{aligned} \frac{d}{dx} \sin |x+2| &= \frac{d}{dx} \sin(-x-2) \quad \text{if } x+2 < 0 \\ &= -\cos(-x-2) \quad x < -2 \end{aligned}$$

Key point: We get two cases based on whether or not what is inside the absolute value is positive. If $x+2 > 0$ then what is inside the absolute value is positive, so $|x+2| = x+2$, while if $x+2 < 0$ then what is inside the absolute value is negative so $|x+2| = -x-2$.



B.0 One Dimensional Calculus - Outline

B.0.1 One-Dimensional Derivatives

B.0.2 1-Dimensional integrals

What is an Integral?

The Fundamental Theorem of Calculus

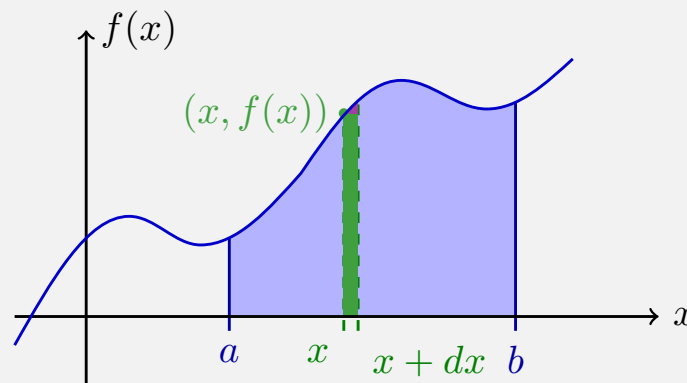
Common Formulas

Substitution

Other Formulas

B.0.3 Integration By Parts

Definition of an Integral



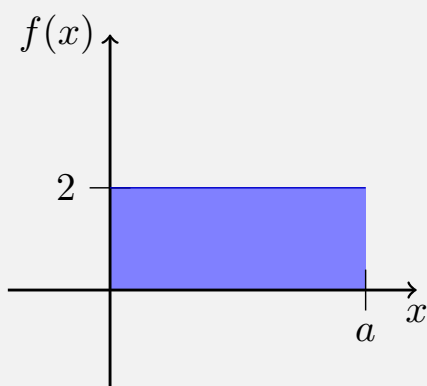
$$\int_a^b f(x) dx = \text{area under curve}$$

In some sense, $f(x) dx$ is the area of an infinitely thin rectangle and the integral says that the area under the curve is the sum of the areas of infinitely many of these thin rectangles.

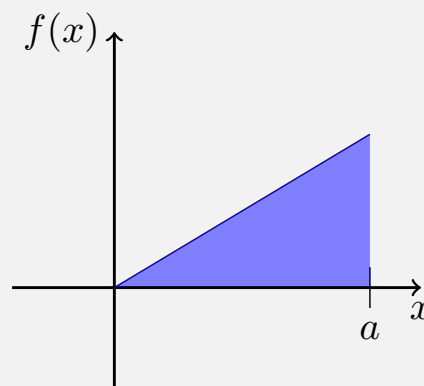


Geometric Examples

Often we can use geometry to find the integral/area under the curve.



$$\int_0^a 2dx = 2a$$



$$\int_0^a xdx = \frac{a^2}{2}$$

Geometric Examples



In that example,

1. The integral of a constant was a linear function.
2. The integral of a line was a quadratic function.

So in these two examples, when we integrated the power of a polynomial increased by 1.

When we differentiate,

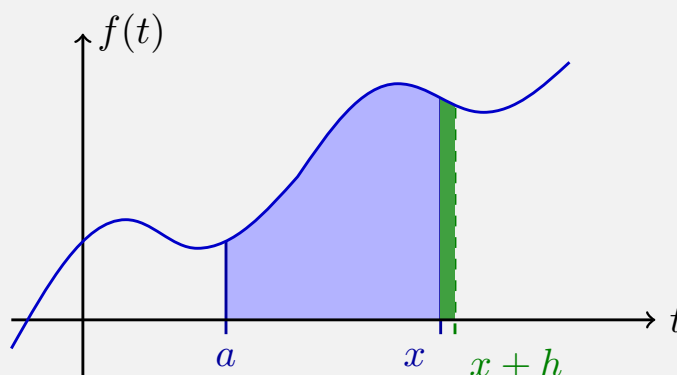
1. The derivative of a linear function is a constant.
2. The derivative of a quadratic function is linear.

More generally, when we differentiate the power of a polynomial decreases by 1. That is the opposite of when we integrate.

Hmmm.....isn't that an interesting coincidence?



The Fundamental Theorem of Calculus



$$\begin{aligned}\frac{d}{dx}F(x) &= \frac{d}{dx} \int_a^x f(t) dt = \lim_{h \rightarrow 0} \frac{\int_a^{x+h} f(t) dt - \int_a^x f(t) dt}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x) \cdot h}{h} = f(x)\end{aligned}$$

The Fundamental Theorem of Calculus



Theorem (Fundamental Theorem of Calculus)

$$\frac{d}{dx} \int_a^x f(t) dx = f(x)$$

Generalization: If v and u are functions,

$$\frac{d}{dx} \int_a^v f(t) dt = f(v) \frac{dv}{dx} \frac{d}{dx} \int_a^u f(t) dt = f(v) \frac{dv}{dx} - f(u) \frac{du}{dx}$$

In words, the Fundamental Theorem of Calculus says that derivatives and integrals are inverse operations.



Evaluating Integrals

The Fundamental Theorem of Calculus says that derivatives and integrals are inverse operations. To find the integral of $f(x)$, we need to find a function whose derivative is $f(x)$.

Examples

$$\int_0^5 x \, dx = \left. \frac{1}{2}x^2 \right|_0^5 = \frac{5^2}{2} - \frac{0^2}{2} = \frac{25}{2}$$

$$\begin{aligned} \int_a^b x^n \, dx &= \left. \frac{1}{n+1} \cdot x^{n+1} \right|_a^b \\ &= \frac{b^{n+1}}{n+1} - \frac{a^{n+1}}{n+1} \end{aligned}$$

Common Formulas



$$\int a \, dx = ax + C$$

$$\int x \, dx = \frac{x^2}{2} + C$$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad \text{for } n \neq -1$$

$$\int e^{bx} \, dx = \frac{1}{b}e^{bx} + C$$

$$\begin{aligned} \int a^x \, dx &= \int e^{x \ln a} \, dx \\ &= \frac{1}{\ln a} a^x + C \end{aligned}$$



Examples

$$\begin{aligned}
 \int_{-2}^5 3x^4 dx &= 3 \cdot \frac{x^5}{5} \Big|_{-2}^5 \\
 &= 3 \cdot \frac{5^5}{5} - 3 \cdot \frac{(-2)^5}{5} \\
 \int_2^\infty \frac{3}{x^4} dx &= \frac{3}{x^3} \cdot \frac{1}{-3} \Big|_2^\infty \\
 &= \frac{-1}{\infty^3} - \frac{-1}{2^3} \\
 &= 0 + \frac{1}{8} = \frac{1}{8}
 \end{aligned}$$

Substitution



When we differentiate, we often have nested functions and need to use the chain rule. For example,

$$\begin{aligned}
 \frac{d}{dx} e^{x^2} &= e^{x^2} \cdot 2x \\
 \frac{d}{dx} (x^2 + 3)^5 &= 5(x^2 + 3)^4 \cdot 2x
 \end{aligned}$$

In both those examples, the $2x$ factor comes from the chain rule. Often when we are doing integration, we will have a term that we need to somehow recognize as a chain rule factor. If we can do that, we can do a substitution to do the chain rule “backwards.”



Substitution

Suppose we want to integrate $2x \cdot e^{(x^2)}$. Let $u = x^2$. Then $\frac{du}{dx} = 2x$ so $du = 2x dx$ and we get

$$\begin{aligned} \int_{x=a}^{x=b} 2x e^{(x^2)} dx &= \int_{u=a^2}^{u=b^2} e^u du \\ &= e^u \Big|_{u=a^2}^{u=b^2} = e^{(x^2)} \Big|_{x=a}^{x=b} \\ &= e^{b^2} - e^{a^2} \end{aligned}$$

Note the limits! Either we convert back to x at the end, or we change the limits to be in terms of u .

Substitution Examples



$$\begin{aligned} \int_2^\infty x e^{-2x^2} dx & \quad u = 2x^2 \quad du = 4x dx \\ x = 2 \quad u &= 2 \cdot 2^2 = 8 \\ x = \infty \quad u &= 2 \cdot \infty^2 = \infty \\ &= \int_8^\infty e^{-u} \cdot \frac{du}{4} \\ &= \frac{1}{4} \cdot (-1) \cdot e^{-u} \Big|_8^\infty = 0 - \frac{-1}{4} e^{-8} \\ &= \frac{1}{4} e^{-8} \end{aligned}$$



Other Formulas

$$\int \frac{dx}{x} = \ln x + C$$

$$\int cf(x) dx = c \int f(x) dx$$

$$\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$$

$$\int \cos x dx = \sin x + C \quad \text{because } \frac{d}{dx} \sin x = \cos x$$

$$\int \sin x dx = -\cos x + C \quad \text{because } \frac{d}{dx} \cos x = -\sin x$$

Examples



$$\begin{aligned} u &= x + 5 & x &= 2, u = 2 + 5 = 7 \\ du &= dx & x &= 5, u = 5 + 5 = 10 \end{aligned}$$

$$\begin{aligned} \int_2^5 \frac{3x}{(x+5)^2} dx &= \int_7^{10} \frac{3(u-5)}{u^2} du \\ &= \int_7^{10} \frac{3}{u} - \frac{15}{u^2} du \\ &= \left(3 \ln u + \frac{15}{u} \right) \Big|_7^{10} \\ &= \left(3 \ln 10 + \frac{15}{10} \right) - \left(3 \ln 7 + \frac{15}{7} \right) \end{aligned}$$



Examples

$$\int_0^\pi (1 + \cos t) dt = t + \sin t \Big|_0^\pi$$

$$= (\pi + 0) - (0 + 0) = \pi$$

$$\int_{-2}^5 |x| dx = \int_{-2}^0 -x dx + \int_0^5 x dx$$

$$= \left. \frac{-x^2}{2} \right|_{-2}^0 + \left. \frac{x^2}{2} \right|_0^5 = \frac{(-2)^2}{2} + \frac{25}{2} = \frac{29}{2}$$

$$\frac{d}{dx} \int_{-2x}^{x^3} e^{5t-5} dt = e^{5x^3-5} \cdot 3x^2 - e^{5(-2x)-5} (-2)$$

B.0 One Dimensional Calculus - Outline



B.0.1 One-Dimensional Derivatives

B.0.2 1-Dimensional integrals

B.0.3 Integration By Parts

Integration By Parts

Tabular integration

The Gamma trick



Integration By Parts

Integration by parts is doing the product rule backwards.

$$\begin{aligned}\frac{d}{dx} uv &= u \cdot \frac{dv}{dx} + \frac{du}{dx} \cdot v \\ \int d(uv) &= uv = \int u dv + \int v du \\ \int u \cdot dv &= uv - \int v du\end{aligned}$$

Integration By Parts



Example

$$\begin{aligned}\int x e^x dx &= \int u dv \\ u &= x & dv &= e^x dx \\ du &= dx & v &= e^x \\ \text{so} \\ \int x e^x dx &= uv - \int v du \\ &= x e^x - \int e^x dx \\ &= x e^x - e^x + C\end{aligned}$$



Logarithms

You can use integration by parts to handle functions whose derivatives are easier to find than their integrals.

$$\begin{aligned}\int \ln x \, dx &= \int u \, dv \\ u &= \ln x & dv &= dx \\ du &= \frac{dx}{x} & v &= x\end{aligned}$$

so

$$\begin{aligned}\int \ln x \, dx &= uv - \int v \, du \\ &= (\ln x)x - \int x \frac{dx}{x} \\ &= x \ln x - x + C\end{aligned}$$

Logarithms



$$\begin{aligned}\int x \ln x \, dx &= \int u \, dv \\ u &= \ln x & dv &= x \, dx \\ du &= \frac{dx}{x} & v &= \frac{x^2}{2}\end{aligned}$$

so

$$\begin{aligned}\int x \ln x \, dx &= uv - \int v \, du \\ &= \frac{x^2}{2} \ln x - \int \frac{x^2}{2x} \, dx \\ &= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C\end{aligned}$$



Which Part to Use

How do we choose u and dv ? We need dv to be something that is easy to integrate and we need u to be something that is easy to differentiate.

Ideally we also want u to become simpler when you differentiate.

Easy to Differentiate		Easy to Integrate
Logs	Polynomials	Exponentials

Integration by parts



Iterated Parts

$$\begin{aligned}
 \int x^2 e^{2x} dx & \quad \text{Let } u = x^2 \quad dv = e^{2x} dx \\
 & \quad du = 2x dx \quad v = \frac{1}{2} e^{2x} \\
 & = x^2 \cdot \frac{1}{2} e^{2x} - \int \frac{1}{2} e^{2x} 2x dx \\
 & = \frac{1}{2} x^2 e^{2x} - \int x e^{2x} dx
 \end{aligned}$$

And to find this, we have to repeat integration by parts.



Tabular integration

Tabular integration is a way to organize our work when doing repeated integration by parts. To integrate $x^2 e^{2x}$,

<u>Derivative column</u>		<u>Integral column</u>
x^2	$+$	e^{2x}
$2x$	$-$	$\frac{1}{2} e^{2x}$
2	$+$	$\frac{1}{4} e^{2x}$
0	$-$	$\frac{1}{8} e^{2x}$

$$\int x^2 e^{2x} dx = x^2 \cdot \frac{1}{2} e^{2x} - 2x \cdot \frac{1}{4} e^{2x} + 2 \cdot \frac{1}{8} e^{2x} - 0$$

The Gamma trick



If we have a definite integral from 0 to infinity, we often can skip using integration by parts.

If $b > 0$ and a is an integer, then

$$\int_0^{\infty} x^a \cdot e^{-bx} dx = \frac{a!}{b^{a+1}}$$

Example

$$\int_0^{\infty} x^2 e^{-2x} dx = \frac{2!}{2^{2+1}} = \frac{1}{4}$$

$$\begin{aligned} -b &= -2 & b &= 2 \\ a &= 2 \end{aligned}$$



The Gamma trick

An example both ways: $\int_0^\infty 4x^2 e^{-x/3} dx$

Derivative column

Integral column

$4x^2$	$+$	$e^{-x/3}$
$8x$	$-$	$-3e^{-x/3}$
8	$+$	$9e^{-x/3}$
0		$-27e^{-x/3}$

So $\int 4x^2 e^{-x/3} dx$ is

$$\begin{aligned}
 & (4x^2)(-3e^{-x/3}) - (8x)(9e^{-x/3}) + 8 \cdot (-27e^{-x/3}) \\
 &= (-12x^2 - 72x - 8 \cdot 27)e^{-x/3}
 \end{aligned}$$

The Gamma trick



$$\begin{aligned}
 \int_0^\infty 4x^2 e^{-x/3} &= (-12x^2 - 72x - 8 \cdot 27)e^{-x/3} \Big|_0^\infty \\
 &= 8 \cdot 27
 \end{aligned}$$

or we can let $b = 1/3$ and $a = 2$ in our formula to get

$$\begin{aligned}
 \int_0^\infty x^a e^{-bx} &= \frac{a!}{b^{a+1}} \\
 \int_0^\infty x^2 e^{-x/3} &= \frac{2!}{(1/3)^{2+1}} = 2 \cdot 27 \\
 \int_0^\infty 4x^2 e^{-x/3} &= 4 \cdot \frac{2!}{\left(\frac{1}{3}\right)^{2+1}} = 8 \cdot 27
 \end{aligned}$$



Other Definite Integrals

$$\int_1^{\infty} 4x^2 e^{-x/3} \quad \begin{array}{l} \text{We want } u = 0 \text{ when } x = 1 \\ u = x - 1, \quad x = u + 1 \end{array}$$

$$= \int_0^{\infty} 4(u+1)^2 e^{(-u-1)/3} du$$

$$= 4e^{-1/3} \int_0^{\infty} (u^2 + 2u + 1) e^{-u/3} du$$

and now we plug into $\int_0^{\infty} x^a e^{-bx} = \frac{a!}{b^{a+1}}$

$$= 4e^{-1/3} \left[\frac{2!}{\left(\frac{1}{3}\right)^3} + 2 \cdot \frac{1}{\left(\frac{1}{3}\right)^2} + \frac{1}{\frac{1}{3}} \right]$$

The Gamma trick



Idea of Proof:

Let $u = x^a$ and $dv = e^{-bx} dx$

$$\begin{aligned} \int_0^{\infty} x^a e^{-bx} dx &= x^a \cdot \frac{-1}{b} e^{-bx} \Big|_0^{\infty} - \int_0^{\infty} ax^{a-1} \left(\frac{-1}{b} \right) e^{-bx} dx \\ &= 0 - 0 + \frac{a}{b} \int_0^{\infty} x^{a-1} e^{-bx} dx \\ &= \frac{a}{b} \cdot \frac{(a-1)!}{b^{a-1+1}} \\ &= \frac{a!}{b^{a+1}} \end{aligned}$$

It also is related to $E[X^a]$ when X is an exponential random variable as well as the density of a Gamma random variable.

Continuous Distributions: Overview

1

Discrete vs Continuous

Mixed Distributions

Exercises

Comparing Uniforms

2

Let N be an *integer* uniformly chosen from $\{1, 2, \dots, 100\}$ and let X be a *real number* uniformly chosen from $(0, 100)$. Then

$$P[N = n] = \frac{1}{100} \quad n = 1, 2, 3, \dots, 100$$

$$P[N \leq n] = \frac{n}{100} \quad n = 1, 2, 3, \dots, 100$$

$$P[X = x] = 0 \quad \text{for all } x$$

$$P[X \leq x] = \frac{x}{100} \quad 0 \leq x \leq 100$$

Point: The cumulative distribution function $F(x) = P[X \leq x]$ still makes sense for continuous distributions, and will still be useful.

$P[X = x] = 0$ for all x for a purely continuous function, and so we'll need a slightly different idea.

Discrete vs Continuous

3

For discrete random variables, we often summed expressions that involved $P[X = x]$, such as

$$E[X] = \sum x \cdot P[X = x]$$

In the continuous case, the sums will become integrals and the “density” of X , denoted $f(x)$, will replace $P[X = x]$ in most formulas. For example,

$$E[X] = \int x \cdot f(x) dx$$

We will define $f(x)$ in the next lesson.

Mixed Distributions

4

Not all distributions are purely discrete or purely continuous.

A mixed distribution is some of each.

On the exam, mixed distributions often come from adding deductibles and limits to continuous loss amounts.

Mixed Example

5

Losses X are uniformly distributed on $(0, 100)$. Let Y be the payment amount after a deductible of 30 is applied to the loss.

The deductible of 30 means that $Y = 0$ if the loss X is less than 30, and $Y = X - 30$ if the loss exceeds 30. That means that

$$P[Y = 0] = P[X \leq 30] = \frac{30}{100}$$

$$P[Y = y] = 0 \quad \text{for } y > 0$$

$$P[Y \leq y] = P[X \leq y + 30] = \frac{y + 30}{100} \quad \text{for } 0 < y < 70$$

so Y has a discrete piece (at 0), a continuous piece (from 0 to 70), and the cdf makes sense everywhere so we can still use it to study Y .

Exercise 1

6

If N is uniform on $\{1, 2, 3, 4, 5\}$, and X is uniform on $(0, 5)$, find $P[N \leq 2.3]$ and $P[X \leq 2.3]$.

Exercise 1

6

If N is uniform on $\{1, 2, 3, 4, 5\}$, and X is uniform on $(0, 5)$, find $P[N \leq 2.3]$ and $P[X \leq 2.3]$.

$$\begin{aligned} P[N \leq 2.3] &= P[N = 1 \text{ or } 2] \\ &= \frac{2}{5} \end{aligned}$$

$$P[X \leq 2.3] = \frac{2.3}{5}$$

Exercise 2

7

Loss amounts are uniform on the interval $(0, 6)$, and insured with a deductible of 1.6. Find the probabilities that:

- a) the payment for a randomly chosen loss is 0
- b) the payment for a randomly chosen loss is less than 2.

Exercise 2

7

Loss amounts are uniform on the interval $(0, 6)$, and insured with a deductible of 1.6. Find the probabilities that:

- a) the payment for a randomly chosen loss is 0
- b) the payment for a randomly chosen loss is less than 2.

Let X denote the amount of a randomly chosen loss, and Y the corresponding payment.

$$\begin{aligned}P[Y = 0] &= P[X \leq 1.6] = \frac{1.6}{6.0} \\P[Y \leq 2] &= P[X \leq 2 + 1.6] \\&= P[X \leq 3.6] = \frac{3.6}{6.0}\end{aligned}$$

Densities and CDFs

1

Definitions of CDFs and Densities

Percentiles

Exercises

Definitions: CDF

2

The *cumulative distribution function* (CDF) of X is given by

$$F(x) = F_X(x) = P[X \leq x]$$

This applies to all random variables, whether they have discrete, continuous, or mixed distributions.

Definitions: Density

3

If X is purely continuous, then the *density* $f(x)$ is given by

$$f(x) = \frac{d}{dx} F(x)$$

The cdf is then $F(x) = \int_{-\infty}^x f(y) dy$

For discrete X , $F(x) = P[X \leq x] = \sum_{y \leq x} P[X = y]$

In most formulas, $f(y) dy$ will take the place of $P[X = y]$.

In some sense, $f(y) dy = P[y < X \leq y + dy]$

Properties of the CDF

4

The cumulative distribution function satisfies

1. $0 \leq F(x) \leq 1$
2. If $x \leq y$ then $F(x) \leq F(y)$
3. $\lim_{x \rightarrow \infty} F(x) = 1$
4. $\lim_{x \rightarrow -\infty} F(x) = 0$

Since $F(x) = P[X \leq x]$, $P[X > x] = 1 - F(x)$

$1 - F(x)$ is called the *survival function*

Properties of Densities

5

For continuous X , the density $f_X(x)$ satisfies:

1. $f(x) \geq 0$
2. There need not be an upper bound for $f(x)$
3. $\int_{-\infty}^{\infty} f(x)dx = 1$
4. $\int_a^b f(x)dx = P[a < X \leq b] = F(b) - F(a)$

Example

6

Continuous Uniform

Suppose that X is uniform on $(0, 0.1)$. What are $F(x)$ and $f(x)$?

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{0.1} = 10x & 0 \leq x \leq 0.1 \\ 1 & 0.1 < x \end{cases}$$

$$f(x) = \begin{cases} 0 & x < 0 \\ 10 & 0 < x < 0.1 \\ 0 & 0.1 < x \end{cases}$$

Percentiles and Medians

7

x is a k -th percentile of X if $F(x) = k\%$.

The median is the 50-th percentile, so $F(x) = 0.5$ at the median.

On the exam, continuous distributions will usually only have one point x such that $F(x) = k\%$. In that case the k -th percentile is uniquely defined and we say that x is *the* k -th percentile.

Exercise 1

8

The CDF of X satisfies:

$$F(x) = \begin{cases} 0 & x < 1 \\ (x-1) - \frac{1}{4}(x-1)^2 & 1 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

Find $P[X \leq 2]$, $P[1.5 < X \leq 2]$ and $f(x)$.

Exercise 1

8

The CDF of X satisfies:

$$F(x) = \begin{cases} 0 & x < 1 \\ (x-1) - \frac{1}{4}(x-1)^2 & 1 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

Find $P[X \leq 2]$, $P[1.5 < X \leq 2]$ and $f(x)$.

$$P[X \leq 2] = F(2) = (2-1) - \frac{1}{4}(2-1)^2 = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\begin{aligned} P[1.5 < X \leq 2] &= F(2) - F(1.5) \\ &= \frac{3}{4} - [(1.5-1) - 0.25(0.5)^2] = 0.3125 \end{aligned}$$

$$f(x) = F'(x) = \begin{cases} 0 & x < 1 \\ 1 - \frac{1}{2}(x-1) & 1 < x < 3 \\ 0 & 3 < x \end{cases}$$

Exercise 2

9

A continuous random variable Y has density $f(y) = 2/y^3$ for $1 < y < \infty$ and $f(y) = 0$ otherwise. Find a formula for the CDF $F(y)$, and find $P[Y \leq 4 \mid Y > 2]$.

Exercise 2

9

A continuous random variable Y has density $f(y) = 2/y^3$ for $1 < y < \infty$ and $f(y) = 0$ otherwise. Find a formula for the CDF $F(y)$, and find $P[Y \leq 4 \mid Y > 2]$.

Key point: The CDF is the definite integral, not any arbitrary anti-derivative. $f(y) = 0$ for $y < 1$, so we can start our integral at 1. More precisely, for $y > 1$,

$$\begin{aligned} F(y) &= \int_{-\infty}^y f(t) dt = \int_{-\infty}^1 0 dt + \int_1^y \frac{2}{t^3} dt \\ &= \int_1^y \frac{2}{t^3} dt = \left. \frac{-1}{t^2} \right|_1^y = \frac{-1}{y^2} - \frac{-1}{1^2} = 1 - \frac{1}{y^2} \end{aligned}$$

$$\begin{aligned} P[Y \leq 4 \mid Y > 2] &= \frac{P[2 < Y \leq 4]}{P[Y > 2]} = \frac{F(4) - F(2)}{1 - F(2)} \\ &= \frac{(1 - (1/4)^2) - (1 - (1/2)^2)}{1 - (1 - (1/2)^2)} = \frac{(1/2^2) - (1/4^2)}{1/2^2} = \frac{3}{4} \end{aligned}$$

Mixed distributions

1

Deductibles and/or Benefit Limits

A Possibility of No Loss

When $F(x)$ isn't continuous

Exercises

Deductibles and Benefit Limits

2

Suppose an insurance policy has a *deductible* of d and a *payment limit* of u . A customer / insured has a loss of L .

- Insured is responsible for the first d of loss.
- Insurance company / insurer pays for portion of loss that exceeds d up to a total payment u
- Insured is responsible for the rest.

If $L < d$ Insurance payment = 0

If $d \leq L \leq d + u$ Insurance payment = $L - d$

If $d + u < L$ Insurance payment = u

Remarks:

- Payment always refers to payment made by insurer
- The *premium* is paid by insured to insurer
- Exam P study note only covers payment limits, other types are on later exams

Deductible example

3

Suppose that loss amounts X have density $f(x) = 0.02x$, $0 < x < 10$. If there is a deductible of 2 and a maximum payment of 6, then what is the probability of a payment of 5 or less? What is the probability of a payment of 6?

$$\begin{aligned} P[\text{Payment} \leq 5] &= P[X \leq 5 + 2] \\ &= \int_0^7 0.02x \, dx \\ &= 0.01 \cdot 7^2 = \boxed{0.49} \end{aligned}$$

$$\begin{aligned} P[\text{Payment} = 6] &= P[X \geq 6 + 2] = P[X \geq 8] \\ &= \int_8^{10} 0.02x \, dx \\ &= 0.01 \cdot (10^2 - 8^2) = \boxed{0.36} \end{aligned}$$

A Possibility of No Loss

4

Losses, if they occur, are uniformly distributed on the interval $(100, 500)$

If there is a 60% probability of no loss and a 40% probability of exactly one loss, what is the cdf of the total loss amount?

Let $L = \text{loss}$. $P[L = 0] = 0.6$

$P[L \leq 33] = P[L = 0] = 0.6$. In fact, $P[L \leq \ell] = 0.6$ for all $\ell < 100$

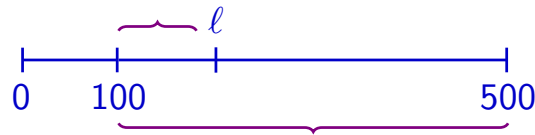
$$F_L(\ell) = \begin{cases} 0 & \ell < 0 \\ 0.6 & 0 \leq \ell < 100 \\ ?? & 100 \leq \ell < 500 \\ 1 & \ell \geq 500 \end{cases}$$

A Possibility of No Loss

5

If $100 \leq \ell < 500$,

$$\begin{aligned}
 P[L \leq \ell] &= P[\text{no loss}] + P[L \leq \ell, \text{have a loss}] \\
 &= P[\text{no loss}] + P[\text{loss}] \cdot P[L \leq \ell \mid \text{have a loss}] \\
 F_L(\ell) &= 0.6 + 0.4 \cdot P[L \leq \ell \mid \text{have a loss}] \\
 &= 0.6 + 0.4 \cdot \frac{\ell - 100}{500 - 100} \\
 &= 0.6 + 0.4 \cdot \frac{\ell - 100}{400}
 \end{aligned}$$

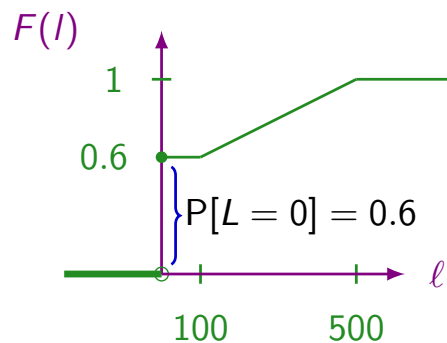


A Possibility of No Loss

6

So we have found that

$$F_L(\ell) = \begin{cases} 0 & \ell < 0 \\ 0.6 & 0 \leq \ell < 100 \\ 0.6 + 0.4 \cdot \frac{\ell - 100}{400} & 100 \leq \ell < 500 \\ 1 & 500 \leq \ell \end{cases}$$



When $F(x)$ isn't continuous

7

If X is purely continuous, then $\int_{-\infty}^{\infty} f(x) dx = 1$, and $F(x)$ is continuous.

If $F(x)$ is defined piecewise, it may not be continuous and we may have a mixed distribution.

Possible jumps are at the beginning and end of cases.

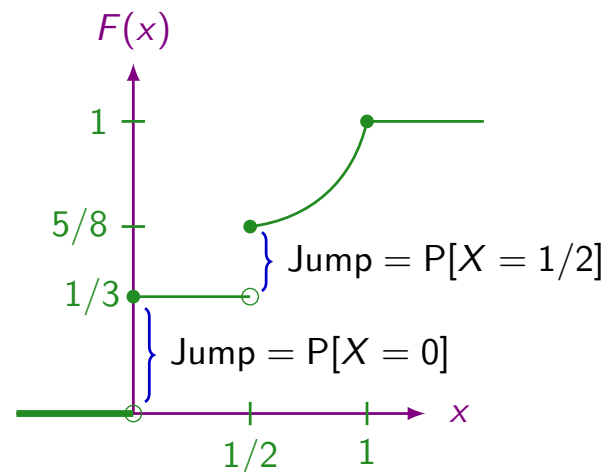
When $F(x)$ isn't continuous

8

Suppose X has CDF

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{3} & 0 \leq x < \frac{1}{2} \\ \frac{x^2 + 1}{2} & \frac{1}{2} \leq x < 1 \\ 1 & 1 \leq x \end{cases}$$

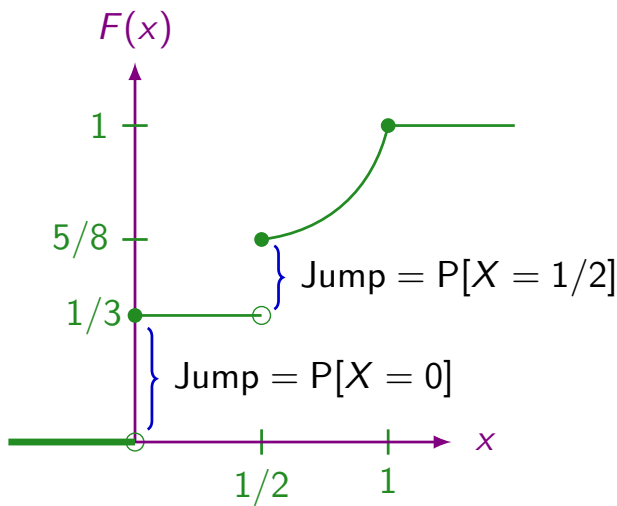
$$F(0) = \frac{1}{3} \quad F(1/2) = \frac{5}{8}$$



There is no jump at 1 even though it is the endpoint of a case because formula for $F(x)$ leading up to 1 matches $F(1)$.

When $F(x)$ isn't continuous

9



Jumps are points with non-zero probability:

$$P[X = 0] = \frac{1}{3} - 0$$

$$P\left[X = \frac{1}{2}\right] = \frac{5}{8} - \frac{1}{3}$$

for $(1/2) < x < 1$ (continuous piece):

$$f(x) = \frac{d}{dx} F(x)$$

$$= \frac{d}{dx} \frac{x^2 + 1}{2}$$

$$= x$$

Exercise 1

10

An insurance policy pays for a random loss X subject to a deductible of d . The loss amount is a continuous random variable with density function

$$f(x) = \begin{cases} 2x & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

For a random loss X , the probability that the insurance payment is less than 0.3 is equal to 0.49. Find d .

Exercise 1

10

An insurance policy pays for a random loss X subject to a deductible of d . The loss amount is a continuous random variable with density function

$$f(x) = \begin{cases} 2x & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

For a random loss X , the probability that the insurance payment is less than 0.3 is equal to 0.49. Find d .

$$P[\text{Payment} \leq 0.3] = P[\text{Loss} \leq 0.3 + d]$$

$$0.49 = \int_0^{0.3+d} f(x) dx = \int_0^{0.3+d} 2x dx$$

$$0.49 = x^2 \Big|_0^{0.3+d} = (0.3 + d)^2$$

$$d = 0.7 - 0.3 = \boxed{0.4}$$

Exercise 2

11

A random variable X has CDF

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{(x-1)^2}{5} & 1 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

Find $P[X = 1]$, $P[X = 3]$ and $f(x)$ for $1 < x < 3$.

Exercise 2

11

A random variable X has CDF

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{(x-1)^2}{5} & 1 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

Find $P[X = 1]$, $P[X = 3]$ and $f(x)$ for $1 < x < 3$.

$$P[X = 1] = F(1) - \lim_{x \uparrow 1} F(x) = \frac{(1-1)^2}{5} - 0 = 0$$

$$P[X = 3] = F(3) - \lim_{x \uparrow 3} F(x) = 1 - \frac{(3-1)^2}{5} = \frac{1}{5}$$

$$f(x) = \frac{d}{dx} F(x) = \frac{d}{dx} \frac{(x-1)^2}{5} = \frac{2(x-1)}{5} \quad \text{for } 1 < x < 3$$

Moments of Continuous Distributions

1

Definition

Continuous Example

Exercises

Definition of Moments

2

In the purely discrete case, we had

$$E[X] = \sum_x x \cdot P[X = x]$$

In the purely continuous case, this becomes

$$E[X] = \int_x x \cdot f(x) dx$$

Definition of Moments

3

For other moments, the discrete formula

$$E[g(X)] = \sum_x g(x) \cdot P[X = x]$$

turns into

$$E[g(X)] = \int_x g(x) \cdot f(x) dx$$

In particular,

$$\begin{aligned} \text{Var}[X] &= E[X^2] - (E[X])^2 \\ &= \int x^2 \cdot f(x) dx - \left(\int x \cdot f(x) dx \right)^2 \end{aligned}$$

Example

4

A random variable X has density $3x^2$ for $0 < x < 1$. Find its mean and variance.

$$\begin{aligned} E[X] &= \int_0^1 x \cdot f(x) dx \\ &= \int_0^1 x \cdot 3x^2 dx = \int_0^1 3x^3 dx \\ &= \frac{3}{4} x^4 \Big|_0^1 \\ &= \frac{3}{4} \end{aligned}$$

Example: Variance

5

For the variance,

$$\begin{aligned} E[X^2] &= \int_0^1 x^2 f(x) dx \\ &= \int_0^1 x^2 \cdot 3x^2 dx = \int_0^1 3x^4 dx \\ &= \frac{3}{5} x^5 \Big|_0^1 = \frac{3}{5} \\ \text{Var}[X] &= E[X^2] - (E[X])^2 \\ &= \frac{3}{5} - \left(\frac{3}{4}\right)^2 \\ &= \frac{3}{80} \end{aligned}$$

Exercise 1

6

If Y has density $f(y) = 1 - 0.5y$ for $0 < y < 2$, and $f(y) = 0$ otherwise, find $E[Y]$.

Exercise 1

6

If Y has density $f(y) = 1 - 0.5y$ for $0 < y < 2$, and $f(y) = 0$ otherwise, find $E[Y]$.

$$\begin{aligned} E[Y] &= \int_0^2 y f(y) dy \\ &= \int_0^2 y(1 - 0.5y) dy \\ &= \int_0^2 y - 0.5y^2 dy \\ &= \left. \frac{y^2}{2} - \frac{0.5y^3}{3} \right|_0^2 \\ &= 2 - \frac{4}{3} = \boxed{\frac{2}{3}} \end{aligned}$$

Exercise 2

7

If Y has density $f(y) = 1 - 0.5y$ for $0 < y < 2$, and $f(y) = 0$ otherwise, find $\text{Var}[Y]$.

Exercise 2

7

If Y has density $f(y) = 1 - 0.5y$ for $0 < y < 2$, and $f(y) = 0$ otherwise, find $\text{Var}[Y]$.

$$E[Y] = \frac{2}{3} \text{ from before}$$

$$\begin{aligned} E[Y^2] &= \int_0^2 y^2 f(y) dy = \int_0^2 y^2(1 - 0.5y) dy \\ &= \int_0^2 y^2 - 0.5y^3 dy \\ &= \left. \frac{y^3}{3} - \frac{y^4}{8} \right|_0^2 \\ &= \frac{8}{3} - \frac{16}{8} = \frac{2}{3} \end{aligned}$$

$$\text{Var}[Y] = E[Y^2] - (E[Y])^2 = \frac{2}{3} - \left(\frac{2}{3}\right)^2 = \boxed{\frac{2}{9}}$$

Moments of Mixed Distributions

1

Mixed Distributions Idea

Example

Exercises

Mixed distributions

2

For discrete variables, $E[X] = \sum_x x \cdot P[X = x]$

For purely continuous variables, $E[X] = \int x \cdot f(x) dx$

For mixed distributions:

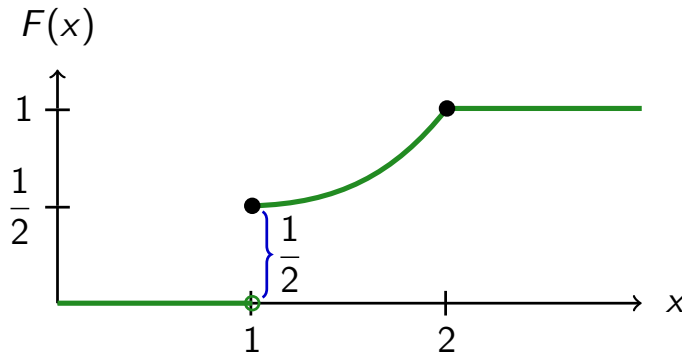
1. Use the discrete formula on discrete piece
2. Use the continuous formula on the continuous piece
3. Then sum the two parts

Example

3

Find the mean and variance of X if

$$F(x) = \begin{cases} 0 & \text{for } x < 1 \\ \frac{x^2 - 2x + 2}{2} & \text{for } 1 \leq x < 2 \\ 1 & \text{for } x \geq 2 \end{cases}$$



$$\begin{aligned} P[X = 1] &= F(1) - \lim_{x \uparrow 1} F(x) \\ &= \frac{1}{2} - 0 \\ &= \frac{1}{2} \end{aligned}$$

Example: Mean

4

For $1 < x < 2$,

$$f(x) = \frac{d}{dx} F(x) = \frac{d}{dx} \frac{x^2 - 2x + 2}{2} = x - 1$$

$$\begin{aligned} E[X] &= 1 \cdot P[X = 1] + \int_1^2 x \cdot (x - 1) dx \\ &= 1 \cdot \frac{1}{2} + \int_1^2 (x^2 - x) dx \\ &= \frac{1}{2} + \left(\frac{x^3}{3} - \frac{x^2}{2} \right) \Big|_1^2 \\ &= \frac{1}{2} + \left(\frac{8}{3} - \frac{4}{2} \right) - \left(\frac{1}{3} - \frac{1}{2} \right) \\ &= \frac{4}{3} \end{aligned}$$

Example: Variance

5

For the variance:

$$\begin{aligned}
 E(X^2) &= 1^2 \cdot P[X = 1] + \int_1^2 x^2 \cdot (x - 1) dx \\
 &= 1^2 \cdot \frac{1}{2} + \int_1^2 (x^3 - x^2) dx \\
 &= \frac{1}{2} + \left(\frac{x^4}{4} - \frac{x^3}{3} \right) \Big|_1^2 \\
 &= \frac{1}{2} + \left(\frac{16}{4} - \frac{8}{3} \right) - \left(\frac{1}{4} - \frac{1}{3} \right) \\
 &= \frac{23}{12} \\
 \text{Var}(X) &= \frac{23}{12} - \left(\frac{4}{3} \right)^2 = \frac{5}{36}
 \end{aligned}$$

Exercise 1

6

Suppose that X is a mixed random variable such that $P[X = 3] = 0.5$ and X has density $f(x) = x$ for $0 < x < 1$, and 0 otherwise. Find $E[X]$.

Exercise 1

6

Suppose that X is a mixed random variable such that $P[X = 3] = 0.5$ and X has density $f(x) = x$ for $0 < x < 1$, and 0 otherwise. Find $E[X]$.

$$\begin{aligned} E[X] &= 3 \cdot P[X = 3] + \int_0^1 x f(x) dx \\ &= 3 \cdot 0.5 + \int_0^1 x \cdot x dx \\ &= \frac{3}{2} + \left. \frac{x^3}{3} \right|_0^1 \\ &= \frac{3}{2} + \frac{1}{3} \\ &= \boxed{\frac{11}{6}} \end{aligned}$$

Exercise 2

7

Suppose that X is a mixed random variable such that $P[X = 3] = 0.5$ and X has density $f(x) = x$ for $0 < x < 1$, and 0 otherwise. Find $E[X^2]$.

Exercise 2

7

Suppose that X is a mixed random variable such that $P[X = 3] = 0.5$ and X has density $f(x) = x$ for $0 < x < 1$, and 0 otherwise. Find $E[X^2]$.

$$\begin{aligned} E[X^2] &= 3^2 \cdot P[X = 3] + \int_0^1 x^2 \cdot f(x) \, dx \\ &= \frac{9}{2} + \int_0^1 x^3 \, dx \\ &= \frac{9}{2} + \left. \frac{x^4}{4} \right|_0^1 \\ &= \frac{9}{2} + \frac{1}{4} \\ &= \boxed{\frac{19}{4}} \end{aligned}$$

The Survival Function Approach

1

Using the survival function

Generalizations

Pros and Cons

Example Revisited

Other Examples

Exercises

Using the survival function

2

Suppose that $P[X \geq 0] = 1$ and X is continuous. Then

$$E(X) = \int_0^{\infty} x \cdot f(x) dx$$

Using integration by parts,

$$u = x \quad dv = f(x) dx$$

$$du = dx \quad v = F(x) - 1$$

$$E(X) = \int_0^{\infty} x \cdot f(x) dx = uv \Big|_0^{\infty} + \int_0^{\infty} (-v) du$$

$$\text{at } 0, \quad uv = 0 \cdot [F(0) - 1] = 0$$

$$\text{at } \infty, \quad v = F(\infty) - 1 = 1 - 1 = 0, \quad uv = 0$$

$$E(X) = 0 + \int_0^{\infty} [1 - F(x)] dx = \int_0^{\infty} P[X > x] dx$$

Generalizations: Discrete Distributions

3

So for continuous, non-negative X ,

$$E[X] = \int_0^{\infty} P[X > x] dx$$

This actually holds for all non-negative random variables, including discrete and mixed distributions. For discrete distributions,

$$\begin{aligned} \int_n^{n+1} P[X > x] dx &= P[X > n] \\ \int_0^{\infty} P[X > x] dx &= \sum_{n=0}^{\infty} P[X > n] \end{aligned}$$

See A.3.2 for an intuitive interpretation of this.

Generalizations: $E[g(x)]$

4

For any non-negative variable (continuous, mixed, or discrete), if $g(0) = 0$ then

$$E[g(X)] = \int_0^{\infty} g'(x) P[X > x] dx$$

For example, if $g(x) = x^2$ then $g'(x) = 2x$ and we get

$$\begin{aligned} E[X^2] &= \int_0^{\infty} 2x \cdot P[X > x] dx \\ E[X^3] &= \int_0^{\infty} 3x^2 \cdot P[X > x] dx \end{aligned}$$

Unfortunately this is rarely useful, main application is with mixed distributions.

Pros and Cons of Survival Method

5

Advantages of survival method:

- Often saves some steps, especially if $F(x)$ is given but $f(x)$ is not.
- Often faster for mixed distributions.
- Often gives nicer integrals (e.g., if $f(x) = e^{-x}$, integrating $x \cdot f(x)dx$ requires integration by parts, but the survival method does not.)

Disadvantages:

- Because the integral starts at 0, it sometimes will be messier.
- If $f(x)$ is directly given, finding $P[X > x]$ can require an extra step.

Example Revisited

6

Find the mean and variance of X if

$$F(x) = \begin{cases} 0 & \text{for } x < 1 \\ \frac{x^2 - 2x + 2}{2} & \text{for } 1 \leq x < 2 \\ 1 & \text{for } x \geq 2 \end{cases}$$

$$\begin{aligned} E(X) &= \int_0^{\infty} P[X > x] dx = \int_0^{\infty} 1 - F(x) dx \\ &= \int_0^1 (1 - 0) dx + \int_1^2 \frac{2x - x^2}{2} dx + \int_2^{\infty} (1 - 1) dx \\ &= 1 + \left(\frac{x^2}{2} - \frac{x^3}{6} \right) \Big|_1^2 + 0 \\ &= 1 + \left(\frac{4}{2} - \frac{8}{6} \right) - \left(\frac{1}{2} - \frac{1}{6} \right) + 0 = \frac{4}{3} \end{aligned}$$

Example: Variance

7

$$\begin{aligned}
E(X^2) &= \int_0^\infty \frac{d}{dx}(x^2) \cdot P[X > x] dx = \int_0^\infty 2x \cdot [1 - F(x)] dx \\
&= \int_0^1 2x \cdot 1 dx + \int_1^2 2x \cdot \frac{2x - x^2}{2} dx + \int_2^\infty 0 dx \\
&= \int_0^1 2x dx + \int_1^2 (2x^2 - x^3) dx \\
&= x^2 \Big|_0^1 + \left(\frac{2x^3}{3} - \frac{x^4}{4} \right) \Big|_1^2 \\
&= (1 - 0) + \left(\frac{2 \cdot 8}{3} - \frac{16}{4} \right) - \left(\frac{2}{3} - \frac{1}{4} \right) = \frac{23}{12} \\
\text{Var}(X) &= \frac{23}{12} - \left(\frac{4}{3} \right)^2 = \frac{5}{36}
\end{aligned}$$

Example

8

Suppose X has density $f(x) = \frac{3 \cdot 100^3}{(x + 100)^4}$ for $0 < x < \infty$ and 0 otherwise. Find $E[X]$.

We have 3 possible approaches:

1. Find $\int_0^\infty \frac{x \cdot 3 \cdot 100^3}{(x + 100)^4} dx$ using u -substitution with $u = x + 100$
2. Find $\int_0^\infty \frac{x \cdot 3 \cdot 100^3}{(x + 100)^4} dx$ using integration by parts
3. Use the survival method

Remark: The last two approaches are mathematically identical. The survival method is just automating the integration by parts.

Survival Approach

9

$$\begin{aligned}
 f(x) &= \frac{3 \cdot 100^3}{(x + 100)^4} \\
 P[X > x] &= \int_x^\infty \frac{3 \cdot 100^3}{(t + 100)^4} dt \\
 &= \frac{-100^3}{(t + 100)^3} \Big|_x^\infty = \frac{100^3}{(x + 100)^3} \\
 E[X] &= \int_0^\infty \frac{100^3}{(x + 100)^3} dx \\
 &= \frac{-1}{2} \frac{100^3}{(x + 100)^2} \Big|_0^\infty \\
 &= \frac{100^3}{2 \cdot 100^2} = \boxed{50}
 \end{aligned}$$

Exercise 1

10

Use the survival approach to find $E[X]$ if the cdf of X is

$$F(x) = \begin{cases} 1 - \frac{100^3}{x^3} & x > 100 \\ 0 & x \leq 100 \end{cases}$$

Exercise 1

10

Use the survival approach to find $E[X]$ if the cdf of X is

$$F(x) = \begin{cases} 1 - \frac{100^3}{x^3} & x > 100 \\ 0 & x \leq 100 \end{cases}$$

$$\begin{aligned} E[X] &= \int_0^{\infty} P[X > x] dx = \int_0^{\infty} [1 - F(x)] dx \\ &= \int_0^{100} (1 - 0) dx + \int_{100}^{\infty} \left[1 - \left(1 - \frac{100^3}{x^3} \right) \right] dx \\ &= 100 + \int_{100}^{\infty} \frac{100^3}{x^3} dx \\ &= 100 + \frac{-1}{2} \frac{100^3}{x^2} \Big|_{100}^{\infty} = \boxed{150} \end{aligned}$$

Exercise 2

11

Use the density to find $E[X]$ if the cdf of X is

$$F(x) = \begin{cases} 1 - \frac{100^3}{x^3} & x > 100 \\ 0 & x \leq 100 \end{cases}$$

Exercise 2

11

Use the density to find $E[X]$ if the cdf of X is

$$F(x) = \begin{cases} 1 - \frac{100^3}{x^3} & x > 100 \\ 0 & x \leq 100 \end{cases}$$

$$f(x) = \frac{3 \cdot 100^3}{x^4}, x > 100$$

$$\begin{aligned} E[X] &= \int_{100}^{\infty} x \cdot \frac{3 \cdot 100^3}{x^4} dx = \int_{100}^{\infty} \frac{3 \cdot 100^3}{x^3} dx \\ &= \frac{-1}{2} \cdot \frac{3 \cdot 100^3}{x^2} \Big|_{100}^{\infty} = \boxed{150} \end{aligned}$$

Continuous Uniforms: Basics

1

Definitions

CDF and Density

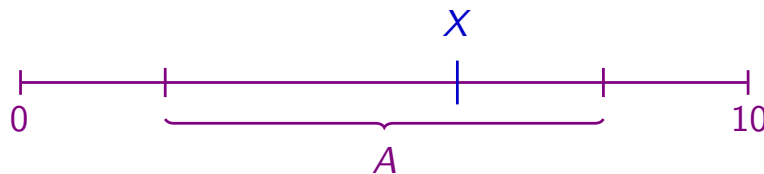
Mean and Variance

Exercises

Continuous Uniforms

2

Pick a point X uniformly between 0 and 10



$$P[X \in A] = \frac{\text{length of } A}{\text{total length}}$$

for $0 < x < 10$ can find the CDF and thus density by

$$P[X \leq x] = F(x) = \frac{x}{10}$$

$$f(x) = \frac{d}{dx} F(x) = \frac{1}{10} = \frac{1}{\text{total length}}$$

CDF and Density

3

More generally, if X is uniform on S ,

$$P[X \in A] = \frac{\text{length (or area) of } A}{\text{length (or area) of } S}$$

$$\text{density} = \frac{1}{\text{length of } S}$$

If X is uniform on (a, b)



$$f(x) = \frac{1}{b-a} \quad F(x) = \frac{x-a}{b-a}$$

Mean and variance of Uniform(0, 1)

4

If $X \sim \text{uniform}(0, 1)$

$$f(x) = \frac{1}{1-0} = 1$$

$$E[X] = \int_0^1 x \cdot 1 \, dx = \left. \frac{1}{2}x^2 \right|_0^1 = \frac{1}{2}$$

$$E[X^2] = \int_0^1 x^2 \cdot 1 \, dx = \left. \frac{x^3}{3} \right|_0^1 = \frac{1}{3}$$

$$\begin{aligned} \text{Var}[X] &= E[X^2] - (E[X])^2 \\ &= \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \end{aligned}$$

General Mean

5

To go from a Uniform(0, 1) to a general uniform,

$$\begin{aligned}
 \text{If } X &\sim \text{Uniform}(a, b) \\
 X - a &\sim \text{Uniform}(0, b - a) \\
 \frac{X - a}{b - a} &\sim \text{Uniform}(0, 1) \\
 E\left[\frac{X - a}{b - a}\right] &= \frac{1}{2} \\
 \frac{1}{b - a} (E[X] - a) &= \frac{1}{2} \\
 E[X] &= \frac{b - a}{2} + a = \frac{b + a}{2}
 \end{aligned}$$

i.e., mean = average of endpoints.

General variance

6

For the variance:

$$\begin{aligned}
 \text{Var}\left[\frac{X - a}{b - a}\right] &= \text{Var}[\text{Uniform}(0, 1)] = \frac{1}{12} \\
 \text{Var}\left[\frac{X - a}{b - a}\right] &= \frac{1}{(b - a)^2} \cdot \text{Var}[X - a] \\
 \frac{1}{12} &= \frac{1}{(b - a)^2} \cdot \text{Var}[X] \\
 \frac{(b - a)^2}{12} &= \text{Var}[X] \\
 \frac{\text{length}^2}{12} &= \text{Var}[X]
 \end{aligned}$$

Discrete vs Uniform

7

So for a continuous uniform on (a, b) ,

$$\begin{aligned}E[X] &= \frac{a+b}{2} = \text{Average of endpoints} \\ \text{Var}[X] &= \frac{(b-a)^2}{12} = \frac{(\text{Length of Interval})^2}{12}\end{aligned}$$

while for a discrete uniform on $a, a+1, \dots, b$

$$\begin{aligned}E[X] &= \frac{a+b}{2} = \text{Average of endpoints} \\ \text{Var}[X] &= \frac{(\text{Number of Possible Values})^2 - 1}{12}\end{aligned}$$

Exercise 1

8

If N is uniform on $\{7, 8, 9, 10, 11, 12, 13\}$, find the mean and variance of N .

Exercise 1

8

If N is uniform on $\{7, 8, 9, 10, 11, 12, 13\}$, find the mean and variance of N .

The key point is that the set of possible values is explicitly listed out as finite, so we have a discrete uniform.

$$\begin{aligned} E[N] &= \frac{7 + 13}{2} = 10 \\ \text{Var}[N] &= \frac{(\# \text{ possible values})^2 - 1}{12} \\ &= \frac{7^2 - 1}{12} = 4 \end{aligned}$$

Exercise 2

9

If X is uniform on $(7, 13)$, find the mean and variance of X .

Exercise 2

9

If X is uniform on $(7, 13)$, find the mean and variance of X .

Now we are uniform on a continuous interval, so have a continuous uniform.

$$E[X] = \frac{7 + 13}{2} = 10$$
$$\text{Var}[X] = \frac{(13 - 7)^2}{12} = 3$$

Continuous Uniforms: Exam Style Questions

1

Mean and Variance Review

Moments of Mixtures

Examples

Exercises

Mean and Variance Review

2

Recall that for a continuous uniform on (a, b) ,

$$\begin{aligned}E[X] &= \frac{a+b}{2} = \text{Average of endpoints} \\ \text{Var}[X] &= \frac{(b-a)^2}{12} = \frac{(\text{Length of Interval})^2}{12}\end{aligned}$$

while for a discrete uniform on $a, a+1, \dots, b$

$$\begin{aligned}E[X] &= \frac{a+b}{2} = \text{Average of endpoints} \\ \text{Var}[X] &= \frac{(\text{Number of Possible Values})^2 - 1}{12}\end{aligned}$$

But exam questions won't just involve a simple uniform distribution and instead will modify it somehow.

Moments with Mixtures

3

Key idea: Raw moments (eg. mean, 2nd moment) can be broken up into pieces.

Danger: Variance cannot be broken up into cases without an extra correction term.

$$E[X] = \sum_x x \cdot P[X = x]$$

and from the law of total probability: if $A_1, A_2 \dots$ is a list of all possible cases

$$E[X] = \sum_{\text{all } A_i} E[X \mid X \in A_i] \cdot P[X \in A_i]$$

$$E[X^2] = \sum_{\text{all } A_i} E[X^2 \mid X \in A_i] \cdot P[X \in A_i]$$

$$E[g(X)] = \sum_{\text{all } A_i} E[g(X) \mid X \in A_i] \cdot P[X \in A_i]$$

We will use this with cases based on deductibles.

Example 1

4

Losses X have a uniform distribution on $[0, 100]$.

Losses are insured with a deductible. At what level must a deductible be set in order for the expected payment to be 40% of what it would be with no deductible?

$$X \sim U(0, 100)$$

d = deductible

Y = Payment after deductible

$$Y = \begin{cases} 0 & \text{if } X \leq d \\ X - d & \text{if } X > d \end{cases}$$

$$E[X] = \frac{0 + 100}{2} = 50$$

and we need d such that $E[Y] = 0.4 \cdot 50 = 20$

Example 1: Calculus Approach

5

The slower approach is to use calculus and set up the relevant integral for $E[Y]$:

$$\begin{aligned}
 P[Y \leq y] &= P[X \leq y + d] \\
 &= \frac{y + d}{100} \\
 f_Y(y) &= \frac{1}{100} \quad \text{for } y > 0 \\
 E[Y] &= 0 \cdot P[Y = 0] + \int_0^{100-d} y \cdot \frac{1}{100} dy \\
 &= \frac{y^2}{2} \cdot \frac{1}{100} \Big|_0^{100-d} = \frac{(100 - d)^2}{200}
 \end{aligned}$$

Example 1: Mixture Approach

6

A faster way to find $E[Y]$ is:

$$\begin{aligned}
 E[Y] &= E[Y \mid X \leq d] \cdot P[X \leq d] + E[Y \mid X > d] \cdot P[X > d] \\
 &\quad \text{if } X \leq d, Y = 0 \\
 &\quad \text{if } X > d, Y \sim \text{Uniform}(d - d = 0, 100 - d) \\
 E[Y] &= 0 \cdot P[X \leq d] + \frac{100 - d}{2} \cdot P[X > d] \\
 20 &= \frac{100 - d}{2} \cdot \frac{100 - d}{100} = \frac{(100 - d)^2}{200} \\
 20 \cdot 200 &= (100 - d)^2 \\
 \sqrt{20 \cdot 200} &= 100 - d \\
 d &= 100 - \sqrt{4,000} = \boxed{36.75}
 \end{aligned}$$

Example 2

7

A homeowner insures their home against storm damage with an insurance policy with a deductible of 50 florins. In the event of storm damage, repair costs are modeled by a uniform random variable on the interval $(0, 300)$.

Find the standard deviation of the insurance payment in the event that the home receives storm damage.

$$\begin{aligned} X &= \text{loss, } Y = \text{payment} \\ \text{if } X \leq 50, Y &= 0 \\ \text{if } X > 50, Y &\sim U(50 - 50, 300 - 50) = U(0, 250) \end{aligned}$$

Example 2 Solution

8

$$\begin{aligned} E[Y] &= 0 \cdot P[X \leq 50] + \frac{0 + 250}{2} \cdot P[X > 50] \\ &= 0 + 125 \cdot \frac{300 - 50}{300} = 125 \cdot \frac{5}{6} = 104.17 \\ E[Y^2] &= E[Y^2 \mid X \leq 50] \cdot P[X \leq 50] + E[Y^2 \mid X > 50] \cdot P[X > 50] \\ &= 0 + \frac{5}{6} \cdot E[U^2, U \sim \text{Uniform}(0, 250)] \\ &\text{for any random variable } U, E[U^2] = \text{Var}[U] + (E[U])^2 \\ &= \frac{5}{6} \cdot \left[\frac{250^2}{12} + \left(\frac{250}{2} \right)^2 \right] = 17,361 \\ \text{Var}[Y] &= 17,361 - 104.17^2 = 6,510 \\ \text{SD}[Y] &= \boxed{80.7} \end{aligned}$$

Exercise 1

9

Loss amounts are uniform on $(0, 20)$, and insured with a deductible of 3 and a payment limit of 12. Find the expected payment amount on a randomly selected loss.

Exercise 1

9

Loss amounts are uniform on $(0, 20)$, and insured with a deductible of 3 and a payment limit of 12. Find the expected payment amount on a randomly selected loss.

Let X denote the loss and Y the payment.

$$Y = \begin{cases} 0 & X \leq 3 \\ (X - 3) & 3 < X \leq 15 \\ 12 & X > 3 + 12 = 15 \end{cases}$$

$$\begin{aligned} E[Y] &= 0 \cdot P[X < 3] + E[U(3 - 3, 15 - 3)] \cdot P[3 < X \leq 15] + 12 \cdot P[X > 15] \\ &= 0 + \frac{0 + 12}{2} \cdot \frac{15 - 3}{20} + 12 \cdot \frac{20 - 15}{20} \\ &= 0 + 6 \cdot 0.6 + 12 \cdot 0.25 = \boxed{6.6} \end{aligned}$$

Exercise 2

10

Loss amounts are uniform on $(0, 20)$, and insured with a deductible of 3 and a payment limit of 12. Find the variance of the payment for a randomly selected loss.

Exercise 2

10

Loss amounts are uniform on $(0, 20)$, and insured with a deductible of 3 and a payment limit of 12. Find the variance of the payment for a randomly selected loss.

Again let X denote the loss and Y the payment. We already know $E[Y] = 6.6$, so finding the 2nd moment will give us the variance.

$$\begin{aligned}
 E[Y^2 \mid 3 < X \leq 15] &= E[U^2, U \sim \text{Unif.}(0, 12)] = \text{Var}[U] + (E[U])^2 \\
 &= \frac{(12 - 0)^2}{12} + \left(\frac{0 + 12}{2}\right)^2 = 12 + 36 = 48 \\
 E[Y^2] &= E[Y^2 \mid X \leq 3] \cdot P[X \leq 3] + E[Y^2 \mid 3 < X \leq 15] \cdot P[3 < X \leq 15] \\
 &\quad + E[Y^2 \mid 15 < X] \cdot P[15 < X] \\
 &= 0 \cdot \frac{3}{20} + 48 \cdot \frac{15 - 3}{20} + 12^2 \cdot \frac{20 - 15}{20} \\
 &= 0 + 48 \cdot 0.6 + 144 \cdot 0.25 = 64.8 \\
 \text{Var}[Y] &= E[Y^2] - (E[Y])^2 = 64.8 - 6.6^2 = \boxed{21.24}
 \end{aligned}$$

Exponential random variables

1

Density and CDF

Mean and variance

Memoryless property

Example

Exercises

Density and CDF of exponentials

2

X is an *exponential random variable* with mean θ if

$$F_X(x) = 1 - e^{-x/\theta}$$
$$1 - F(x) = e^{-x/\theta}$$

Sometimes will be called a rate $\lambda = 1/\theta$ exponential instead.
Exam questions may also give density rather than name.

$$F_X(x) = 1 - e^{-x/\theta} = 1 - e^{-\lambda x}$$
$$f(x) = \frac{d}{dx} F(x) = \frac{1}{\theta} e^{-x/\theta} = \lambda e^{-\lambda x} \text{ for } x > 0$$

Exponentials often are used to model waiting times (e.g., time between hits of a web page, time between rain drops, etc.)

Mean of an exponential

3

$$f(x) = \frac{1}{\theta} e^{-x/\theta} \quad F(x) = 1 - e^{-x/\theta} \text{ for } 0 < x < \infty$$

$$P[X > x] = 1 - F(x) = e^{-x/\theta} = \text{survival function}$$

$$E[X] = \int_0^{\infty} x \cdot f(x) dx = \int_0^{\infty} P[X > x] dx$$

$$\begin{aligned} E[X] &= \int_0^{\infty} e^{-x/\theta} dx \\ &= -\theta e^{-x/\theta} \Big|_0^{\infty} \\ &= -\theta e^{-\infty} - (-\theta e^{-0}) \\ &= \theta \end{aligned}$$

Variance

4

$$\text{Variance} = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_0^{\infty} x^2 \cdot f(x) dx = \int_0^{\infty} x^2 \frac{1}{\theta} e^{-x/\theta} dx$$

Using Tabular Integration

x^2	$+$	$\frac{1}{\theta} e^{-x/\theta}$
$2x$	$-$	$- e^{-x/\theta}$
2	$+$	$\theta e^{-x/\theta}$
0	$-$	$-\theta^2 e^{-x/\theta}$
$\uparrow$		$\uparrow$
differentiate left		integrate right

Variance

5

Plugging in:

$$\begin{aligned} E(X^2) &= (x^2)(-e^{-x/\theta}) - (2x)(\theta e^{-x/\theta}) + 2(-\theta^2)(e^{-x/\theta}) \Big|_0^\infty \\ &= -0 - 0 - 0 - (-0 - 0 - 2\theta^2 e^{-0}) \\ &= 2\theta^2 \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= 2\theta^2 - \theta^2 \\ &= \theta^2 \end{aligned}$$

You won't need to rederive this. Just remember $\text{Var}(X) = [E(X)]^2$ for exponentials

Variance of Geometrics

6

If X is exponential, then $\text{Var}[X] = (E[X])^2$.

We can use this to remember the variance formula for geometrics. Geometrics are the discrete analog to exponentials, and

$$\text{If } N \sim \text{Geo}\{1, 2, 3, \dots\} \text{ then } E[N] = \frac{1}{p}, \quad \text{Var}[N] = \frac{(1-p)}{p^2}$$

$$\text{If } N \sim \text{Geo}\{0, 1, 2, \dots\} \text{ then } E[N] = \frac{1-p}{p}, \quad \text{Var}[N] = \frac{(1-p)}{p^2}$$

For a geometric, instead of $\text{Variance} = \text{Mean}^2$ we have

$$\text{Var}[\text{Geo.}] = E[\text{Geo. starting at 0}] \cdot E[\text{Geo. starting at 1}]$$

Memoryless property

7

Suppose that X is an exponential random variable with mean θ

$$P[X > x] = e^{-x/\theta}$$

What is $P[X > x + a \mid X > a]$?

$$\begin{aligned} P[X > x + a \mid X > a] &= \frac{P[X > x + a, X > a]}{P[X > a]} \\ &= \frac{e^{-(x+a)/\theta}}{e^{-a/\theta}} \\ &= e^{(-x-a)/\theta} e^{a/\theta} = e^{-x/\theta} \end{aligned}$$

i.e., $P[X > x + a \mid X > a] = P[X > x]$

Applications of the Memoryless Property

8

Since $P[X > x + a \mid X > a] = P[X > x]$, we have $P[X - a > x \mid X > a] = P[X > x]$.

In other words, given that $X > a$, $X - a$ has the same distribution as the original variable X . For example, if the time between buses is exponential with mean 15 minutes, the amount of time I need to wait ($X - a$) is an exponential with mean 15 minutes no matter how long it has been (a minutes) since the last bus.

For us, key application is payment amounts $X - d$ with a deductible d conditioned on a payment being made (i.e., given $X > d$) have same distribution as losses X .

$$\begin{aligned} E[X - a \mid X > a] &= E[X] = \theta \\ E[X \mid X > a] &= E[X - a \mid X > a] + a = \theta + a \\ \text{Var}[X \mid X > a] &= \text{Var}[X - a \mid X > a] = \text{Var}[X] = \theta^2 \end{aligned}$$

Example

9

Loss amounts are exponential with rate 0.02. If losses are insured with a deductible of 10, find the probability of a loss exceeding 40 given that a positive payment is made.

Let X denote the loss amount. X is exponential with mean $\theta = 1/\lambda = 1/0.02 = 50$.

$$\begin{aligned} P[X > 40 \mid X > 10] &= P[X - 10 > 30 \mid X > 10] \\ &= P[X > 30] \text{ by memoryless property} \\ &= e^{-30/\theta} = e^{-3/5} \end{aligned}$$

Note: Exponentials are the only continuous distribution that follow the memoryless property.

Exercise 1

10

Losses are exponential with mean 50, and are insured with a deductible of 10. Find the median loss amount given that a positive payment is made.

Exercise 1

10

Losses are exponential with mean 50, and are insured with a deductible of 10. Find the median loss amount given that a positive payment is made.

We want $P[X \leq x \mid X > 10] = 0.5$. We know $(X - 10 \mid X > 10) \sim \text{Exponential}(50)$

$$P[X \leq x \mid X > 10] = 1 - P[X > x \mid X > 10]$$

$$0.5 = 1 - P[X - 10 > x - 10 \mid X > 10]$$

$$1 - 0.5 = P[X > x - 10] = e^{-(x-10)/50}$$

$$\ln(0.5) = \frac{-(x - 10)}{50}$$

$$x = -50 \ln(0.5) + 10$$

$$= 44.66$$

Exercise 2

11

Losses have density $f(x) = 0.1e^{-0.1x}$ for $x > 0$, and 0 otherwise. If losses are insured with a deductible of 3, find the expected payment for a randomly selected loss.

Exercise 2

11

Losses have density $f(x) = 0.1e^{-0.1x}$ for $x > 0$, and 0 otherwise. If losses are insured with a deductible of 3, find the expected payment for a randomly selected loss.

Let X denote our loss, and Y the payment. X is exponential with mean $1/0.1 = 10$, and

$$\begin{aligned} E[Y] &= E[Y \mid X \leq 3] \cdot P[X \leq 3] + E[Y \mid X > 3] \cdot P[X > 3] \\ &= 0 \cdot (1 - e^{-3/10}) + E[X - 3 \mid X > 3] \cdot e^{-3/10} \\ &= 0 + 10 \cdot e^{-3/10} \\ &= \boxed{7.41} \end{aligned}$$

Gamma, Exponential & Poisson

1

Density and Link with Exponential

The Gamma CDF

Mean and Variance

Examples

Exercises

General α (Optional)

The Gamma CDF – Poisson Connection (Optional)

The Gamma Density

2

Suppose that $X \sim \exp(\theta)$. Then

$$f(x) = \frac{1}{\theta} \cdot e^{-x/\theta} \quad \text{for } x > 0$$

the $1/\theta$ in front is so $\int_0^{\infty} f(x) dx = 1$

To generalize this, we can add a factor of $x^{\alpha-1}$ in front of the exponential, so

$$f(x) = c x^{\alpha-1} e^{-x/\theta}$$

where c is chosen to make this a density. The resulting distribution is called a $\text{Gamma}(\alpha, \theta)$ distribution.

When α is an integer, it turns out that a Gamma is the sum of α iid exponentials.

The Gamma Density: Integer α

3

Key points to remember: $\alpha = 1$ is an exponential. Lead term has factor of $x^{\alpha-1}$.

Unlikely to need $\alpha = 3$. Shouldn't need $\alpha > 3$. For $x > 0$,

$$\alpha = 1 : \quad f(x) = \frac{1}{\theta} e^{-x/\theta}$$

$$\alpha = 2 : \quad f(x) = \frac{x}{\theta^2} e^{-x/\theta}$$

$$\alpha = 3 : \quad f(x) = \frac{1}{2} \cdot \frac{x^2}{\theta^3} e^{-x/\theta}$$

General form: for $x > 0$, α an integer:

$$f(x) = \frac{1}{(\alpha - 1)!} \cdot \frac{x^{\alpha-1}}{\theta^\alpha} e^{-x/\theta}$$

The Gamma CDF

4

If $\alpha = 1$ then X is exponential, so CDF is $F(x) = 1 - e^{-x/\theta}$.

For other integer α , the CDF is still nice.

Again, unlikely to need $\alpha > 2$ and would be shocking to need $\alpha > 3$.

$$\alpha = 1 : \quad F(x) = 1 - e^{-x/\theta}$$

$$\alpha = 2 : \quad F(x) = 1 - e^{-x/\theta} - \frac{x}{\theta} e^{-x/\theta}$$

$$\alpha = 3 : \quad F(x) = 1 - e^{-x/\theta} - \frac{x}{\theta} e^{-x/\theta} - \left(\frac{x}{\theta}\right)^2 \cdot \frac{1}{2} e^{-x/\theta}$$

$$P[X < x] = P[X \leq x] = F(x) = 1 - \sum_{i=0}^{\alpha-1} P[\text{Poisson}(x/\theta) = i]$$

$$P[X \geq x] = P[X > x] = \sum_{i=0}^{\alpha-1} P[\text{Poisson}(x/\theta) = i]$$

Mean and Variance

5

Let $X \sim \exp(\theta)$ and $Y \sim \text{Gamma}(\alpha, \theta)$

If α is an integer, then Y is sum of α iid $\exp(\theta)$ variables.

$$\begin{aligned} E[Y] &= \alpha \cdot E[X] = \alpha\theta \\ \text{Var}[Y] &= \alpha \cdot \text{Var}[X] = \alpha\theta^2 \end{aligned}$$

If α is an integer, these formulas are basic properties of sums.

If α is not an integer, it turns out that they still hold.

Example 1

6

If X is Gamma distributed with mean 10 and variance 50, find $P[X > 10]$.

$$\begin{aligned} E[X] &= \alpha\theta = 10 \\ \text{Var}[X] &= \alpha\theta^2 = 50 \\ \frac{\text{Var}[X]}{E[X]} &= \theta = 5 \\ \alpha &= \frac{10}{5} = 2 \\ P[X > 10] &= e^{-10/\theta} + \frac{10}{\theta} e^{-10/\theta} \\ &= e^{-2} + 2e^{-2} \\ &= \boxed{0.406} \end{aligned}$$

Example 2

7

A company has two electric generators. The time until failure for each generator follows an exponential distribution with mean 10. The company will begin using the second generator immediately after the first one fails.

What is the probability that the total time that the generators produce electricity is less than 30 hours?

$$X_1 \sim \exp(10), \quad X_2 \sim \exp(10)$$

$$Y = X_1 + X_2 \sim \text{Gamma}(\alpha = 2, \theta = 10)$$

$$\text{Want: } P[Y \leq 30]$$

$$= P\left[\text{Poisson}\left(\frac{30}{10}\right) \geq 2\right]$$

$$= 1 - e^{-3} - 3e^{-3}$$

$$= 0.80$$

Exercise 1

8

A random variable X has density $f(x) = 4xe^{-2x}$ for $0 < x < \infty$, and 0 otherwise. Find the variance of X .

Exercise 1

8

A random variable X has density $f(x) = 4xe^{-2x}$ for $0 < x < \infty$, and 0 otherwise. Find the variance of X .

The density is of the form (constant)·(variable to a power)·exp(- c · variable).

That makes it a Gamma. We get α from the power, θ from what is in the exponential.

$$\text{Looking at the power: } x = x^1 = x^{\alpha-1}$$

$$1 = \alpha - 1$$

$$\alpha = 2$$

$$\text{Looking inside exponent: } -2x = -x/\theta$$

$$\theta = 1/2$$

$$\text{Var}[X] = \alpha\theta^2 = 2 \cdot \left(\frac{1}{2}\right)^2 = \frac{1}{2}$$

Exercise 2

9

An insured has 3 losses. If loss amounts are independent and exponentially distributed with mean 5, find the probability that the sum of the 3 losses is no more than 11.2.

Exercise 2

9

An insured has 3 losses. If loss amounts are independent and exponentially distributed with mean 5, find the probability that the sum of the 3 losses is no more than 11.2.

The sum of independent exponentials is a Gamma.

$\alpha = 3 = \#$ of variables in sum.

$\theta = 5 =$ mean of each exponential.

Total is Gamma($\alpha = 3, \theta = 5$).

$$F(11.2) = 1 - e^{-11.2/5} - \frac{11.2}{5} e^{-11.2/5} - \frac{1}{2} \left(\frac{11.2}{5} \right)^2 e^{-11.2/5} = 0.388$$

General α (Optional)

10

X is a gamma (α, θ) random variable if for $x > 0$ the density is

$$\begin{aligned} & \frac{1}{(\alpha - 1)!} \cdot \frac{x^{\alpha-1}}{\theta^\alpha} e^{-x/\theta} \quad \text{for } \alpha \text{ an integer} \\ & \frac{1}{\Gamma(\alpha)} \cdot \frac{x^{\alpha-1}}{\theta^\alpha} e^{-x/\theta} \quad \text{for general } \alpha \end{aligned}$$

where $\Gamma(\alpha)$ is the number such that $\int_0^\infty f(x) dx = 1$

Facts about $\Gamma(\alpha)$:

1. $\Gamma(\alpha) = (\alpha - 1) \cdot \Gamma(\alpha - 1)$ if $\alpha - 1 > 0$
2. $\Gamma(\alpha) = (\alpha - 1)!$ for α a positive integer
3. $\Gamma(1/2) = \sqrt{\pi}$, $\Gamma(3/2) = (3/2 - 1) \cdot \Gamma(1/2) = \frac{1}{2}\sqrt{\pi}$, etc.

Exponential Waiting Times

11

How else can we visualize the Poisson term in the CDF?

The following is not on the exam syllabus, but may be helpful for remembering the equations.

$X \sim \exp(\theta)$ often used to model time between visits to a webpage

Exponential Waiting Times

12

Suppose $N = \#$ of visits by time t .

$X_1 =$ time of first visit

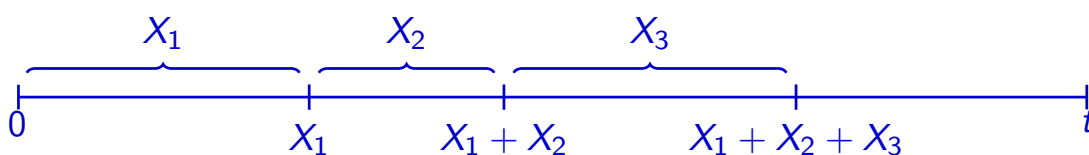
$X_1 + X_2 =$ time of 2nd visit

$X_1 + \cdots + X_n =$ time of n^{th} visit

$\theta =$ average time between visits

Intuitively, $E(N) = \frac{t}{\theta}$

In fact, $N \sim \text{Poisson}\left(\frac{t}{\theta}\right)$



First 2 Waiting Times

13

Claim: $N \sim \text{Poisson}\left(\frac{t}{\theta}\right)$, that is N has mean $\lambda = \frac{t}{\theta}$

$$\begin{aligned} P[N = 0] &= P[\text{No hits before time } t] \\ &= P[\text{first hit after time } t] \\ &= P[X > t] = e^{-t/\theta} = e^{-\lambda} \end{aligned}$$

$$P[N = 1] = P[\text{Exactly 1 visit by time } t]$$

Integrating over possible cases x for when that visit occurred:

$$\begin{aligned} P[N = 1] &= \int_0^t f(x) \cdot P[X_2 > t - x] dx = \int_0^t \frac{1}{\theta} e^{-x/\theta} e^{-(t-x)/\theta} dx \\ &= \int_0^t \frac{1}{\theta} e^{-t/\theta} dx = \frac{t}{\theta} e^{-t/\theta} = \lambda e^{-\lambda} \end{aligned}$$

General Result

14

So the claim holds for $P[N = 0]$ and $P[N = 1]$.

To complete the proof, you can proceed by induction. We have seen the base case and integration by parts goes from n to $n + 1$.

Neither mathematical induction nor Poisson processes are on syllabus, so we will omit the details.

Beta and Pareto Distributions

1

Syllabus Coverage

Beta

Pareto

Single Parameter Pareto

Exercises

Syllabus Coverage

2

Beta and Pareto distributions are explicitly listed on syllabus.

But recommended readings aren't consistent on notation for Beta or even definition of Pareto.

Historically, both have been tested purely by having density given to avoid ambiguities. Presumably this will continue in future.

Point: All problems can be done from first principles, material in this lesson is optional

Beta

3

X is Beta(a, b) if $f(x) = cx^{a-1}(1-x)^{b-1}$ for $0 < x < 1$, and 0 otherwise.

Don't need to memorize c b/c density must be given.

$$\text{Fact: } c = \frac{(a+b-1)!}{(a-1)!(b-1)!}$$

Key idea: Most likely $b = 2$ or 3 , can expand $(1-x)^{b-1}$ in doing integrals.

Moments (Optional):

$$E[X] = \frac{a}{a+b}$$

$$E[X^2] = \frac{a(a+1)}{(a+b)(a+b+1)}$$

Easy to do from scratch, tested rarely, not worth memorizing anything.

Beta Example

4

Find $E[X]$ given $f(x) = 6x(1-x)$ for $0 < x < 1$ and 0 otherwise.

$$\begin{aligned} E[X] &= \int_0^1 x \cdot f(x) \, dx \\ &= \int_0^1 x \cdot 6x(1-x) \, dx \\ &= \int_0^1 6x^2 - 6x^3 \, dx \\ &= \frac{6}{3} - \frac{6}{4} \\ &= \frac{6}{12} = \frac{1}{2} \end{aligned}$$

Or $a = b = 2$ and $E[X] = a/(a+b) = 2/(2+2) = 1/2$

Pareto Distributions

5

Suppose $X \geq 0$ but unbounded.

We need $\int_0^\infty f(x) dx = 1$, which requires $\lim_{x \rightarrow \infty} f(x) = 0$.

An exponential distribution does this with $f(x) = \lambda e^{-\lambda x}$

But x^{-p} also goes to 0. Can it be basis for a density?

Problem: x^{-p} is horrible at 0.

2 solutions to avoid division by 0 at 0.

- Have $f(x) \propto (x + \theta)^{-p}$ for $x > 0$, and 0 otherwise
- Have $f(x) \propto x^{-p}$ but require $x > \theta > 0$

Some texts call first distribution Pareto, second single parameter Pareto. Others call second distribution Pareto, don't consider first. We will use Pareto / Single parameter Pareto names.

Pareto Distribution

6

For $\alpha > 0, \theta > 0$, X is Pareto(α, θ) if $f(x) = 0$ for $x < 0$ and for $x > 0$,

$$f(x) = \frac{\alpha \theta^\alpha}{(x + \theta)^{\alpha+1}}$$

Don't need to memorize! Density or CDF must be given.

Have seen examples in previous lessons.

If $\alpha > 1$ then

$$\begin{aligned} 1 - F(x) &= \int_x^\infty f(t) dt = \int_x^\infty \theta^\alpha \cdot \frac{\alpha}{(t + \theta)^{\alpha+1}} dt \\ &= \theta^\alpha \cdot \frac{-1}{(t + \theta)^\alpha} \Big|_x^\infty = \frac{\theta^\alpha}{(x + \theta)^\alpha} \\ E[X] &= \int_0^\infty [1 - F(x)] dx = \int_0^\infty \frac{\theta^\alpha}{(x + \theta)^\alpha} dx \\ &= \frac{-\theta^\alpha}{(\alpha - 1)(x + \theta)^{\alpha-1}} \Big|_0^\infty = \frac{\theta}{\alpha - 1} \end{aligned}$$

Single Parameter Pareto

7

A single parameter Pareto avoids division by 0 by starting at $\theta > 0$.

X is a single parameter Pareto(α, θ) if for $x > \theta$,

$$f(x) = \frac{\alpha\theta^\alpha}{x^{\alpha+1}}$$

and $f(x) = 0$ for $x < \theta$.

The simpler denominator makes calculations easier. E.g.,

$$\begin{aligned} E[X] &= \int_{\theta}^{\infty} x \cdot \frac{\alpha\theta^\alpha}{x^{\alpha+1}} dx \\ &= \int_{\theta}^{\infty} \frac{\alpha\theta^\alpha}{x^\alpha} dx \\ &= \frac{-\alpha}{\alpha-1} \cdot \frac{\theta^\alpha}{x^{\alpha-1}} \Big|_{\theta}^{\infty} = \frac{\alpha\theta}{\alpha-1} \end{aligned}$$

Exercise 1

8

If $f(x) = 12x^2(1-x)$ for $0 < x < 1$ and 0 otherwise, find $\text{Var}[X]$

Exercise 1

8

If $f(x) = 12x^2(1 - x)$ for $0 < x < 1$ and 0 otherwise, find $\text{Var}[X]$

$$\begin{aligned}
 E[X] &= \int_0^1 x \cdot 12x^2(1 - x) dx = \int_0^1 12x^3 - 12x^4 dx \\
 &= \frac{12}{4} - \frac{12}{5} = \frac{12}{20} = \frac{3}{5} \\
 E[X^2] &= \int_0^1 x^2 \cdot 12x^2(1 - x) dx = \int_0^1 12x^4 - 12x^5 dx \\
 &= \frac{12}{5} - \frac{12}{6} = \frac{12}{30} = \frac{2}{5} \\
 \text{Var}[X] &= E[X^2] - (E[X])^2 \\
 &= \frac{2}{5} - \left(\frac{3}{5}\right)^2 = \frac{1}{25}
 \end{aligned}$$

Exercise 1: Alternate Solution

9

Alternatively, $f(x) = 12x^2(1 - x)$ is a Beta with $x^{a-1} = x^2$ so $a - 1 = 2$ and $a = 3$.
 $(1 - x)^{b-1} = (1 - x)^1$ so $b = 2$.

$$\begin{aligned}
 E[X] &= \frac{a}{a + b} \\
 &= \frac{3}{3 + 2} = \frac{3}{5} \\
 E[X^2] &= \frac{a(a + 1)}{(a + b)(a + b + 1)} \\
 &= \frac{3 \cdot 4}{5 \cdot 6} = \frac{12}{30} = \frac{2}{5} \\
 \text{Var}[X] &= \frac{2}{5} - \left(\frac{3}{5}\right)^2 \\
 &= \frac{2}{5} - \frac{9}{25} = \frac{1}{25}
 \end{aligned}$$

Exercise 2

10

Y has density $f(y) = \frac{2 \cdot 100^2}{(y + 100)^3}$ for $0 < y < \infty$, and $f(y) = 0$ otherwise. Find the 75th percentile of Y .

Exercise 2

10

Y has density $f(y) = \frac{2 \cdot 100^2}{(y + 100)^3}$ for $0 < y < \infty$, and $f(y) = 0$ otherwise. Find the 75th percentile of Y .

Let t be the 75th percentile. So $F(t) = 0.75$ and $1 - F(t) = 1 - 0.75 = 0.25$.

$$\begin{aligned}
 1 - F(t) &= \int_t^{\infty} f(y) dy \\
 0.25 &= \int_t^{\infty} \frac{2 \cdot 100^2}{(y + 100)^3} dy \\
 0.25 &= \frac{-100^2}{(y + 100)^2} \Big|_t^{\infty} = \frac{100^2}{(t + 100)^2} \\
 0.5 &= \frac{100}{t + 100} \\
 0.5t + 50 &= 100 \\
 t &= \boxed{100}
 \end{aligned}$$

Normal Distribution

1

Density (not tested)

The cdf (very important) Using tables

Linear Interpolation

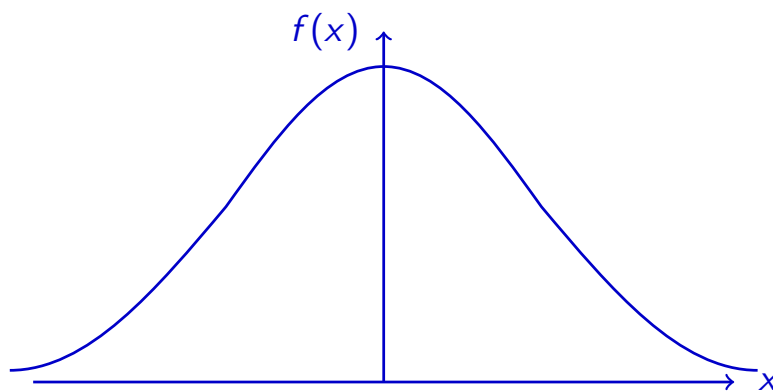
SOA # 80

Sums of independent normals

SOA # 18

The normal density

2



For a standard normal, $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$

↑ makes std. deviation is 1

↑ makes $\int_{-\infty}^{\infty} f(x) dx = 1$

Other Normals

3

A standard normal Z has mean 0 and variance 1.

What if we want Y to be normal with mean μ and variance σ^2 ?

We can construct such a Y by rescaling Z :

$$\begin{aligned} Z &\sim N(0, 1) \\ Y &= \sigma \cdot Z + \mu \\ E[Y] &= \sigma \cdot E[Z] + \mu = \mu \\ \text{Var}[Y] &= \text{Var}[\sigma Z] = \sigma^2 \cdot \text{Var}[Z] = \sigma^2 \end{aligned}$$

It turns out that Y is still normal, so

$$Y = \sigma Z + \mu \sim \text{Normal}(\mu, \sigma^2)$$

Other Normals

4

What is the density of $Y \sim N(\mu, \sigma^2)$? (Optional)

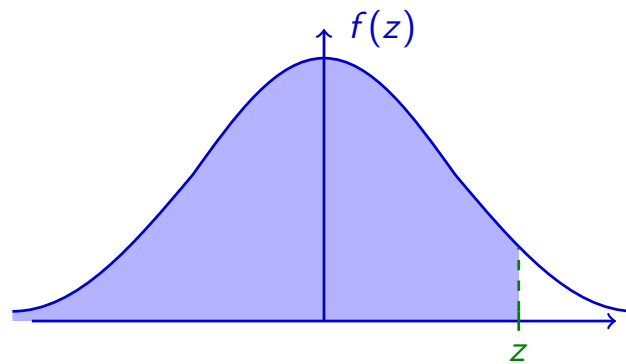
$$\begin{aligned} f_Z(z) &= \frac{1}{\sqrt{2\pi}} \cdot e^{-z^2/2} \\ Y &= \sigma Z + \mu \quad y = g(z) = \sigma z + \mu \\ z &= g^{-1}(y) = \frac{y - \mu}{\sigma} \end{aligned}$$

If $Y = g(Z)$

$$\begin{aligned} f_Y(y) &= f_Z(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| \\ f_Y(y) &= \frac{1}{\sqrt{2\pi}} \left[\exp \left\{ -\frac{1}{2} \left(\frac{y - \mu}{\sigma} \right)^2 \right\} \right] \cdot \frac{1}{\sigma} \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/(2\sigma^2)} \end{aligned}$$

The Normal cdf

5



Suppose that Z is a standard normal, so

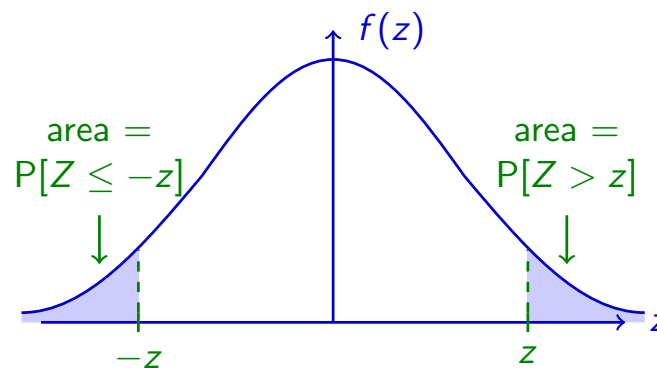
$$Z \sim \text{Normal}(0, 1)$$

$$\Phi(z) = P[Z \leq z]$$

denotes the CDF of Z (so it is the shaded area). Fact: No elementary formula for $\Phi(z)$ exists, so we will look it up on tables.

The Normal cdf

6



Tables for $\Phi(z)$ only include $z \geq 0$. To find the cdf for negative values, we compare $\Phi(-z)$ with $\Phi(z)$.

$$P[Z > z] = P[Z \leq -z]$$

$$1 - \Phi(z) = \Phi(-z)$$

NORMAL DISTRIBUTION TABLE

7

Entries represent the area under the standardized normal distribution from $-\infty$ to z , i.e., $\Phi(z) = \Pr(Z \leq z)$ is the cdf. The value of z to the first decimal is given in the left column. The second decimal place is given in the top row.

z	0.00	0.01	0.02	0.03	0.04	...
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	
$\vdots$			$\vdots$			

$$\Phi(0.12) = 0.5478$$

$$\begin{aligned}\Phi(-0.33) &= 1 - \Phi(0.33) \\ &= 1 - 0.6293\end{aligned}$$

Probabilities Example

8

X is normal with mean $\mu = 2$ and variance $\sigma^2 = 9$.

Find $P[X > 3.86]$, $P[X > 1.49]$ and the 95th percentile of X .

Idea: $X = \sigma Z + \mu$ for a standard normal Z , so $Z = (X - \mu)/\sigma$.

Will convert to Z and use tables.

$$\begin{aligned}P[X > 3.86] &= P\left[\frac{X - \mu}{\sigma} > \frac{3.86 - 2}{3}\right] = P[Z > 0.62] \\ &= 1 - \Phi(0.62) \\ &= 1 - 0.7324 = 0.2676\end{aligned}$$

$$\begin{aligned}P[X > 1.49] &= P\left[\frac{X - \mu}{\sigma} > \frac{1.49 - 2}{3}\right] = P[Z > -0.17] \\ &= 1 - \Phi(-0.17) = \Phi(0.17) \\ &= 0.5675\end{aligned}$$

Percentile Example

9

X is normal with mean $\mu = 2$ and variance $\sigma^2 = 9$. Find the 95th percentile of X .

First, let's find z , the 95th percentile of Z , then convert to t , the 95th percentile of X .

$$0.95 = P[Z \leq z] = \Phi(z)$$

$$z = \Phi^{-1}(0.95)$$

$$z = 1.6449$$

$$X = \sigma Z + \mu$$

$$t = \sigma z + \mu = 3 \cdot 1.6449 + 2$$

$$t = \boxed{6.935}$$

Note the list of select inverse values at bottom of table.

Sums of Normals

10

If X and Y are independent normals, then it turns out that $X + Y$ is also normal.

$$E(X + Y) = E(X) + E(Y)$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

$$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

If we want averages,

$$\frac{X + Y}{2} \sim \text{Normal} \left(\frac{E(X) + E(Y)}{2}, \frac{1}{4}(\text{Var}(X) + \text{Var}(Y)) \right)$$

More generally, for any c , we have

$$cX \sim \text{Normal}(cE(X), c^2 \text{Var}(X))$$

Exercise 1

11

X is normal with mean -2.47 and variance 1.69 . Find $P[X > 0]$

Exercise 1

11

X is normal with mean -2.47 and variance 1.69 . Find $P[X > 0]$

$$\begin{aligned} P[X > 0] &= P\left[\frac{X - \mu}{\sigma} > \frac{0 - (-2.47)}{\sqrt{1.69}}\right] \\ &= P[Z > 1.9] \\ &= 1 - \Phi(1.9) \\ &= 1 - 0.9713 \\ &= \boxed{0.0287} \end{aligned}$$

To avoid sign errors, can do reasonability check. We want the probability that X is much bigger than its mean, so should have a small value as the answer.

Exercise 2

12

If X and Y are independent normal random variables with $E[X] = 0.6$, $E[Y] = 3.2$, $\text{Var}[X] = 1.08$ and $\text{Var}[Y] = 1.17$, find the 75th percentile of the average of X and Y .

Exercise 2

12

If X and Y are independent normal random variables with $E[X] = 0.6$, $E[Y] = 3.2$, $\text{Var}[X] = 1.08$ and $\text{Var}[Y] = 1.17$, find the 75th percentile of the average of X and Y .

Let $W = (X + Y)/2$ and t the 75th percentile of W . Also let Z be a standard normal, and z the 75th percentile of Z .

$$\begin{aligned}
 E[W] &= \frac{E[X] + E[Y]}{2} = \frac{0.6 + 3.2}{2} = 1.9 \\
 \text{Var}[W] &= \text{Var}\left[\frac{X + Y}{2}\right] = \frac{1}{4}\text{Var}[X + Y] \\
 \text{Var}[W] &= \frac{1}{4}(1.08 + 1.17) = \frac{2.25}{4} \\
 \text{SD}[W] &= \sqrt{2.25/4} = 0.75 \\
 z &= 0.6745 \\
 t &= \text{SD}[W] \cdot z + E[W] = 0.75 \cdot 0.6745 + 1.9 = \boxed{2.4}
 \end{aligned}$$

Interpolation

1

Basic Idea

Linear Interpolation

Inverse Values

Exercises

Example

2

Let $X \sim N(\mu = 4, \sigma^2 = 2.2)$. Find $P[X > 3.3]$ to the nearest thousandth.

$$\begin{aligned} P[X > 3.3] &= P\left[\frac{X - \mu}{\sigma} > \frac{3.3 - 4}{\sqrt{2.2}}\right] \\ &= 1 - P\left[Z \leq \frac{-0.7}{\sqrt{2.2}}\right] \\ &= 1 - \Phi(-0.472) = \Phi(0.472) \end{aligned}$$

In our tables, $\Phi(0.47) = 0.6808$ and $\Phi(0.48) = 0.6844$.

What is $\Phi(0.472)$?

To nearest hundredth, it is 0.68.

To nearest thousandth, it is at least 0.681, at most 0.684, but can't immediately tell.

Exam Rules

3

Question wants $\Phi(0.472)$ to the nearest thousandth.

Know $0.6808 = \Phi(0.47) < \Phi(0.472) < \Phi(0.48) = 0.6844$

True value (via computer, not tables) $\Phi(0.472) = 0.6815 \approx 0.682$

Exam rule: Out of values that are possible based on tables, here 0.681, 0.682, 0.683 and 0.684, the only one included as an answer choice will be true value of 0.682.

Exam strategy options:

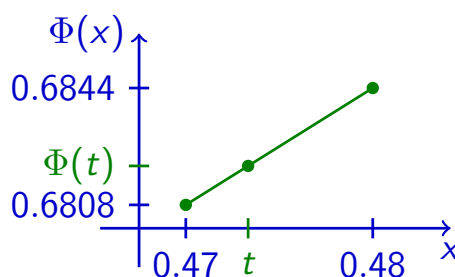
1. Accept that you may not match exam choices on normal distribution questions
2. Guesstimate. 0.472 is closer to 0.47 than 0.48. $\Phi(0.472)$ will be in the middle, but closer to $\Phi(0.47)$, so probably 0.682
3. Use linear interpolation to get better approximation

You can use any of these approaches.

Linear Interpolation Example

4

From tables, $\Phi(0.47) = 0.6808$, $\Phi(0.48) = 0.6844$. Want $\Phi(0.472)$.



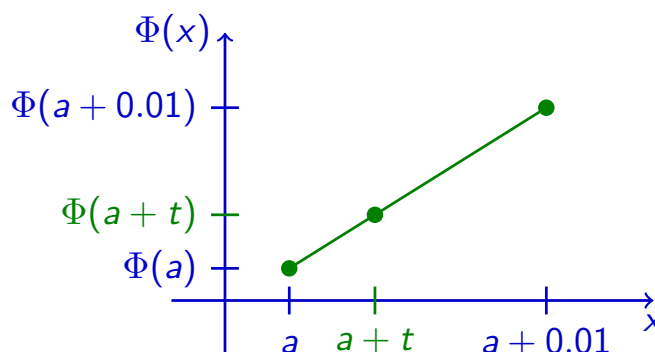
Comparing slopes:

$$\begin{aligned}\frac{\Phi(t) - 0.6808}{t - 0.47} &\approx \frac{0.6844 - 0.6808}{0.48 - 0.47} \\ \Phi(0.472) &\approx \frac{0.472 - 0.47}{0.48 - 0.47} \cdot (0.6844 - 0.6808) + 0.6808 \\ \Phi(0.472) &\approx 0.2(0.6844 - 0.6808) + 0.6808 = 0.6815\end{aligned}$$

Linear Interpolation: Formulas

5

From tables, have $\Phi(a)$ and $\Phi(a + 0.01)$. Want $\Phi(a + t)$.



$$\frac{\Phi(a + t) - \Phi(a)}{(t + a) - a} = \frac{\Phi(a + 0.01) - \Phi(a)}{0.01}$$

$$\Phi(a + t) = \Phi(a) + \frac{t}{0.01} [\Phi(a + 0.01) - \Phi(a)]$$

Percentile Example

6

Still have $X \sim N(\mu = 4, \sigma^2 = 2.2)$. Find the 13th percentile of X .

For a standard normal, the 13th percentile z is:

$$z = \Phi^{-1}(0.13) = -\Phi^{-1}(1 - 0.13) = -\Phi^{-1}(0.87).$$

From tables, $\Phi(1.12) = 0.8686$, $\Phi(1.13) = 0.8708$, so $-1.13 < z < -1.12$.

Let t denote the 13th percentile of X

$$X = \sigma Z + \mu = \sqrt{2.2}Z + 4$$

$$t = \sqrt{2.2}z + 4$$

$$-1.13\sqrt{2.2} + 4 < t < -1.12\sqrt{2.2} + 4$$

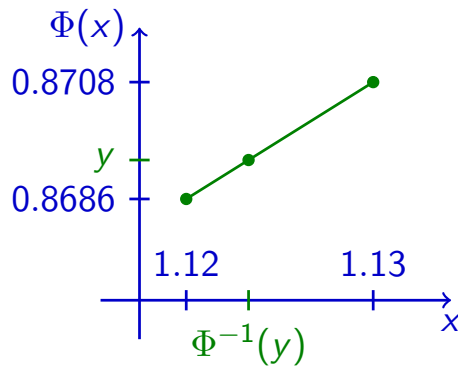
$$2.324 < t < 2.339$$

$\Phi(1.13) = 0.8708$ is closer to 0.87 than $\Phi(1.12) = 0.8686$, so exact value is closer to 2.324, which corresponds to $z = 1.13$, than 2.339. Exact value is around 2.33.

Inverse Values: Linear Interpolation

7

We want $t = 4 - \sqrt{2.2} \cdot \Phi^{-1}(0.87)$



$$\frac{y - 0.8686}{\Phi^{-1}(y) - 1.12} \approx \frac{0.8708 - 0.8686}{1.13 - 1.12}$$

$$\frac{0.87 - 0.8686}{\Phi^{-1}(0.87) - 1.12} \approx \frac{0.8708 - 0.8686}{1.13 - 1.12}$$

$$\frac{0.87 - 0.8686}{0.8708 - 0.8686} = \frac{\Phi^{-1}(0.87) - 1.12}{1.13 - 1.12}$$

$$\frac{14}{22} = \frac{\Phi^{-1}(0.87) - 1.12}{0.01}$$

$$\Phi^{-1}(0.87) = 1.12 + \frac{14}{22} \cdot 0.01 = 1.1264$$

$$t = 4 - \sqrt{2.2} \cdot 1.1264 = 2.329$$

Exercise 1

8

If X is normal with mean 0.6 and variance 1.3, find $P[X \leq 0.8]$

Exercise 1

8

If X is normal with mean 0.6 and variance 1.3, find $P[X \leq 0.8]$

$$\begin{aligned}P[X \leq 0.8] &= P\left[\frac{X - \mu}{\sigma} \leq \frac{0.8 - 0.6}{\sqrt{1.3}}\right] \\&= P[Z \leq 0.1754] \\P[Z \leq 0.17] &\approx 0.5675 \\P[Z \leq 0.18] &\approx 0.5714 \\P[Z \leq 0.1754] &\approx 0.5675 + 0.54(0.5714 - 0.5675) \\&= 0.5696\end{aligned}$$

Exercise 2

9

Find the 60th percentile of X if $E[X] = 81.2$ and $\text{Var}[X] = 33.8$.

Exercise 2

9

Find the 60th percentile of X if $E[X] = 81.2$ and $\text{Var}[X] = 33.8$.

Let z denote the 60th percentile of a standard normal, and t the 60th percentile of X .

$$z = \Phi^{-1}(0.6)$$

$$\Phi^{-1}(0.5987) = 0.25$$

$$\Phi^{-1}(0.6026) = 0.26$$

$$\begin{aligned}\Phi^{-1}(0.6) &= 0.25 + \frac{0.6 - 0.5987}{0.6026 - 0.5987}(0.26 - 0.25) \\ &= 0.25 + \frac{13}{39} \cdot 0.01 = 0.25333\end{aligned}$$

$$x = \sigma z + \mu$$

$$= \sqrt{33.8} \cdot z + 81.2 = 82.67$$

The Central Limit Theorem

1

Basic CLT

Normal Approximation

Example

Averages

Exercises

CLT: Central Limit Theorem

2

Key idea: Sums of a large number of random variables are often approximately normal

Theorem (CLT: Exam version)

If $X_1, \dots, X_n$ are iid random variables, then

$$\frac{(X_1 + \dots + X_n) - n E[X_1]}{\sqrt{n \operatorname{Var}[X_1]}} \Rightarrow N(0, 1)$$

Alternatively, let $S_n = X_1 + \dots + X_n$

$$E[S_n] = nE[X_1] = n\mu$$

$$\operatorname{SD}[S_n] = \sqrt{n \operatorname{Var}[X_1]} = \sigma\sqrt{n}$$

$$\frac{S_n - n\mu}{\sqrt{n}\sigma} \Rightarrow N(0, 1)$$

Main Application: Normal Approximations

3

This lets us approximate the distribution of sums of random variables.

Let $S_n = X_1 + \cdots + X_n$. Finding the exact distribution of S_n is hard, but

$$\begin{aligned}
 P[S_n \leq x] &= P\left[\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq \frac{x - n\mu}{\sigma\sqrt{n}}\right] \\
 &\quad \begin{array}{cc} \uparrow & \uparrow \\ \text{Std normal} & \text{"z-value"} \end{array} \\
 &\approx \Phi\left(\frac{x - n\mu}{\sigma\sqrt{n}}\right)
 \end{aligned}$$

and we can find that on a normal table

Example

4

An insurance company pays claims on 625 losses. Losses are independent and exponentially distributed with mean 3. Find the approximate probability that the total payment is between 1800 and 2010.

$$\begin{aligned}
 S_n &= X_1 + \cdots + X_{625} & X_i &\sim \text{Exponential}(3) \\
 E[X_i] &= 3 & \text{Var}[X_i] &= 3^2 = 9 \\
 E[S_n] &= 625 \cdot 3 = 1,875 \\
 \text{Var}[S_n] &= 625 \cdot 9 = 5,625 \\
 \text{SD}[S_n] &= 75 \\
 P[1800 < S_n < 2010] &= P\left[\frac{1800 - 1875}{75} < \frac{S_n - E[S_n]}{\text{SD}[S_n]} < \frac{2010 - 1875}{75}\right] \\
 &= P[-1 < Z < 1.8] \\
 &= \Phi(1.8) - \Phi(-1) = \Phi(1.8) - [1 - \Phi(1)] \\
 &= 0.9641 - [1 - 0.8413] = 0.8054
 \end{aligned}$$

Sums versus Products

5

Suppose $X_1, X_2, \dots, X_{100}$ are iid random variables with $P[X_i = 1] = P[X_i = -1] = 1/2$. Think of X_i as the amount that we win in bet number i .

$100 X_1$ is the payoff if we bet \$100 on the first bet. It is either -100 or 100 .

$\sum_{i=1}^{100} X_i$ is the net payoff after all 100 bets. We will win some and lose some, so $\sum_{i=1}^{100} X_i$ will probably be close to 0. In particular, it is closer to 0 than $100X_1$ and therefore has smaller variance.

$$\text{Var}(100 X_1) = 100^2 \text{Var}(X_1) = 100^2$$

$$\text{Var}(\sum X_i) = 100 \text{Var}(X) = 100$$

Averages

6

Suppose that $X_1, \dots, X_n$ are iid random variables, and let $\bar{X}$ be their average. What is the distribution of $\bar{X}$?

$$\begin{aligned} S &= X_1 + X_2 + \dots + X_n \\ \bar{X} &= \frac{S}{n} = \frac{1}{n} (X_1 + X_2 + \dots + X_n) \\ E[\bar{X}] &= \frac{1}{n} (E(X_1) + E(X_2) + \dots + E(X_n)) \\ &= \frac{1}{n} \cdot n \cdot E[X] \\ &= E[X] \end{aligned}$$

Averages

7

$$\begin{aligned}
 \text{Var}[\bar{X}] &= \text{Var} \left[\frac{1}{n} (X_1 + \cdots + X_n) \right] \\
 &= \frac{1}{n^2} (\text{Var}(X_1) + \cdots + \text{Var}(X_n)) \\
 &= \frac{1}{n^2} \cdot n \text{Var} X \\
 &= \frac{\text{Var}(X)}{n}
 \end{aligned}$$

Since S is roughly normal, so is S/n , so we have that

$$\bar{X} \approx N\left(E(X), \frac{\text{Var}(X)}{n}\right)$$

This comes up often enough that it is worth knowing.

Exercise 1

8

Losses have mean 4 and standard deviation 3. If losses are independent, use a normal approximation to estimate the probability that the sum of 30 losses is at least 100.

Exercise 1

8

Losses have mean 4 and standard deviation 3. If losses are independent, use a normal approximation to estimate the probability that the sum of 30 losses is at least 100.

Let X_i denote a loss amount, S the sum, and Z a standard normal.

$$\begin{aligned}
 E[S] &= 30 \cdot E[X] = 30 \cdot 4 = 120 \\
 \text{Var}[S] &= 30 \cdot \text{Var}[X] = 30 \cdot 3^2 = 270 \\
 \text{SD}[S] &= \text{SD}[X]\sqrt{30} = \sqrt{270} = 16.43 \\
 P[S \geq 100] &= P\left[\frac{S - E[S]}{\text{SD}[S]} \geq \frac{100 - 120}{16.43}\right] \\
 &\approx P[Z > -1.217] = 1 - \Phi(-1.217) = \Phi(1.217) \\
 &\approx \Phi(1.21) + 0.7[\Phi(1.22) - \Phi(1.21)] \\
 &= 0.8869 + 0.7[0.8888 - 0.8869] = 0.8882
 \end{aligned}$$

Could have said 0.89 without interpolation.

Exercise 2

9

Losses are independent, each with density $0.2e^{-0.2x}$ for $x > 0$ and 0 otherwise. Losses are insured with a deductible of 5. The first 60 randomly selected positive payments are averaged. Using a normal approximation, estimate the 88th percentile of the average.

Exercise 2

9

Losses are independent, each with density $0.2e^{-0.2x}$ for $x > 0$ and 0 otherwise. Losses are insured with a deductible of 5. The first 60 randomly selected positive payments are averaged. Using a normal approximation, estimate the 88th percentile of the average.

X is exponential with mean $1/0.2 = 5$. By the memoryless property, the payments Y_i have the same distribution as the losses, so are also exponential with mean 5.

$$E[\bar{Y}] = E[Y] = 5$$

$$\text{Var}[\bar{Y}] = \text{Var}[Y]/60 = 5^2/60 = 0.4167$$

$$\text{SD}[\bar{Y}] = \sqrt{0.4167} = 0.6455$$

$$\Phi(1.17) = 0.8790$$

$$\Phi(1.18) = 0.8810$$

$$\Phi^{-1}(0.88) = 1.17 + \frac{10}{20} \cdot 0.01 = 1.175$$

$$\text{88th \%ile of } \bar{Y} = 0.6455 \cdot 1.175 + 5 = \boxed{5.76}$$

Continuity Correction

1

Example

Formulas

Example

Exercises

Example: Binomial Version

2

Suppose X is binomial with $n = 25$ and $p = 0.2$. Find:

- $E[X]$ and $\text{Var}[X]$
- Give an expression for the probability that X is at least 8
- Given an expression for the probability that X is more than 8.

$$E[X] = np = 25 \cdot 0.2 = 5$$

$$\text{Var}[X] = np(1 - p) = 25 \cdot 0.2 \cdot 0.8 = 4$$

$$P[X \geq 8] = \sum_{k=8}^{25} P[X = k] = \sum_{k=8}^{25} \binom{25}{k} 0.2^k (1 - 0.2)^{25-k}$$

$$P[X > 8] = \sum_{k=9}^{25} P[X = k] = \sum_{k=9}^{25} \binom{25}{k} 0.2^k (1 - 0.2)^{25-k}$$

By computer, $P[X \geq 8] = 0.109$ and $P[X > 8] = 0.047$

Example: Normal Version

3

Suppose W is normal with mean 5 and variance 4. Find:

- a) The probability that W is at least 8
- b) The probability that W is more than 8

a) is asking for $P[W \geq 8]$ while b) wants $P[W > 8]$.

But normals are continuous, so $P[W = 8] = 0$ and $P[W \geq 8] = P[W > 8]$.

$$\begin{aligned} P[W > 8] &= P\left[\frac{W - E[W]}{SD[W]} > \frac{8 - 5}{\sqrt{4}}\right] \\ &= 1 - \Phi(1.5) \\ &= 1 - 0.9332 \\ &= 0.0668 \end{aligned}$$

Example: Finding Approximation

4

In those examples, X and W had the same mean and variance, but

$$P[X \geq 8] = 0.1091 \text{ and } P[X > 8] = 0.0468$$

$$P[W \geq 8] = 0.0668 = P[W > 8] = 0.0668.$$

Shouldn't they be closer if we are using normals as approximations?

The problem is X must be an integer, while W does not.

W can be 7.7, 8.233, 8.6479, etc.

A better approximation to $P[X \geq 8]$ is $P[W \text{ rounds to something } \geq 8]$.

I.e., $P[X \geq 8] \approx P[W > 7.5]$. $P[X > 8] = P[W > 8.5]$, etc.

$$P[W > 7.5] = 1 - \Phi[(7.5 - 5)/2] = 0.1056$$

$$P[W > 8.5] = 1 - \Phi[(8.5 - 5)/2] = 0.0401$$

Still not perfect (b/c n is small) but much better.

Continuity Correction

5

A *continuity correction* is used with normal approximations of integer valued variables.

It corrects for the normal being continuous while the variable we care about is not.

Used when approximating a sum S_n of *discrete* variables with a normal W .

$$P[S_n \in A] \approx P[W \text{ rounds to something} \in A].$$

1. $P[S_n < x] \approx P[W < x - 0.5]$
2. $P[S_n \leq x] \approx P[W < x + 0.5]$
3. $P[S_n \geq x] \approx P[W > x - 0.5]$
4. $P[S_n > x] \approx P[W > x + 0.5]$

Try to think these through rather than memorize.

Example

6

A roulette player betting on black has an $18/38$ probability of winning on each spin of the wheel. Using a normal approximation with a continuity correction, find the approximate probability of betting on black and winning in at least 45 out of 100 spins.

$$S_n = \# \text{ wins} \sim \text{Binomial} \left(100, \frac{18}{38} \right)$$

$$E[S_n] = 100 \cdot \frac{18}{38} \approx 47.368$$

$$SD[S_n] = \sqrt{100 \cdot \frac{18}{38} \cdot \frac{20}{38}} \approx 4.993$$

$$\begin{aligned} P[S_n \geq 45] &\approx P[W > 44.5] \\ &= P \left[\frac{W - E[W]}{SD[W]} > \frac{44.5 - 47.368}{4.993} \right] \\ &= P[Z > -0.5745] = 1 - \Phi(-0.5745) \\ &= 1 - (1 - \Phi(0.5745)) = \Phi(0.5745) \approx 0.72 \end{aligned}$$

Exercise 1

7

Using a normal approximation with a continuity correction, estimate $P[X \leq 4]$ if X is Poisson with mean 2.3.

Exercise 1

7

Using a normal approximation with a continuity correction, estimate $P[X \leq 4]$ if X is Poisson with mean 2.3.

Let W be normal with the same mean and variance as X .

$$\begin{aligned} E[X] &= 2.3 \\ SD[X] &= \sqrt{\text{Var}[X]} = \sqrt{2.3} \\ P[X \leq 4] &\approx P[W < 4.5] \\ &= \Phi\left(\frac{4.5 - 2.3}{\sqrt{2.3}}\right) \\ &= \Phi(1.45) \\ &= 0.9265 \end{aligned}$$

Exercise 2

8

Using a normal approximation with a continuity correction, estimate $P[X = 4]$ if X is Poisson with mean 2.3. How does that compare to the true value?

Exercise 2

8

Using a normal approximation with a continuity correction, estimate $P[X = 4]$ if X is Poisson with mean 2.3. How does that compare to the true value?

Let W be normal with the same mean and variance as X .

$$\begin{aligned}
 E[X] &= 2.3 & SD[X] &= \sqrt{2.3} \\
 P[X = 4] &\approx P[3.5 < W < 4.5] \\
 &= P[W \leq 4.5] - P[W \leq 3.5] \\
 &= \Phi\left(\frac{4.5 - 2.3}{\sqrt{2.3}}\right) - \Phi\left(\frac{3.5 - 2.3}{\sqrt{2.3}}\right) \\
 &= \Phi(1.45) - \Phi(0.79) \\
 &= 0.9265 - 0.7852 = 0.1413 \\
 \text{Actual } P[X = 4] &= e^{-2.3} \cdot \frac{2.3^4}{4!} = 0.1169
 \end{aligned}$$

Lognormal Distributions

1

Example of lognormal

Definition of lognormal

Moments

Mean and Variance of lognormal

Exercises

Example of lognormal

2

Losses in a year have a lognormal distribution, $Y = e^X$, where X is a normal random variable with mean 3 and variance 0.5. What is the probability that losses exceed 80?

$$\begin{aligned}
 P[Y > 80] &= P[e^X > 80] \\
 &= P[X > \ln 80] = P[X > 4.38] \\
 &= P\left[\frac{X - E(X)}{\underset{\substack{\uparrow \\ \text{Std normal}}}{SD(X)}} > \frac{4.38 - 3}{\underset{\substack{\uparrow \\ \text{z-value}}}{\sqrt{0.5}}}\right] \\
 &= 1 - \Phi\left(\frac{4.38 - 3}{\sqrt{.5}}\right) \\
 &= 1 - \Phi(1.95) \\
 &= 1 - 0.97 = 0.03
 \end{aligned}$$

Lognormal Definition

3

Definition (lognormal)

Y is a lognormal if $Y = e^X$, X is normal

In words, the log of a lognormal distribution is normal.

Notes: if X is normal, then the range of X is $-\infty$ to ∞
So e^X is well-defined, but $\ln(X)$ is not.

Lognormal Moments

4

Suppose that X is a $\text{Normal}(\mu, \sigma^2)$ random variable and $Y = e^X$ is the corresponding lognormal. Then

$$E[Y] = E[e^X] = e^{\mu + (\sigma^2)/2}$$

$$E[Y] \neq e^\mu$$

$$\begin{aligned} E[Y^2] &= E[(e^X)^2] \\ &= E[e^{2X}] \\ &= e^{2\mu + 2\sigma^2} \end{aligned}$$

More generally (optional formula, may help with memorizing)

$$E[e^{tX}] = e^{t\mu + (t\sigma)^2/2}$$

Key point: For finding moments, we use these formulas about the lognormal. For finding probabilities, we take logs and work with the underlying normal.

Mean and Variance of Lognormal

5

Losses Y have a lognormal distribution with mean 50 and variance 400. What is the probability that losses exceed 75?

$$\begin{aligned}
 Y &= e^X \\
 E(Y) &= E(e^X) \\
 50 &= e^{\mu + \sigma^2/2} \\
 E(Y^2) &= 50^2 + 400 = E(e^{2X}) \\
 &= e^{2\mu + 2\sigma^2} \\
 \ln(50) &= \mu + \frac{\sigma^2}{2} \\
 \ln(50^2 + 400) &= 2\mu + 2\sigma^2 \\
 \sigma^2 &= 0.148 \quad \mu = 3.84
 \end{aligned}$$

Remark: To solve, double $\ln[E(Y)]$ equation and subtract from 2nd moment equation.

Mean and Variance of Lognormal

6

Losses Y have a lognormal distribution with mean 50 and variance 400. What is the probability that losses exceed 75?

$$\begin{aligned}
 \sigma^2 &= 0.148 \quad \mu = 3.84 \\
 P[Y > 75] &= P[X > \ln(75)] \\
 &= P\left[\frac{X - 3.84}{0.148^{1/2}} > \frac{\ln(75) - 3.84}{0.148^{1/2}}\right] \\
 &= 1 - \Phi(1.245) \\
 &\approx 1 - [0.8925 + 0.5(0.8944 - 0.8925)] \approx 0.11
 \end{aligned}$$

Exercise 1

7

Suppose that X is normal with $P[X > 5] = 0.5$ and $P[X > 8] = 0.05$. Find $E[e^{2X}]$.

Exercise 1

7

Suppose that X is normal with $P[X > 5] = 0.5$ and $P[X > 8] = 0.05$. Find $E[e^{2X}]$.

$$\begin{aligned}
 \mu &= 5 \\
 \frac{8 - 5}{\sigma} &= 1.645 \\
 \sigma &= 1.8237 \\
 e^X &\sim \text{LN}(\mu = 5, \sigma^2 = 3.3259) \\
 E[e^{2X}] &= E[(e^X)^2] \\
 &= e^{2\mu + 2\sigma^2} \\
 &= e^{16.6518} \\
 &= \boxed{17,052,000}
 \end{aligned}$$

Exercise 2

8

X is lognormal with mean 10 and variance 200. Find $P[X > 10]$.

Exercise 2

8

X is lognormal with mean 10 and variance 200. Find $P[X > 10]$.

$$E[X] = 10 = e^{\mu + \sigma^2/2}$$

$$E[X^2] = 10^2 + 200 = e^{2\mu + 2\sigma^2}$$

$$2\mu + \sigma^2 = 2\ln(10)$$

$$2\mu + 2\sigma^2 = \ln(300)$$

$$\sigma^2 = \ln(300) - 2\ln(10) = 1.099$$

$$\mu = 1.753$$

$$P[X > 10] = P[\ln(X) > \ln(10)]$$

$$= P\left[\frac{\ln(X) - \mu}{\sigma} > \frac{\ln(10) - 1.753}{\sqrt{1.099}}\right]$$

$$= 1 - \Phi(0.52) = \Phi(-0.52) = \boxed{0.3015}$$

Warning: Lognormals are very sensitive to rounding. Carry as many digits as possible.

Deductibles: Calculus Approach

1

Percentiles

Expected Values

Payment Limits

Losses Only

2

Loss amounts X have density $f(x) = 0.02x$ for $0 < x < 10$ and 0 otherwise. Find the 10th and 90th percentiles of X .

$$\begin{aligned}P[X \leq x] &= \int_0^x 0.02t \, dt \\&= 0.01x^2\end{aligned}$$

Let s and t denote the 10th and 90th percentiles of X . Then

$$P[X \leq s] = 0.10$$

$$0.01s^2 = 0.10$$

$$s = 3.16$$

$$0.01t^2 = 0.90$$

$$t = 9.49$$

Deductible Example

3

Loss amounts have density $f(x) = 0.02x$ for $0 < x < 10$ and 0 otherwise. Losses are insured subject to a deductible of 4. Let Y denote the payment amount corresponding to a randomly selected loss. Find the 90th percentile of Y .

We want to find y such that $P[Y \leq y] = 0.90$.

$$\begin{aligned} P[Y \leq y] &= P[X \leq y + 4] = \int_0^{y+4} 0.02x \, dx \\ &= 0.01(y + 4)^2 \\ 0.90 &= 0.01(y + 4)^2 \\ y &= 5.49 \end{aligned}$$

Note that this equals the 90th percentile of X minus the deductible of 4.

Deductible Example

4

Loss amounts have density $f(x) = 0.02x$ for $0 < x < 10$ and 0 otherwise. Losses are insured subject to a deductible of 4. Let Y denote the payment amount corresponding to a randomly selected loss. Find the 10th percentile of Y .

Trying the approach from last time:

$$\begin{aligned} P[Y \leq y] &= P[X \leq y + 4] = 0.01(y + 4)^2 \\ 0.10 &= 0.01(y + 4)^2 \\ y &= -0.83 \text{ which isn't possible} \end{aligned}$$

This happened because $P[Y = 0] = P[X \leq 4] = 0.16 > 0.10$, so the 16th percentile, and every smaller percentile, of Y is 0.

The 10th percentile of X was 3.16. A loss of 3.16 results in a payment of 0, which is the 10th percentile of Y . The p -th percentile of Y is the payment corresponding to the p -th percentile of X .

Conditional Payment Example

5

Loss amounts have density $f(x) = 0.02x$ for $0 < x < 10$ and 0 otherwise. Losses are insured subject to a deductible of 4. Find the 90th percentile of a loss that exceeds the deductible.

We are conditioning on $X > 4$, so

$$\begin{aligned} 0.9 &= P[X \leq t \mid X > 4] = \frac{P[4 < X \leq t]}{P[X > 4]} \\ &= \frac{0.01 \cdot t^2 - 0.01 \cdot 4^2}{1 - 0.01 \cdot 4^2} \\ t &= \boxed{9.57} \end{aligned}$$

Or, using the survival function

$$\begin{aligned} 1 - 0.9 &= P[X > t \mid X > 4] = \frac{P[X > t]}{P[X > 4]} \\ 0.1 &= \frac{1 - 0.01 \cdot t^2}{1 - 0.01 \cdot 4^2}, \quad t = \boxed{9.57} \end{aligned}$$

Deductible: Expected Payment

6

Loss amounts have density $f(x) = 0.02x$ for $0 < x < 10$ and 0 otherwise. Losses are insured subject to a deductible of 4.

What is the expected payment amount?

Let Y denote the payment. So $Y = X - 4$ for $X > 4$, and $Y = 0$ for $X \leq 4$.

$$\begin{aligned} E[Y] &= \int_0^4 0 \cdot f_X(x) dx + \int_4^{10} (x - 4) \cdot f_X(x) dx \\ &= 0 \cdot P[X \leq 4] + \int_4^{10} (x - 4)(0.02x) dx \\ &= 0 + \int_4^{10} 0.02x^2 - 0.08x dx \\ &= \frac{0.02}{3}x^3 - 0.04x^2 \Big|_4^{10} = \boxed{2.88} \end{aligned}$$

Adding a Payment Limit

7

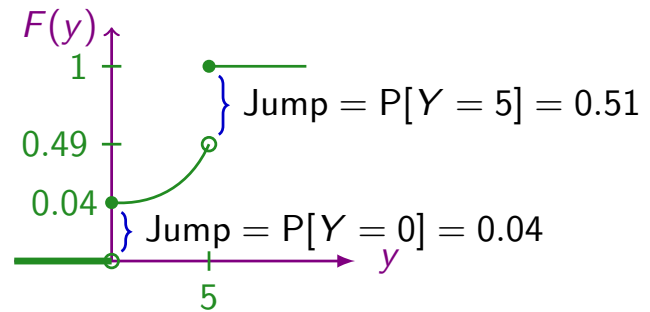
Loss amounts have density $f(x) = 0.02x$ for $0 < x < 10$, and $f(x) = 0$ otherwise. Losses are insured subject to a deductible of 2 and a maximum payment limit of 5. Find the cdf of the payment amount.

Let Y denote the payment. Then $Y = 0$ if $X \leq 2$, so $P[Y = 0] = P[X \leq 2]$.

$P[Y \leq y] = P[X \leq y + 2]$ for $0 < y < 5$.

$P[Y \leq 5] = 1$ since 5 is the maximum possible payment.

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ 0.04 & y = 0 \\ 0.01(y + 2)^2 & 0 < y < 5 \\ 1 & y > 5 \end{cases}$$



Expected Payment with Limit: Densities

8

Loss amounts have density $f(x) = 0.02x$ for $0 < x < 10$, and $f(x) = 0$ otherwise. Losses are insured subject to a deductible of 2 and a maximum payment limit of 5. Find the expected payment amount.

Y has a mixed distribution: $P[Y = 0] = 0.04$ and $P[Y = 5] = P[X > 7] = 0.51$.

$$E[Y] = 0 \cdot P[Y = 0] + 5 \cdot P[Y = 5] + \int_0^5 y \cdot f_Y(y) dy$$

So one approach would be to find $f_Y(y)$.

But sticking with X :

$$\begin{aligned} E[Y] &= \int_0^\infty \text{payment} \cdot f_X(x) dx \\ &= \int_0^2 0 \cdot 0.02x dx + \int_2^7 (x - 2) \cdot 0.02x dx + \int_7^{10} 5 \cdot 0.02x dx \\ &= 0 + \frac{0.02}{3} (7^3 - 2^3) - 0.02 (7^2 - 2^2) + 5 \cdot 0.51 = \boxed{3.8833} \end{aligned}$$

Expected Payment with Limit: Survival Function

9

Loss amounts have density $f(x) = 0.02x$ for $0 < x < 10$, and $f(x) = 0$ otherwise. Losses are insured subject to a deductible of 2 and a maximum payment limit of 5. Find the expected payment amount.

Using the survival method, $P[Y > y] = P[X > y + 2]$ for $0 < y < 5$. $P[Y > 5] = 0$.

$$\begin{aligned}
 \int_0^{\infty} P[Y > y] dy &= \int_0^5 P[X > y + 2] dy + \int_5^{\infty} 0 dy \\
 &= \int_2^7 P[X > x] dx \\
 &= \int_2^7 (1 - 0.01x^2) dx \\
 &= 5 - \frac{0.01 \cdot 7^3}{3} + \frac{0.01 \cdot 2^3}{3} \\
 &= \boxed{3.8833}
 \end{aligned}$$

Deductibles: Cases Approach

1

Overview

Uniform Examples

Exponential Examples

Splitting Into Cases

2

Let X be a loss. With a deductible of d , there are two cases for the payment Y :

$$Y = \begin{cases} 0 & X \leq d \\ X - d & X > d \end{cases}$$

$$\begin{aligned} E[Y] &= E[Y | X \leq d] \cdot P[X \leq d] + E[Y | X > d] \cdot P[X > d] \\ &= 0 \cdot P[X \leq d] + E[X - d | X > d] \cdot P[X > d] \end{aligned}$$

Sometimes $E[X - d | X > d]$ can be found without calculus.

If $X \sim U(0, a)$, then $(X - d | X > d) \sim U(0, a - d)$ and $E[X - d | X > d] = \frac{a - d}{2}$;
if $X \sim \exp(\theta)$ then $(X - d | X > d) \sim \exp(\theta)$ and $E[X - d | X > d] = \theta$.

Uniform Example

3

An industrial worker's loss due to injury is uniformly distributed on $(0, 100)$. An insurance company will reimburse the worker for 60% of the amount of the loss that exceeds a deductible of 20. What is the expected reimbursement for a randomly selected loss?

There are two cases: either the loss exceeds 20, or it doesn't. Let X denote the loss, and Y the reimbursement.

$$\begin{aligned}
 E[Y] &= E[Y \mid X > 20] \cdot P[X > 20] + E[Y \mid X \leq 20] \cdot P[X \leq 20] \\
 &= E[0.6 \cdot U(0, 80)] \cdot \frac{8}{10} + 0 \cdot \frac{2}{10} \\
 &= 0.6 \cdot 40 \cdot 0.8 \\
 &= \boxed{19.2}
 \end{aligned}$$

or: $E[Y] = \int_0^{20} 0 \cdot \frac{1}{100} dx + \int_{20}^{100} 0.6 \cdot (x - 20) \cdot \frac{1}{100} dx$

Adding Limits

4

Losses X are uniform on $(0, 100)$. An insurance company will pay a reimbursement of 60% of the amount of the loss that exceeds a deductible of 20, up to a maximum payment of 30. Find the expected reimbursement for a randomly selected loss.

Again let $Y = \text{Payment}$. We now have 3 cases:

no payment ($X < 20$), a payment of $0.6(X - 20)$, or a max payment of 30.

The max kicks in when $0.6(X - 20) = 30$ or $X = 70$.

$$\begin{aligned}
 E[Y] &= 0 \cdot P[X \leq 20] + E[Y \mid 20 < X \leq 70] \cdot P[20 < X \leq 70] \\
 &\quad + 30 \cdot P[X > 70] \\
 &= 0 \cdot \frac{2}{10} + E[0.6 \cdot U(0, 50)] \cdot \frac{5}{10} + 30 \cdot \frac{3}{10} \\
 &= 0 + 0.6 \cdot 25 \cdot 0.5 + 9 = \boxed{16.5}
 \end{aligned}$$

Or: $E[Y] = \int_0^{20} 0 \cdot \frac{1}{100} dx + \int_{20}^{70} \frac{0.6(x - 20)}{100} dx + \int_{70}^{100} \frac{30}{100} dx$

Exponential Example

5

Losses X are exponentially distributed with mean 50. An insurance company will pay a reimbursement of 60% of the amount of the loss that exceeds a deductible of 20. Find the mean and variance of a randomly selected reimbursement Y .

$$\begin{aligned}E[Y] &= 0 \cdot P[X \leq 20] + E[Y \mid X > 20] \cdot P[X > 20] \\&= 0 + 0.6 \cdot E[X - 20 \mid X > 20] \cdot P[X > 20] \\&= 0.6 \cdot 50 \cdot e^{-20/50} = \boxed{20.1}\end{aligned}$$

$$\begin{aligned}E[Y^2] &= E[0^2] \cdot P[X \leq 20] + E[(0.6(X - 20))^2 \mid X > 20] \cdot P[X > 20] \\&= 0 + 0.6^2 E[(X - 20)^2 \mid X > 20] \cdot P[X > 20] \\&= 0.36 \cdot (50^2 + 50^2) \cdot e^{-20/50} = 1,206.6\end{aligned}$$

$$\text{Var}[Y] = 1,206.6 - 20.1^2 = \boxed{802.2}$$

Review of Other Continuous Ideas

1

Key Formulas

Distribution Review

Densities and CDFs

2

$$F(x) = P[X \leq x]$$

$$F(\infty) = 1 \quad F(-\infty) = 0$$

If X is continuous, then $F(x)$ is continuous and

$$f(x) = \frac{d}{dx} F(x)$$

$$P[a < X \leq b] = \int_a^b f(x) dx = F(b) - F(a)$$

If X has a mixed distribution, then $F(x)$ has some jumps. Jump sizes determine probabilities, i.e.,

$$P[X = a] = F(a) - \lim_{x \uparrow a} F(x)$$

Moments

3

If X is continuous, then

$$E[X] = \int x \cdot f(x) dx$$

$$E[g(X)] = \int g(x) \cdot f(x) dx$$

For mixed distributions, sum the discrete and continuous pieces

If X is non-negative, and $g(0) = 0$, then

$$E[X] = \int_0^\infty P[X > x] dx$$

$$E[g(X)] = \int_0^\infty g'(x) \cdot P[X > x] dx$$

These last two equations hold for mixed distributions as well.

Normal Approximation

4

Suppose $X_1, X_2, \dots, X_n$ are i.i.d. (independent, identically distributed)

Let $\mu = E[X_i]$ and $\sigma^2 = \text{Var}[X_i]$

Let $S = X_1 + X_2 + \dots + X_n$

The *Central Limit Theorem* says S is approximately normal.

$$\frac{S - E[S]}{\text{SD}[S]} = \frac{S - nE[X]}{\sqrt{n \text{Var}[X]}} = \frac{S - n\mu}{\sigma\sqrt{n}} \approx N(0, 1)$$

Key Distributions

5

	Uniform(a, b)	Exponential	Gamma, $\alpha = 2$	Gamma, integer α
$f(x)$	$\frac{1}{b-a}$	$\frac{1}{\theta}e^{-x/\theta}$	$\frac{x}{\theta^2}e^{-x/\theta}$	$\frac{x^{\alpha-1}}{\theta^\alpha(\alpha-1)!}e^{-x/\theta}$
$F(x)$	$\frac{x-a}{b-a}$	$1 - e^{-x/\theta}$	$1 - e^{-x/\theta} - \frac{x}{\theta}e^{-x/\theta}$	$1 - \sum_{i=0}^{\alpha-1} \frac{(x/\theta)^i}{i!}e^{-x/\theta}$
$E[X]$	$\frac{a+b}{2}$	θ	$\alpha\theta$	$\alpha\theta$
$\text{Var}[X]$	$\frac{(b-a)^2}{12}$	θ^2	$\alpha\theta^2$	$\alpha\theta^2$

Joint Distributions and CDFs

1

Joint Distributions

Joint CDFs

Exercises

Joint Distribution

2

In 1-dimension, for a random variable X we had $\sum_x \Pr[X = x] = 1$

In 2 dimensions, suppose we have 2 random variables X and Y .

The joint probability mass function is $\Pr[X = x, Y = y]$.

Total probability is still 1 $\Rightarrow \sum_x \sum_y \Pr[X = x, Y = y] = 1$

Note: Current syllabus only includes discrete joint distributions.

Continuous joint distributions no longer tested due to calculus requirements.

Example

3

The joint distribution of X and Y is given by

		X		
		0	1	2
Y	1	0.1	0.2	0.3
	2	0.1	0.1	0.2

$$P[X = 0, Y = 1] = 0.1$$

$$P[X = 2, Y = 2] = 0.2$$

Total probability needs to sum to 1:

$$\begin{aligned} \sum_{x,y} P[X = x, Y = y] &= 0.1 + 0.2 + 0.3 + 0.1 + 0.1 + 0.2 \\ &= 1 \end{aligned}$$

Uniform Example

4

Suppose X and Y are integer valued random variables with $1 \leq X \leq 4$ and $1 \leq Y \leq X$. If every possible outcome is equally likely, find $P[X = Y = 3]$.

Possible outcomes are:

$$\{X = 1, Y = 1\}$$

$$\{X = 2, Y = 1\}, \{X = 2, Y = 2\}$$

$$\{X = 3, Y = 1\}, \{X = 3, Y = 2\}, \{X = 3, Y = 3\}$$

$$\{X = 4, Y = 1\}, \{X = 4, Y = 2\}, \{X = 4, Y = 3\}, \{X = 4, Y = 4\}$$

That is 10 possible outcomes, all with the same probability.

$$\text{Total probability} = 1 = 10 \cdot P[X = Y = 3]$$

$$\text{so } P[X = Y = 3] = \boxed{0.1}$$

Joint CDFs

5

In 1-d, the cdf of X is $F_X(x) = P[X \leq x]$

In 2-d, the joint cdf of X and Y is

$$F_{X,Y}(x, y) = P[X \leq x, Y \leq y]$$

Properties:

1. $0 \leq F(x, y) \leq 1$
2. $F(x, \infty) = P[X \leq x, Y \leq \infty] = P[X \leq x] = F_X(x)$
3. $F(\infty, y) = P[X \leq \infty, Y \leq y] = P[Y \leq y] = F_Y(y)$
4. $F(\infty, \infty) = 1$
5. $F(-\infty, y) = 0 = F(x, -\infty)$

CDF Example

6

The joint distribution of X and Y is given by

		X		
		0	1	2
Y	1	0.1	0.2	0.3
	2	0.1	0.1	0.2

Some values of the CDF:

$$F_{X,Y}(0, 1) = P[X \leq 0, Y \leq 1] = 0.1$$

$$\begin{aligned} F_{X,Y}(1, 1) &= P[X \leq 1, Y \leq 1] \\ &= 0.1 + 0.2 = 0.3 \end{aligned}$$

$$\begin{aligned} F_{X,Y}(1, 2) &= P[X \leq 1, Y \leq 2] \\ &= 0.1 + 0.2 + 0.1 + 0.1 = 0.5 \end{aligned}$$

Exercise 1

7

$P[X = x, Y = y] = c(x + y)$ for x and y integers such that $1 \leq x \leq 3$ and $1 \leq y \leq x$.
Find c .

Exercise 1

7

$P[X = x, Y = y] = c(x + y)$ for x and y integers such that $1 \leq x \leq 3$ and $1 \leq y \leq x$.
Find c .

The point is that the total probability must equal 1.

If $x = 1$ then $y = 1$.

If $x = 2$ then y can be 1 or 2

If $x = 3$, then y can be 1, 2, or 3.

That gives 6 possible cases to sum, and

$$1 = c(1 + 1) + c(2 + 1) + c(2 + 2) + c(3 + 1) + c(3 + 2) + c(3 + 3)$$

$$1 = c[2 + 3 + 4 + 4 + 5 + 6] = 24c$$

$$c = \boxed{\frac{1}{24}}$$

Exercise 2

8

X and Y are integer valued random variables. If you know that $F_{X,Y}(3,3) = 1$, $F_{X,Y}(3,2) = 0.7$, $F_{X,Y}(2,3) = 0.6$, $F_{X,Y}(2,2) = 0.4$, $F_{X,Y}(1,3) = 0.3$, and $F_{X,Y}(1,2) = 0.2$, find $P[X = 2]$.

Exercise 2

8

X and Y are integer valued random variables. If you know that $F_{X,Y}(3,3) = 1$, $F_{X,Y}(3,2) = 0.7$, $F_{X,Y}(2,3) = 0.6$, $F_{X,Y}(2,2) = 0.4$, $F_{X,Y}(1,3) = 0.3$, and $F_{X,Y}(1,2) = 0.2$, find $P[X = 2]$.

Since X is integer valued, $P[X = 2] = P[X \leq 2] - P[X \leq 1] = F_X(2) - F_X(1)$.

$F_X(x) = F_{X,Y}(x, \infty)$ so we wish we could plug in $y = \infty$. But $F_{X,Y}(3,3) = 1$, so $X \leq 3$ and $Y \leq 3$. In particular, 3 is the largest Y can be, so $F_X(x) = F_{X,Y}(x, 3)$, and

$$\begin{aligned} P[X = 2] &= F_X(2) - F_X(1) \\ &= F_{X,Y}(2, 3) - F_{X,Y}(1, 3) \\ &= 0.6 - 0.3 = \boxed{0.3} \end{aligned}$$

Marginal and Conditional Distributions

1

Marginal Distributions

Discrete Example

Conditional Distributions

Key Points

Exercises

Marginal distributions

2

For discrete variables,

$$P[X = x] = \sum_y P[X = x, Y = y]$$

is marginal distribution of X .

In words, it's the distribution of X without knowing Y .

SOA/CAS 1999 Practice Exam # 13

3

Let X and Y be discrete random variables with joint probability function

$$p(x, y) = \frac{2x + y}{12}$$

for $(x, y) = (0, 1), (0, 2), (1, 2), (1, 3)$, and 0 otherwise. Determine the marginal probability function of X .

X has two possible values, 0 or 1. Finding the marginal probability function means finding $P[X = 0]$ and $P[X = 1]$.

SOA/CAS 1999 Practice Exam # 13

4

$$p(x, y) = \frac{2x + y}{12}$$

for $(x, y) = (0, 1), (0, 2), (1, 2), (1, 3)$, and 0 otherwise.

$$\begin{aligned} P[X = 0] &= P[(X, Y) = (0, 1)] + P[(X, Y) = (0, 2)] \\ &= \frac{2 \cdot 0 + 1}{12} + \frac{2 \cdot 0 + 2}{12} = \frac{3}{12} = \frac{1}{4} \end{aligned}$$

$$\begin{aligned} P[X = 1] &= P[(X, Y) = (1, 2)] + P[(X, Y) = (1, 3)] \\ &= \frac{2 \cdot 1 + 2}{12} + \frac{2 \cdot 1 + 3}{12} = \frac{9}{12} = \frac{3}{4} \end{aligned}$$

$$P[X = x] = 0 \quad \text{otherwise}$$

To check our work, note that $P[X = 0] + P[X = 1] = 1$

Variation on Discrete Example

5

Let X and Y be discrete random variables with joint probability function

$$p(x, y) = \frac{2x + y}{12}$$

for $(x, y) = (0, 1), (0, 2), (1, 2), (1, 3)$, and 0 otherwise. Find $P[X = 0 \mid Y = 2]$

$$P[X = 0 \mid Y = 2] = \frac{P[X = 0, Y = 2]}{P[Y = 2]}$$

$$P[X = 0, Y = 2] = \frac{2 \cdot 0 + 2}{12} = \frac{2}{12}$$

$$\begin{aligned} P[Y = 2] &= P[X = 0, Y = 2] + P[X = 1, Y = 2] \\ &= \frac{2 \cdot 0 + 2}{12} + \frac{2 \cdot 1 + 2}{12} = \frac{6}{12} \end{aligned}$$

$$P[X = 0 \mid Y = 2] = \frac{2/12}{6/12} = \frac{1}{3}$$

Conditional distributions

6

For discrete variables,

$$\begin{aligned} P[X = x \mid Y = y] &= \frac{P[X = x, Y = y]}{P[Y = y]} \\ &= \frac{P[X = x, Y = y]}{\sum_x P[X = x, Y = y]} \end{aligned}$$

Note that the role of the denominator is so that the conditional distribution is itself a probability distribution, i.e., $\sum_x P[X = x \mid Y = y] = 1$.

Independence

7

X and Y are independent if $P[X = x \mid Y = y] = P[X = x]$

For discrete variables,

$$P[X = x \mid Y = y] = \frac{P[X = x, Y = y]}{P[Y = y]}$$

$$\begin{aligned} P[X = x, Y = y] &= P[Y = y] \cdot P[X = x \mid Y = y] \\ &= P[Y = y] \cdot P[X = x] \text{ if } X \text{ and } Y \text{ are independent} \end{aligned}$$

In particular, X and Y are independent if and only if the joint probability mass function factors as a function of x times a function of y and the range of $\{X, Y\}$ is a discrete rectangle.

Example

8

The joint distribution of X and Y is:

		X		
		0	1	2
Y	1	0.1	0.2	0.3
	2	0.1	0.1	0.2

$$P[X = 0] = 0.1 + 0.1 = 0.2$$

$$P[X = 1] = 0.2 + 0.1 = 0.3$$

$$P[X = 2] = 0.3 + 0.2 = 0.5$$

$$P[Y = 1] = 0.1 + 0.2 + 0.3 = 0.6$$

$$P[Y = 2] = 0.1 + 0.1 + 0.2 = 0.4$$

$$P[X = 0 \mid Y = 1] = \frac{0.1}{0.6} = \frac{1}{6}$$

$$P[X = 1 \mid Y = 1] = \frac{0.2}{0.6} = \frac{2}{6}$$

$$P[X = 2 \mid Y = 1] = \frac{0.3}{0.6} = \frac{3}{6}$$

$$P[Y = 1 \mid X = 2] = 0.6$$

$$P[Y = 2 \mid X = 2] = 0.4$$

Key Points

9

The marginal distribution $P[X = x]$ of X can only involve x . It cannot involve any other variable.

The conditional distribution $P[X = x \mid Y = y]$ can involve x and y .

If X and Y are independent, $P[X = x \mid Y = y] = P[X = x]$. In particular, the conditional distribution of X will only involve x . Even the range cannot depend on y .

Exercise 1

10

Let X and Y be discrete random variables with joint probability function

$$p(x, y) = \frac{x^2 y}{23}$$

for $(x, y) = (1, 1), (1, 2), (2, 2), (2, 3)$, and 0 otherwise. Determine the marginal probability function of X .

Exercise 1

10

Let X and Y be discrete random variables with joint probability function

$$p(x, y) = \frac{x^2 y}{23}$$

for $(x, y) = (1, 1), (1, 2), (2, 2), (2, 3)$, and 0 otherwise. Determine the marginal probability function of X .

$$\begin{aligned} P[X = 1] &= P[X = 1, Y = 1] + P[X = 1, Y = 2] \\ &= \frac{1}{23}(1^2 \cdot 1 + 1^2 \cdot 2) = \frac{3}{23} \\ P[X = 2] &= P[X = 2, Y = 2] + P[X = 2, Y = 3] \\ &= \frac{1}{23}(2^2 \cdot 2 + 2^2 \cdot 3) = \frac{20}{23} \end{aligned}$$

To check, note that $P[X = 1] + P[X = 2] = (3/23) + (20/23) = 1$

Exercise 2

11

Let X and Y be discrete random variables with joint probability function

$$p(x, y) = \frac{x^2 y}{23}$$

for $(x, y) = (1, 1), (1, 2), (2, 2), (2, 3)$, and 0 otherwise. Find $P[X = 1 \mid Y = 2]$

Exercise 2

11

Let X and Y be discrete random variables with joint probability function

$$p(x, y) = \frac{x^2 y}{23}$$

for $(x, y) = (1, 1), (1, 2), (2, 2), (2, 3)$, and 0 otherwise. Find $P[X = 1 \mid Y = 2]$

$$\begin{aligned} P[X = 1 \mid Y = 2] &= \frac{P[X = 1, Y = 2]}{P[Y = 2]} \\ &= \frac{P[X = 1, Y = 2]}{P[X = 1, Y = 2] + P[X = 2, Y = 2]} \\ &= \frac{\frac{1}{23}(1^2 \cdot 2)}{\frac{1}{23}(1^2 \cdot 2 + 2^2 \cdot 2)} \\ &= \frac{2}{2 + 8} = \boxed{\frac{1}{5}} \end{aligned}$$

Joint Moments

1

Examples

Definitions

Exercises

Discrete Example

2

Let X denote that number of years I will use my current computer, and Y the number of years that I will use my tablet. The joint distribution of X and Y is

		X		
		1	2	3
Y	1	0.05	0.16	0.19
	2	0.10	0.13	0.12
	3	0.07	0.10	0.08

What is the average number of years that I will use the tablet?

One approach is to first find the marginal distribution of Y .

$$P[Y = 1] = 0.05 + 0.16 + 0.19 = 0.40$$

$$P[Y = 2] = 0.10 + 0.13 + 0.12 = 0.35$$

$$P[Y = 3] = 0.07 + 0.10 + 0.08 = 0.25$$

$$E[Y] = 1 \cdot 0.40 + 2 \cdot 0.35 + 3 \cdot 0.25 = 1.85$$

Discrete Example

3

The joint distribution of X and Y is

		X		
		1	2	3
Y	1	0.05	0.16	0.19
	2	0.10	0.13	0.12
	3	0.07	0.10	0.08

What is $E(Y)$?

A second approach is to sum over all 9 cases.

$$\begin{aligned}
 E[Y] &= 1 \cdot 0.05 + 1 \cdot 0.16 + 1 \cdot 0.19 \\
 &\quad + 2 \cdot 0.10 + 2 \cdot 0.13 + 2 \cdot 0.12 \\
 &\quad + 3 \cdot 0.07 + 3 \cdot 0.10 + 3 \cdot 0.08 \\
 &= 1.85
 \end{aligned}$$

Discrete Example

4

The joint distribution of X and Y is

		X		
		1	2	3
Y	1	0.05	0.16	0.19
	2	0.10	0.13	0.12
	3	0.07	0.10	0.08

What is $E(X + Y)$?

When we are dealing with $X + Y$, the first approach no longer works (easily), but we can still use the second.

$$\begin{aligned}
 E[X + Y] &= (1 + 1) \cdot 0.05 + (2 + 1) \cdot 0.16 + (3 + 1) \cdot 0.19 \\
 &\quad + (1 + 2) \cdot 0.10 + (2 + 2) \cdot 0.13 + (3 + 2) \cdot 0.12 \\
 &\quad + (1 + 3) \cdot 0.07 + (2 + 3) \cdot 0.10 + (3 + 3) \cdot 0.08 \\
 &= 4.02
 \end{aligned}$$

Definitions

5

For discrete random variables X and Y ,

$$E[g(X, Y)] = \sum_x \sum_y g(x, y) \cdot P[X = x, Y = y]$$

Some special cases

$$E[X] = \sum_x \sum_y x \cdot P[X = x, Y = y]$$

$$E[Y^2] = \sum_x \sum_y y^2 \cdot P[X = x, Y = y]$$

Exercise 1

6

Let X and Y be discrete random variables with joint probability function

$$p(x, y) = \frac{2x + y}{12}$$

for $(x, y) = (0, 1), (0, 2), (1, 2), (1, 3)$, and 0 otherwise. Find $E[X]$.

Exercise 1

6

Let X and Y be discrete random variables with joint probability function

$$p(x, y) = \frac{2x + y}{12}$$

for $(x, y) = (0, 1), (0, 2), (1, 2), (1, 3)$, and 0 otherwise. Find $E[X]$.

$$\begin{aligned} E[X] &= 0 \cdot P[X = 0, Y = 1] + 0 \cdot P[X = 0, Y = 2] \\ &\quad + 1 \cdot P[X = 1, Y = 2] + 1 \cdot P[X = 1, Y = 3] \\ &= 0 + 0 + \frac{2 \cdot 1 + 2}{12} + \frac{2 \cdot 1 + 3}{12} \\ &= \frac{9}{12} = \boxed{\frac{3}{4}} \end{aligned}$$

Or we could say:

$$E[X] = 0 \cdot P[X = 0] + 1 \cdot P[X = 1] = 0 \cdot \frac{1}{4} + 1 \cdot \frac{3}{4}$$

with $P[X = 0]$ and $P[X = 1]$ coming from an earlier lesson.

Exercise 2

7

Let X and Y be discrete random variables with joint probability function

$$p(x, y) = \frac{2x + y}{12}$$

for $(x, y) = (0, 1), (0, 2), (1, 2), (1, 3)$, and 0 otherwise. Find $E[(X + 1)Y]$.

Exercise 2

7

Let X and Y be discrete random variables with joint probability function

$$p(x, y) = \frac{2x + y}{12}$$

for $(x, y) = (0, 1), (0, 2), (1, 2), (1, 3)$, and 0 otherwise. Find $E[(X + 1)Y]$.

Key difference: now we want the mean of something that involves both X and Y . We must use the joint probability mass.

$$\begin{aligned} E[(X + 1)Y] &= (0 + 1) \cdot 1 \cdot P[X = 0, Y = 1] + (0 + 1) \cdot 2 \cdot P[X = 0, Y = 2] \\ &\quad + 2 \cdot 2 \cdot P[X = 1, Y = 2] + 2 \cdot 3 \cdot P[X = 1, Y = 3] \\ &= 1 \cdot \frac{2 \cdot 0 + 1}{12} + 2 \cdot \frac{2 \cdot 0 + 2}{12} + 4 \cdot \frac{2 \cdot 1 + 2}{12} + 6 \cdot \frac{2 \cdot 1 + 3}{12} \\ &= \frac{1 + 4 + 16 + 30}{12} = \frac{51}{12} = \boxed{\frac{17}{4}} \end{aligned}$$

Covariances and Correlations

1

Definitions

Example

Using independence

SOA # 75; S.03.15

Exercises

Covariance and correlation

2

$$\text{Var}[X] = E[(X - \mu_X)^2] = E[X^2] - (E[X])^2$$

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E(XY) - E(X) \cdot E(Y)\end{aligned}$$

$$\text{Note: Cov}(X, b) = E[bX] - bE[X] = 0$$

Covariance is “bilinear,” i.e.,

$$\begin{aligned}\text{Cov}(aX + bY, cZ) &= \text{Cov}(aX, cZ) + \text{Cov}(bY, cZ) \\ &= ac \text{Cov}(X, Z) + bc \text{Cov}(Y, Z)\end{aligned}$$

Variance of Sums

3

$$\text{Cov}(X, X) = E(X \cdot X) - E(X) \cdot E(X) = \text{Var}(X)$$

$$\begin{aligned}\text{Var}(X + Y) &= \text{Cov}(X + Y, X + Y) \\ &= \text{Cov}(X, X) + 2\text{Cov}(X, Y) + \text{Cov}(Y, Y) \\ &= \text{Var}(X) + 2\text{Cov}(X, Y) + \text{Var}(Y)\end{aligned}$$

$$\text{Likewise } \text{Var}(aX + bY) = a^2\text{Var}(X) + 2ab\text{Cov}(X, Y) + b^2\text{Var}(Y)$$

$$\begin{aligned}\text{Key Example } \text{Var}(X - Y) &= \text{Var}(X + (-1)Y) \\ &= \text{Var}(X) - 2\text{Cov}(X, Y) + \text{Var}(Y)\end{aligned}$$

If X and Y are independent then $E(XY) = E(X)E(Y)$ so $\text{Cov}(X, Y) = 0$. That means that if X and Y are independent,

$$\begin{aligned}\text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) \\ \text{Var}(X - Y) &= \text{Var}(X) + \text{Var}(Y)\end{aligned}$$

Correlation

4

Definition (Correlation)

The correlation of X and Y is given by

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \cdot \text{SD}(Y)}$$

Note that if X and Y are independent then $\text{Corr}(X, Y) = 0$ since $\text{Cov}(X, Y) = 0$.

Properties of Correlation

5

Suppose that $Y = aX + b$. Then

$$\begin{aligned}\text{Cov}(X, Y) &= \text{Cov}(X, aX + b) \\ &= a \cdot \text{Cov}(X, X) + \text{Cov}(X, b) \\ &= a \cdot \text{Var}[X]\end{aligned}$$

So if $a > 0$, then $\text{Cov}(X, Y) > 0$, and if $a < 0$, $\text{Cov}(X, Y) < 0$

$$\begin{aligned}\text{SD}(Y) &= |a| \text{SD}(X) \\ \text{so } \text{Corr}(X, Y) &= \frac{a \cdot \text{Var}X}{[\text{SD}(X)]^2 |a|} = \begin{cases} 1 & a > 0 \\ -1 & a < 0 \end{cases}\end{aligned}$$

$$\text{Fact : } -1 \leq \text{Corr}(X, Y) \leq 1$$

Example

6

X and Y are integer valued random variables with $0 < X \leq 3$ and $1 \leq Y \leq 2$.
 $P[X = x, Y = y] = kxy$ when positive, for some constant k . Find $\text{Cov}[X, Y]$.

$$1 = k[1 \cdot 1 + 1 \cdot 2 + 2 \cdot 1 + 2 \cdot 2 + 3 \cdot 1 + 3 \cdot 2] = 18k$$

$$k = 1/18$$

$$E[X] = \frac{1 \cdot (1 \cdot 1 + 1 \cdot 2) + 2 \cdot (2 \cdot 1 + 2 \cdot 2) + 3 \cdot (3 \cdot 1 + 3 \cdot 2)}{18} = \frac{42}{18} = \frac{7}{3}$$

$$E[Y] = \frac{1 \cdot (1 \cdot 1 + 2 \cdot 1 + 3 \cdot 1) + 2 \cdot (1 \cdot 2 + 2 \cdot 2 + 3 \cdot 2)}{18} = \frac{30}{18} = \frac{5}{3}$$

$$E[XY] = \frac{(1 \cdot 1) \cdot (1 \cdot 1) + (1 \cdot 2)^2 + (2 \cdot 1)^2 + (2 \cdot 2)^2 + (3 \cdot 1)^2 + (3 \cdot 2)^2}{18} = \frac{70}{18} = \frac{35}{9}$$

$$\text{Cov}[X, Y] = E[XY] - E[X]E[Y] = \frac{35}{9} - \frac{7}{3} \cdot \frac{5}{3} = 0$$

When are random variables independent?

7

Theorem

X and Y are independent if both

1. The support of (X, Y) , i.e., the points such that $P[X = x, Y = y] > 0$, is a rectangular lattice
2. $P[X = x, Y = y] = P[X = x] \cdot P[Y = y]$

Remark: The Soviet notation for independence was $X \perp Y$, partly because of this requirement that the support be a rectangle.

Applications of Independence:

8

If X and Y are independent, then $E[XY] = (E[X]) \cdot (E[Y])$.

More generally,

$$E[g(X) h(Y)] = E[g(X)] \cdot E[h(Y)]$$

so $E[X^2 Y^3] = (E[X^2]) (E[Y^3])$, etc.

$$\begin{aligned} \text{Cov}(X, Y) &= E[XY] - (E[X]) \cdot (E[Y]) \\ &= (E[X]) \cdot (E[Y]) - (E[X]) \cdot (E[Y]) = 0 \end{aligned}$$

Note: Independence implies that $\text{Cov}(X, Y) = 0$, but the converse is false.

Example of Independence

9

X and Y are integer valued random variables with $0 < X \leq 3$ and $1 \leq Y \leq 2$.
 $P[X = x, Y = y] = kxy$ when positive, for some constant k . Find $\text{Cov}[X, Y]$.

We could have simply observed that the support is a rectangle, and $f(x, y)$ factors as a function of x times a function of y , so X and Y are independent and therefore $\text{Cov}(X, Y) = 0$.

SOA # 75; S.03.15

10

An insurance policy pays a total medical benefit consisting of two parts for each claim. Let X represent the part of the benefit that is paid to the surgeon, and let Y represent the part that is paid to the hospital. The variance of X is 5,000, the variance of Y is 10,000, and the variance of the total benefit, $X + Y$, is 17,000.

Due to increasing medical costs, the company that issues the policy decides to increase X by a flat amount of 100 per claim and to increase Y by 10% per claim. Calculate the variance of the total benefit after these revisions have been made.

Total benefit = $X + 100 + 1.1Y$.

$$\begin{aligned}
 \text{Var}(X + 100 + 1.1Y) &= \text{Var}(X + 1.1Y) \\
 &= \text{Cov}(X + 1.1Y, X + 1.1Y) \\
 &= \text{Var}(X) + 2 \cdot 1.1\text{Cov}(X, Y) + 1.1^2 \text{Var}(Y) \\
 &= 5,000 + 2.2 \text{Cov}(X, Y) + 1.21 \cdot 10,000
 \end{aligned}$$

SOA # 75; S.03.15

11

So we just need to find $\text{Cov}(X, Y)$ and plug it in.

$$\text{Var}(X + Y) = \text{Var}(X) + 2 \text{Cov}(X, Y) + \text{Var}(Y)$$

$$17,000 = 5,000 + 2 \text{Cov}(X, Y) + 10,000$$

$$1,000 = \text{Cov}(X, Y)$$

$$\text{Var}(X + 1.1Y) = 5,000 + 2.2 \text{Cov}(X, Y) + (1.21) \cdot 10,000$$

$$= 5,000 + 2,200 + 12,100$$

$$= 19,300$$

Exercise 1

12

Suppose $P[X = Y = 0] = 1/2$ and $P[X = 1, Y = 1] = P[X = -1, Y = 1] = 1/4$. Find $\text{Cov}[X, Y]$. Are X and Y independent?

Exercise 1

12

Suppose $P[X = Y = 0] = 1/2$ and $P[X = 1, Y = 1] = P[X = -1, Y = 1] = 1/4$. Find $\text{Cov}[X, Y]$. Are X and Y independent?

Finding covariance first,

$$E[X] = 0 \cdot 1/2 + 1 \cdot 1/4 + (-1) \cdot 1/4 = 0$$

$$E[Y] = 0 \cdot 1/2 + 1 \cdot (1/4 + 1/4) = 1/2$$

$$E[XY] = 0 \cdot 0 \cdot 1/2 + 1 \cdot 1 \cdot 1/4 + (-1) \cdot 1 \cdot 1/4 = 0$$

$$\text{Cov}[X, Y] = E[XY] - E[X]E[Y]$$

$$= 0 - 0 \cdot (1/2) = \boxed{0}$$

Not independent because support isn't rectangular. Or

$$P[X = 0] = 1/2 = P[Y = 0]$$

$$P[X = Y = 0] = 1/2 \neq P[X = 0] \cdot P[Y = 0]$$

Exercise 2

13

X and Y are random variables with $\text{Corr}(X, Y) = 0.6$, $\text{Var}[X] = 64$ and $\text{Var}[Y] = 100$. Find $\text{Var}[2X - 3Y]$.

Exercise 2

13

X and Y are random variables with $\text{Corr}(X, Y) = 0.6$, $\text{Var}[X] = 64$ and $\text{Var}[Y] = 100$. Find $\text{Var}[2X - 3Y]$.

$$\text{Corr}(X, Y) = \frac{\text{Cov}[X, Y]}{\text{SD}[X]\text{SD}[Y]}$$

$$0.6 = \frac{\text{Cov}[X, Y]}{\sqrt{64}\sqrt{100}}$$

$$48 = \text{Cov}[X, Y]$$

$$\begin{aligned}\text{Var}[2X - 3Y] &= 2^2\text{Var}[X] + 2(2)(-3)\text{Cov}[X, Y] + (-3)^2\text{Var}[Y] \\ &= 4 \cdot 64 - 12 \cdot 48 + 9 \cdot 100 \\ &= \boxed{580}\end{aligned}$$

Conditional Moments

1

Example

Double Expectation Theorem

Examples

Law of Total Variation

Exercises

Example

2

		X		
		0	1	2
Y	1	0.1	0.2	0.3
	2	0.1	0.1	0.2

Previously we saw

$$P[X = 0 \mid Y = 1] = \frac{1}{6} \quad P[X = 1 \mid Y = 1] = \frac{2}{6} \quad P[X = 2 \mid Y = 1] = \frac{3}{6}$$

Conditional Moments:

$$E[X \mid Y = y] = \sum_{\text{all } x} x \cdot P[X = x \mid Y = y]$$

$$E[X \mid Y = 1] = 0 \cdot \frac{1}{6} + 1 \cdot \frac{2}{6} + 2 \cdot \frac{3}{6} = \frac{4}{3}$$

$$E[X \mid Y = 2] = 0 \cdot \frac{0.1}{0.4} + 1 \cdot \frac{0.1}{0.4} + 2 \cdot \frac{0.2}{0.4} = \frac{5}{4}$$

Example Continued

3

$$E[X \mid Y = 1] = \frac{4}{3} \quad E[X \mid Y = 2] = \frac{5}{4}$$

Key point: $E[X \mid Y]$ is a function of Y . *As a result, it is also a random variable.*

$$P[E[X \mid Y] = 4/3] = P[Y = 1] = 0.6$$

$$P[E[X \mid Y] = 5/4] = P[Y = 2] = 0.4$$

$$\begin{aligned} E[E[X \mid Y]] &= 0.6 \cdot \frac{4}{3} + 0.4 \cdot \frac{5}{4} \\ &= 1.3 \\ &= E[X] \end{aligned}$$

That is not a coincidence.

Remember: $E[X \mid Y]$ is random. $E[X]$ is a (non-random) number.

Double Expectation Theorem

4

Note that the conditional expectation of X is a function of y :

$$E[X \mid Y = y] = \sum_{\text{all } x} x \cdot P[X = x \mid Y = y]$$

When $E[X \mid Y = y]$ is a nice function of y , the unconditional expectations are also closely related:

Theorem (Double Expectation)

$$\begin{aligned} E[X] &= E[E[X \mid Y = y]] \\ E[X] &= \sum_{\text{all } y} E[X \mid Y = y] \cdot P[Y = y] \end{aligned}$$

This is useful when the distribution of X is complicated, but the conditional distribution $(X \mid Y)$ and distribution of Y are both nice.

Example

5

Let N be the value rolled by a fair six-sided die. Suppose that I then flip N independent fair coins. What is the expected number of heads? What is the variance in the number of heads?

The key to this example is that if we know N then it is easy to find the first and second moment

$$E[\text{Heads} \mid N] = \frac{N}{2}$$

So by double expectation,

$$\begin{aligned} E[\text{Heads}] &= E[E[\text{Heads} \mid N]] \\ &= E\left[\frac{N}{2}\right] = \frac{7/2}{2} = \boxed{\frac{7}{4}} \end{aligned}$$

Example (Cont)

6

For the second moment, if we know N then the number of heads (H) is binomial with N trials and $p = 1/2$.

That means that

$$\begin{aligned} E[H \mid N] &= \frac{N}{2}, \quad \text{Var}[H \mid N] = \frac{N}{4} \\ E[H^2 \mid N] &= \text{Var}[H \mid N] + (E[H \mid N])^2 \\ &= \frac{N}{4} + \frac{N^2}{4} \\ E[H^2] &= E[E[H^2 \mid N]] \\ &= E\left[\frac{N}{4} + \frac{N^2}{4}\right] \end{aligned}$$

Example (cont)

7

Since N is uniform on $\{1, 2, \dots, 6\}$,

$$\begin{aligned} E[N] &= \frac{1+6}{2} \\ \text{Var}[N] &= \frac{6^2-1}{12} = \frac{35}{12} \\ E[N^2] &= \frac{35}{12} + \left(\frac{7}{2}\right)^2 = \frac{91}{6} \\ E[H^2] &= E\left[\frac{N}{4} + \frac{N^2}{4}\right] \\ &= \frac{7}{2 \cdot 4} + \frac{91}{6 \cdot 4} = \frac{14}{3} \\ \text{Var}[H] &= \frac{14}{3} - \left(\frac{7}{4}\right)^2 = \frac{77}{48} \end{aligned}$$

Law of Total Variation

8

$$\begin{aligned} E[X] &= E[E[X \mid Y]] \\ E[X^2] &= E[E[X^2 \mid Y]] \end{aligned}$$

but the analogous statement **is not true for variances** which is why we used $\text{Var}[H] = E[H^2] - (E[H])^2$.

For the variance you need an additional term:

Definition (Law of Total Variation)

$$\text{Var}(X) = E[\text{Var}(X \mid Y)] + \text{Var}[E(X \mid Y)]$$

This is heavily tested on later exams, but we only care about one specific application.

Variances of Random Sums

9

Suppose $S = X_1 + \cdots + X_N$, where $X_1, X_2, \dots$ are i.i.d. and N is an independent integer valued random variable. What is $\text{Var}[S]$?

Because we would understand it if N was fixed, we will use total variation.

$$\begin{aligned} E[S \mid N] &= NE[X] \\ \text{Var}[S \mid N] &= N \cdot \text{Var}[X] \\ \text{Var}[S] &= E[\text{Var}[S \mid N]] + \text{Var}[E[S \mid N]] \\ &= E[N\text{Var}[X]] + \text{Var}[NE[X]] \\ &= E[N]\text{Var}[X] + \text{Var}[N](E[X])^2 \end{aligned}$$

The final result is perhaps important enough to be worth memorizing.

Exercise 1

10

The number of losses N is Poisson with mean 3. Loss amounts are mutually independent, and also independent of the number of losses, and have mean 5 and variance 20. What is the expected value of the sum of all the losses?

Exercise 1

10

The number of losses N is Poisson with mean 3. Loss amounts are mutually independent, and also independent of the number of losses, and have mean 5 and variance 20. What is the expected value of the sum of all the losses?

Let X denote a loss amount.

$$\begin{aligned} E[S] &= E[E[S \mid N]] \\ &= E[NE[X]] \\ &= E[N]E[X] \\ &= 3 \cdot 5 = \boxed{15} \end{aligned}$$

Exercise 2

11

The number of losses N is Poisson with mean 3. Loss amounts are mutually independent, and also independent of the number of losses, and have mean 5 and variance 20. What is the variance of the sum of all the losses?

Exercise 2

11

The number of losses N is Poisson with mean 3. Loss amounts are mutually independent, and also independent of the number of losses, and have mean 5 and variance 20. What is the variance of the sum of all the losses?

$$E[N] = 3$$

$$\text{Var}[N] = E[N] = 3$$

$$E[X] = 5$$

$$\text{Var}[X] = 20$$

$$\text{Var}[S] = E[N]\text{Var}[X] + \text{Var}[N](E[X])^2$$

$$= 3 \cdot 20 + 3 \cdot 5^2$$

$$= \boxed{135}$$

Order Statistics

1

Maximums

Minimums

Discrete Example

General Formulas

Exercises

Maximum Example

2

Claim amounts for flood damage are independent random variables with common density function

$$f(x) = \begin{cases} \frac{4}{x^5} & \text{for } x > 1 \\ 0 & \text{otherwise} \end{cases}$$

where x is the amount of a claim in thousands.

Suppose 3 such claims X_1, X_2, X_3 will be made. Find the CDF and density of the largest of the 3 claims.

Let M denote the maximum loss amount. The key idea is that $M \leq x$ if and only if each of the 3 claims are at most x .

Maximum Example (cont.)

3

For one claim, $f(x) = \frac{4}{x^5}$ for $x > 1$

$$\begin{aligned}
 P[M \leq x] &= P[\text{all 3 losses} \leq x] \\
 &= (P[X_1 \leq x])^3 \\
 &= \left(\int_1^x \frac{4}{t^5} dt \right)^3 \\
 &= \left(-\frac{1}{t^4} \Big|_1^x \right)^3 \\
 &= \left(1 - \frac{1}{x^4} \right)^3 \quad x > 1
 \end{aligned}$$

Max Example (Density)

4

What about the density? For $x > 1$,

$$\begin{aligned}
 F_M(x) &= P[\text{all 3 losses} \leq x] \\
 &= \left(1 - \frac{1}{x^4} \right)^3, \quad x > 1 \\
 f_M(x) &= 3 \left(1 - \frac{1}{x^4} \right)^2 \frac{4}{x^5}
 \end{aligned}$$

and the density is 0 for $x < 1$ as the loss amounts must all be at least 1 so their maximum must also be at least 1. (And note $F_M(1) = 0$)

Minimum Example

5

Claim amounts are independent random variables with common density function

$$f(x) = \begin{cases} \frac{4}{x^5} & \text{for } x > 1 \\ 0 & \text{otherwise} \end{cases}$$

where x is the amount of a claim in thousands.

Suppose 3 such claims will be made. What is the expected value of the *smallest* of the three claims?

Let Y denote the *minimum* loss amount. For minimums, the key idea is $Y > x$ only if all the individual losses are $> x$. Note inequalities are reversed from maxes.

We can find $E[Y]$ in one of two ways:

$$E[Y] = \int_0^{\infty} P[Y > y] dy = \int_1^{\infty} y \cdot f_Y(y) dy$$

Minimum Example (Cont).

6

For one claim, $f(x) = \frac{4}{x^5}$ for $x > 1$

$$\begin{aligned} P[Y > y] &= P[\text{all 3 losses} > y] \\ &= \left(\int_y^{\infty} \frac{4}{t^5} dt \right)^3 \\ &= \left(-\frac{1}{t^4} \Big|_y^{\infty} \right)^3 \\ &= \left(\frac{1}{y^4} \right)^3 \quad y > 1 \\ &= \frac{1}{y^{12}} \quad y > 1 \end{aligned}$$

Minimum Example: Expected Value Computation

7

$$P[Y > y] = \frac{1}{y^{12}} \quad y > 1$$

$$\begin{aligned} E[Y] &= \int_0^1 1 \, dx + \int_1^\infty \frac{1}{y^{12}} \, dy \\ &= 1 + \frac{1}{11} = \boxed{\frac{12}{11}} \end{aligned}$$

$$F_Y(y) = 1 - \frac{1}{y^{12}} \quad y > 1$$

$$f_Y(y) = \frac{12}{y^{13}} \quad y > 1$$

$$E[Y] = \int_1^\infty y \cdot \frac{12}{y^{13}} \, dy = \left. \frac{-12}{11y^{11}} \right|_1^\infty = \boxed{\frac{12}{11}}$$

Discrete Example

8

If I roll a fair die 5 times, what is the probability that the maximum roll is 4?

This is harder because we could have exactly one 4 and four smaller rolls or two 4s and three smaller, or ...

But dealing with inequalities is easier.

$$\begin{aligned} P[\text{Max roll} \leq 4] &= P[\text{All rolls are} \leq 4] \\ &= \left(\frac{4}{6}\right)^5 \end{aligned}$$

$$\begin{aligned} P[\text{Max roll} = 4] &= P[\text{Max roll} \leq 4] - P[\text{Max roll} \leq 3] \\ &= \left(\frac{4}{6}\right)^5 - \left(\frac{3}{6}\right)^5 = \boxed{0.100} \end{aligned}$$

General Max/Min Formulas

9

Suppose $X_1, \dots, X_n$ are iid random variables. Let $Y_1 = \min\{X_1, \dots, X_n\}$ and let $Y_n = \max\{X_1, \dots, X_n\}$. Then

$$\begin{aligned} P[\max\{X_1, \dots, X_n\} \leq x] &= P[Y_n \leq x] = [F_X(x)]^n \\ P[\min\{X_1, \dots, X_n\} > x] &= P[Y_1 > x] = (P[X > x])^n \end{aligned}$$

If $X_1, \dots, X_n$ are iid discrete random variables, then

$$\begin{aligned} P[Y_n = x] &= P[Y_n \leq x] - P[Y_n \leq x - 1] \\ &= [F_X(x)]^n - [F_X(x - 1)]^n \\ P[Y_1 = x] &= P[Y_1 \geq x] - P[Y_1 \geq x + 1] \\ &= (P[X \geq x])^n - (P[X \geq x + 1])^n \end{aligned}$$

Exercise 1

10

Suppose that X_1, X_2, X_3, X_4 are iid exponential random variables, each with mean 3. Find the probability that at least one of them exceeds 5.

Exercise 1

10

Suppose that X_1, X_2, X_3, X_4 are iid exponential random variables, each with mean 3. Find the probability that at least one of them exceeds 5.

Since we are talking about maxes, it is easier to work with the CDF. Let M denote the maximum of our X_i . Then

$$\begin{aligned} P[M > 5] &= 1 - P[M \leq 5] \\ &= 1 - P[X_1 \leq 5, X_2 \leq 5, \dots] \\ &= 1 - [F_X(5)]^4 \\ &= 1 - (1 - e^{-5/3})^4 \\ &= \boxed{0.567} \end{aligned}$$

Exercise 2

11

Let $N_1, N_2, \dots, N_5$ be 5 iid Poisson random variables with mean 1.2. Find the probability that the maximum of these 5 variables is 2.

Exercise 2

11

Let $N_1, N_2, \dots, N_5$ be 5 iid Poisson random variables with mean 1.2. Find the probability that the maximum of these 5 variables is 2.

$$\begin{aligned} P[\max\{N_1, \dots, N_5\} = 2] &= P[\max\{N_1, \dots, N_5\} \leq 2] - P[\max\{N_1, \dots, N_5\} \leq 1] \\ &= (P[N_1 \leq 2])^5 - (P[N_1 \leq 1])^5 \\ &= [e^{-1.2}(1 + 1.2 + 1.2^2/2)]^5 - [e^{-1.2}(1 + 1.2)]^5 \\ &= 0.8795^5 - 0.6626^5 \\ &= \boxed{0.3985} \end{aligned}$$

General Order Stats

1

Medians

Distribution and density of order stats

Exercises

Finding the Median

2

Suppose we have some data, and want to know whether or not the median m of the sample is ≤ 4 .

Sample: 1.4, 5.3, 3.8 : $m = 3.8 \leq 4$

Sample: 3.7, 2.3, 1.8 : $m = 2.3 \leq 4$

Sample: 4.4, 5.3, 3.8 : $m = 4.4 > 4$

Sample: 4.1, 6.2, 5.5 : $m = 5.5 > 4$

Sample: 2.6, 1.5, 3.2 : $m = 2.6 \leq 4$

Note that $m \leq 4$ if 2 or 3 of the data points are ≤ 4 , but $m > 4$ if 0 or 1 data point are ≤ 4 .

Example

3

Claim amounts for wind damage to insured homes are independent random variables with common density function

$$f(x) = \begin{cases} \frac{3}{x^4} & \text{for } x > 1 \\ 0 & \text{otherwise} \end{cases}$$

where x is the amount of a claim in thousands.

Suppose 3 such claims will be made. Find the density and CDF of the *median* of the 3 claims.

Let Y denote the median. $Y \leq y$ if either 2 or 3 claims are $\leq y$.

Median Example: CDF

4

$$f(x) = \frac{3}{x^4}, x > 1, Y = \text{Median}$$

$$P[X \leq y] = \int_1^y \frac{3}{x^4} dx = 1 - \frac{1}{y^3}$$

$$\begin{aligned} P[Y \leq y] &= P[\text{at least 2 claims} \leq y] \\ &= P[\text{exactly 2 claims} \leq y] + P[\text{all 3 claims} \leq y] \\ &= \binom{3}{2} \left(1 - \frac{1}{y^3}\right)^2 \frac{1}{y^3} + \left(1 - \frac{1}{y^3}\right)^3 \\ &= \left(1 - \frac{1}{y^3}\right)^2 \left(\frac{3}{y^3} + 1 - \frac{1}{y^3}\right) \\ &= \left(1 - \frac{1}{y^3}\right)^2 \left(1 + \frac{2}{y^3}\right) \end{aligned}$$

Median Example: Density

5

$$\begin{aligned}
 F_Y(y) &= \left(1 - \frac{1}{y^3}\right)^2 \left(1 + \frac{2}{y^3}\right) \\
 f_Y(y) &= 2 \left(1 - \frac{1}{y^3}\right) \frac{3}{y^4} \left(1 + \frac{2}{y^3}\right) + \left(1 - \frac{1}{y^3}\right)^2 \left(\frac{2 \cdot (-3)}{y^4}\right) \\
 &= \frac{6}{y^4} \left(1 - \frac{1}{y^3}\right) \left[1 + \frac{2}{y^3} - 1 - \frac{-1}{y^3}\right] \\
 &= \frac{18}{y^7} \left(1 - \frac{1}{y^3}\right)
 \end{aligned}$$

Distributions of order stats

6

Suppose that we have n data points, denoted as $X_1, X_2, \dots, X_n$, where the X_i are iid random variables. We can sort the n data points from smallest to largest:

$$\begin{aligned}
 X_{(1)} &< X_{(2)} < X_{(3)} < \dots < X_{(n)} \\
 Y_1 &< Y_2 < Y_3 < \dots < Y_n
 \end{aligned}$$

The i -th smallest data point is called the i -th order statistic, and is denoted either as $X_{(i)}$ or Y_i .

Distributions of order stats

7

So Y_1 is the smallest data value, aka the minimum.

$$\begin{aligned}
 P[Y_1 \leq y] &= P[\text{at least one } X_i \text{ is } \leq y] \\
 &= 1 - P[\text{all } X_i > y] \\
 &= 1 - (1 - P(X_1 \leq y))^n \\
 &= 1 - (1 - F_X(y))^n
 \end{aligned}$$

Y_n is the largest data value (aka the max), and

$$\begin{aligned}
 P[Y_n \leq y] &= P[\text{all } n \text{ points are } \leq y] \\
 &= [F_X(y)]^n
 \end{aligned}$$

Distributions of Order Stats

8

The other order statistics are messier (and are tested less):

$$\begin{aligned}
 P[Y_n \leq y] &= [F_X(y)]^n \\
 P[Y_{n-1} \leq y] &= P[\text{at least } n-1 \text{ are } \leq y] \\
 &= P[\text{exactly } n-1 \text{ are } \leq y] + P[\text{all } n \text{ are } \leq y] \\
 &= \binom{n}{n-1} [F_X(y)]^{n-1} [1 - F_X(y)] + [F_X(y)]^n \\
 P[Y_{n-2} \leq y] &= P[\text{at least } n-2 \text{ are } \leq y] \\
 &= P[\text{exactly } n-2 \leq y] + P[\text{at least } n-1 \text{ are } \leq y] \\
 &= \binom{n}{n-2} [F_X(y)]^{n-2} (1 - F_X(y))^2 + P[Y_{n-1} \leq y] \\
 &\vdots
 \end{aligned}$$

Densities of Order Stats

9

The density of a general order stat has a nicer formula:

$$f_i(y) = \text{density of } Y_i$$

$$f_i(y)dy = P[y \leq Y_i \leq y + dy]$$

For that to happen, we need $i - 1$ data values to be less than y , one that is between y and $y + dy$, and $n - i$ that are greater than y

$$f_i(y)dy = i \cdot \binom{n}{n-i} (1 - F_X(y))^{n-i} \cdot (F_X(y))^{i-1} \cdot P[y \leq X \leq y + dy]$$

$$f_i(y) = i \cdot \frac{n!}{i!(n-i)!} [1 - F_X(y)]^{n-i} F_X(y)^{i-1} \cdot f_X(y)$$

$$= \frac{n!}{(i-1)!(n-i)!} [1 - F_X(y)]^{n-i} F_X(y)^{i-1} \cdot f_X(y)$$

Exercise 1

10

Suppose that X_1, X_2, X_3, X_4, X_5 are iid exponential random variables, each with mean 4. Find the density of the median of those variables.

Exercise 1

10

Suppose that X_1, X_2, X_3, X_4, X_5 are iid exponential random variables, each with mean 4. Find the density of the median of those variables.

Let Y_3 denote the median order statistic. From our formula,

$$\begin{aligned} f_3(y) &= 3 \cdot \binom{5}{5-3} [1 - F_X(y)]^{5-3} F_X(y)^{3-1} f_X(y) \\ &= 3 \cdot 10 [e^{-y/4}]^2 (1 - e^{-y/4})^2 \frac{1}{4} e^{-y/4} \\ &= \frac{15}{2} e^{-3y/4} (1 - e^{-y/4})^2 \end{aligned}$$

Exercise 2

11

Let $W_1, W_2, \dots, W_4$ be 4 iid Poisson random variables with mean 3.2. Find the probability that the minimum of these 4 variables is 2.

Exercise 2

11

Let $W_1, W_2, \dots, W_4$ be 4 iid Poisson random variables with mean 3.2. Find the probability that the minimum of these 4 variables is 2.

$$\begin{aligned} P[\min\{W_1, \dots, W_4\} = 2] &= P[\min\{W_1, \dots, W_4\} \geq 2] - P[\min\{W_1, \dots, W_4\} \geq 3] \\ &= (P[W_1 \geq 2])^4 - (P[W_1 \geq 3])^4 \\ &= (1 - P[W_1 = 0] - P[W_1 = 1])^4 - (P[W_1 \geq 3])^4 \\ &= (1 - \exp(-3.2)(1 + 3.2))^4 \\ &\quad - (1 - \exp(-3.2)(1 + 3.2 + 3.2^2/2))^4 \\ &= 0.8288^4 - 0.6201^4 \\ &= \boxed{0.324} \end{aligned}$$

Multivariate Review

1

Joint Distributions

Marginal Distributions

Conditional Distributions

Uniforms

Moments

Order Stats

Joint Distributions

2

$$P[(X, Y) \in A] = \sum_{(x,y) \in A} P[(X, Y) = (x, y)]$$

$$\sum_{x,y} P[(X, Y) = (x, y)] = 1$$

$$F(x, y) = F_{X,Y}(x, y) = P[X \leq x, Y \leq y]$$

$$F_{X,Y}(x, \infty) = F_X(x)$$

$$F_{X,Y}(\infty, y) = F_Y(y)$$

Marginal Distributions

3

In the discrete case

$$P[X = x] = \sum_y P[(X, Y) = (x, y)]$$

$$P[Y = y] = \sum_x P[(X, Y) = (x, y)]$$

Essentially we are breaking the probability up into all the possible cases, with the second variable counting the cases.

The marginal of X can only depend on x , not y . Likewise, the marginal of Y can only depend on y , not x .

Conditional Distributions

4

In the discrete case

$$P[Y = y | X = x] = \frac{P[X = x, Y = y]}{P[X = x]}$$

$$= \frac{P[X = x, Y = y]}{\sum_y P[X = x, Y = y]}$$

$$P[X = x, Y = y] = P[X = x] \cdot P[Y = y | X = x]$$

X and Y are independent if 1) the joint distribution factors and 2) the support is a rectangle.

The conditional distribution of Y given $X = x$ can depend on both x and y .

Example

5

The joint distribution of X and Y is given by $P[X = x, Y = y] = \frac{(x^2 + 1)(y + 1)}{73}$ for x and y integers such that $0 \leq y \leq |x| \leq 2$, and is 0 otherwise. Find the marginal distribution of Y .

$$\begin{aligned}
 P[Y = 2] &= P[Y = 2, X = 2] + P[Y = 2, X = -2] \\
 &= \frac{1}{73} [(2^2 + 1)(2 + 1) + ((-2)^2 + 1)(2 + 1)] \\
 &= \frac{15 + 15}{73} = \frac{30}{73} \\
 P[Y = 1] &= 2(P[Y = 1, X = 2] + P[Y = 1, X = 1]) \\
 &= \frac{2}{73} [(2^2 + 1)(1 + 1) + (1^2 + 1)(1 + 1)] \\
 &= \frac{2}{73} (10 + 4) = \frac{28}{73} \\
 P[Y = 0] &= 1 - P[Y = 1] - P[Y = 2] = \frac{73 - 28 - 30}{73} = \frac{15}{73}
 \end{aligned}$$

Example

6

The joint distribution of X and Y is given by $P[X = x, Y = y] = \frac{(x^2 + 1)(y + 1)}{73}$ for x and y integers such that $0 \leq y \leq |x| \leq 2$, and is 0 otherwise. Find the conditional probability that $X = 1$ given $Y = 1$.

$$\begin{aligned}
 P[X = 1 \mid Y = 1] &= \frac{P[X = 1, Y = 1]}{P[Y = 1]} \\
 &= \frac{\frac{1}{73} [(1^2 + 1)(1 + 1)]}{\frac{28}{73}} \\
 &= \frac{4}{28} \\
 &= \frac{1}{7}
 \end{aligned}$$

Uniforms

7

(X, Y) are jointly uniform if all possible (x, y) are equally likely, i.e., the joint probability function is constant.

If (X, Y) are jointly uniform, then the conditional distributions are uniform. The marginals need not be uniform.

If $P[(X, Y) = (x, y)]$, when positive, only depends on x , then the conditional distribution of Y is constant and hence uniform. If $P[(X, Y) = (x, y)]$ only depends on y , then the conditional distribution of X is uniform.

Moments

8

$$E[g(X, Y)] = \sum_{x,y} g(x, y) \cdot P[(X, Y) = (x, y)]$$

$$E[X] = \sum_{x,y} x \cdot P[(X, Y) = (x, y)]$$

$$E[Y^2] = \sum_{x,y} y^2 \cdot P[(X, Y) = (x, y)]$$

$$E[Y | X = x] = E[Y | X] = \sum y \cdot P[Y = y | X = x]$$

$$E[Y^k] = E[E[Y^k | X]]$$

$$\text{Var}[Y] = E[\text{Var}[Y | X]] + \text{Var}[E[Y | X]]$$

If $S = \sum_{i=1}^N X_i$, where X_i are iid and N is independent,

$$\text{Var}[S] = E[N] \cdot \text{Var}[X] + \text{Var}[N] \cdot (E[X])^2$$

Example

9

The joint distribution of X and Y is $P[X = x, Y = y] = x/14$ for x and y integers with $1 \leq y \leq x \leq 3$. Find $E[X]$ and $E[Y]$.

$$\begin{aligned} E[X] &= \sum_{x=1}^3 \sum_{y=1}^x x \cdot \frac{x}{14} = 1 \cdot \frac{1}{14} + 2 \cdot \frac{2}{14} + 2 \cdot \frac{2}{14} + 3 \cdot \frac{3}{14} \cdot 3 \\ &= \frac{1 + 8 + 27}{14} = \frac{18}{7} \end{aligned}$$

$P[X = x, Y = y]$ doesn't involve y , so $(Y \mid X = x)$ is uniform on $\{1, \dots, x\}$

$$\begin{aligned} E[Y \mid X] &= \frac{1 + X}{2} \\ E[Y] &= E[E[Y \mid X]] = E\left[\frac{1 + X}{2}\right] \\ &= \frac{1}{2} + \frac{1}{2} \cdot \frac{18}{7} = \frac{25}{14} \end{aligned}$$

Covariance

10

$$\begin{aligned} \text{Var}[X] &= \sigma^2 = E[X^2] - (E[X])^2 = E[(X - \mu_X)^2] \\ \text{Cov}(X, Y) &= E[XY] - E[X] \cdot E[Y] = E[(X - \mu_X)(Y - \mu_Y)] \\ \text{Var}[X + Y] &= \text{Var}[X] + 2\text{Cov}(X, Y) + \text{Var}[Y] \\ \text{Var}[aX + bY] &= a^2 \text{Var}[X] + 2ab \text{Cov}(X, Y) + b^2 \text{Var}[Y] \\ \text{Var}[X - Y] &= \text{Var}[X] - 2\text{Cov}(X, Y) + \text{Var}[Y] \\ \text{Corr}(X, Y) &= \rho = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \cdot \text{SD}(Y)} \end{aligned}$$

If X and Y are independent, then $\text{Cov}(X, Y) = 0$.

Example

11

Suppose that X and Y are random variables with $\text{Corr}(X, Y) = 0.3$, $E[X] = E[Y] = 1$ and $\text{SD}[X] = \text{SD}[Y] = 2$. What are $E[2X - 3Y]$ and $\text{Var}[2X - 3Y]$?

$$\begin{aligned} E[2X - 3Y] &= 2 \cdot E[X] - 3 \cdot E[Y] \\ &= -1 \\ \text{Var}[2X - 3Y] &= 4\text{Var}[X] - 12\text{Cov}[X, Y] + 9\text{Var}[Y] \\ &= 4 \cdot 4 - 12 \cdot 0.3 \cdot 2 \cdot 2 + 9 \cdot 4 \\ &= 37.6 \end{aligned}$$

Order Stats

12

Suppose $X_1, \dots, X_n$ are iid and integer valued.
Then for integer k ,

$$\begin{aligned} P[\max\{X_1, \dots, X_n\} = k] &= P[\max\{X_1, \dots, X_n\} \leq k] - P[\max\{X_1, \dots, X_n\} \leq k - 1] \\ &= (P[X \leq k])^n - (P[X \leq k - 1])^n \\ P[\min\{X_1, \dots, X_n\} = k] &= P[\min\{X_1, \dots, X_n\} \geq k] - P[\min\{X_1, \dots, X_n\} \geq k + 1] \\ &= (P[X \geq k])^n - (P[X \geq k + 1])^n \end{aligned}$$